Lecture Notes for Math 243:
Calculus III

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Contents

About this document

These lecture notes were prepared by Max Hill for a 16-week Calculus III course (MATH 243) at University of Hawaii at Manoa in Spring 2025. This course emphasizes differentiation, as UH Manoa has a Calclulus IV devoted to integration theory.

The textbook used is James Stewart's *Calculus* (8th edition), in which we cover primarily chapters 10, 12, 13, and 14. In addition to this text, the material in these lecture notes is also based on a number of other sources as well, including:

- Stephen Abbott's *Understanding Analysis* (2001)
- Gilbert Strang's *Calculus*, freely available at `https://ocw.mit.edu/ans7870/resources/Strang/Edited/Calculus/Calculus.pdf`
- *Supplementary Notes for Calculus II*, written by Michael Olinick with the assistance of John D. Emerson. These are course notes for a calculus class from 2012.
- Sigurd B. Angenent's *Math 222 - 2nd Semester Calculus* lecture notes (2013) `https://people.math.wisc.edu/~angenent/222.2013s/UWMath222_2013s.pdf`
- Steve Butler's calculus notes available at `https://www.calc2.org/previous-examsquizzes`
- Michael Fowler's text at `https://galileoandeinstein.phys.virginia.edu/lectures/momentum.html`, for physics stuff

Most of the figures were produced with ChatGPT using the tikz package.

0 Tentative Course Outline

- **Weeks 1-3:** Parametric equations and Polar coordinates
  1. Section 8.1: Arc length
  2. Section 10.1: Curves defined by parametric equations.
  3. Section 10.2: Calculus with plane curves.
  4. Section 10.3: Polar coordinates
  5. Section 10.4: Areas and lengths in polar coordinates
  6. ~~Section 10.5: Conic sections.~~
- **Weeks 4-6:** Vectors and the geometry of space.
  1. Section 12.1: Three-dimensional coordinate systems.
  2. Section 12.2: Vectors.
  3. Section 12.3: The dot product.
  4. Section 12.4: The cross product.
  5. Section 12.5: Equations of lines and planes.
  6. Section 12.6: Cylinders and quadric surfaces.
- **Weeks 7-9:** Vector functions.
  1. Section 13.1: Vector functions and space curves
  2. Section 13.2: Derivatives and integrals of vector functions.
  3. Section 13.3: Arc length and curvature.
  4. Section 13.4: Motion in space: velocity and acceleration.
- **Weeks 10-15:** Partial derivatives.
  1. Section 14.1: Functions of several variables.
  2. Section 14.2: Limits and continuity
  3. Section 14.3: Partial derivatives.
  4. Section 14.4: Tangent planes and linear approximations
  5. Section 14.5: The chain rule.
  6. Section 14.6: Directional derivatives and the gradient vector.
  7. Section 14.7: Maximum and minimum values.
  8. Discovery project p.1010: Quadratic approximation and critical points (Taylors formula for two variables).
  9. Section 14.8: Lagrange multipliers.

1 2025-01-13 | Week 1 | Lecture 1

This lecture is based on section 8.1 in the textbook.

1.1 Arc Length

First, recall the definition of derivative:

$$f'(x) := \lim_{h\to 0} \frac{f(x+h) - f(x)}{h}$$

or equivalently,

$$f'(x) = \lim_{\tilde{x}\to x} \frac{f(\tilde{x}) - f(x)}{\tilde{x} - x} =$$

In other words, the derivative is obtained by computing

$$\frac{\text{change in } y\text{-value}}{\text{change in } x\text{-value}} = \frac{\Delta y}{\Delta x}$$

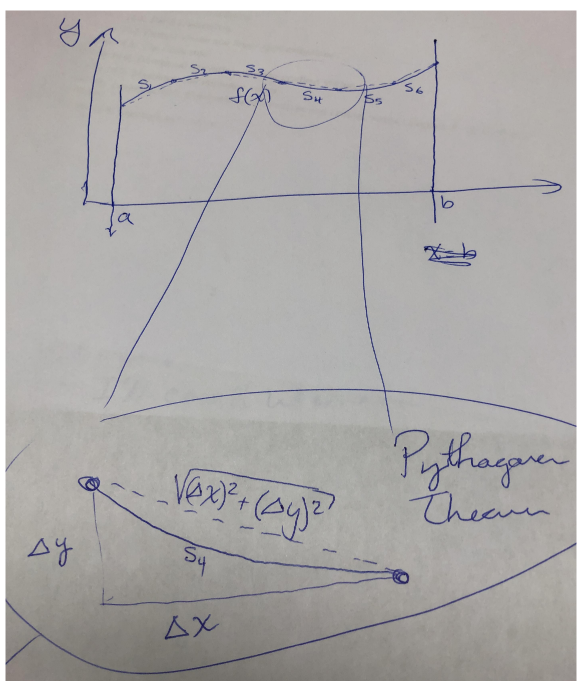
as $\Delta x \to 0$.

One of the central ideas in calculus is to approximate curvy things with flat things. For example, if faced with a curve, approximate it with straight lines. For example,

Problem: I am an ant, and I want to travel from a to b by walking on the graph of $f(x)$. How far must I walk?

Strategy: Cheat a little: think of the ant walking in a bunch of straight lines that are pretty close to the curve.

Here's the picture



Here,

$$L = s_1 + s_2 + \ldots + s_6$$

where, for example,

$$
\begin{aligned}
s_4 &\approx \sqrt{(\Delta x)^2 + (\Delta y)^2} && \text{since the line segment length approximates the curve length} \\
&= \sqrt{\left[1 + \left(\frac{\Delta y}{\Delta x}\right)^2\right](\Delta x)^2} \\
&= \sqrt{1 + \left(\frac{\Delta y}{\Delta x}\right)^2}\,\Delta x \\
&\approx \sqrt{1 + \left(\frac{dy}{dx}\right)^2}\,dx && \text{this holds when } \Delta x \approx 0\text{, so long as the curve is smooth} \\
&= \sqrt{1 + [f'(x)]^2}\,dx && \text{by change of notation}
\end{aligned}
$$

Theorem 1 (Arc length formula)**.** *Let f be a differentiable function on the interval $[a, b]$, and assume that f' is continuous on $[a, b]$. Then the arc length of f from $x = a$ to $x = b$ is*

$$
\textit{Arc Length} = \int_a^b \sqrt{1 + [f'(x)]^2}\,dx
$$

Example 2 (Arc length)**. Problem:** Find the length of the curve defined by $f(x) = 2x^{3/2}$, where x ranges from $x = 0$ t0 $x = 2$.

Solution: $f'(x) = 3x^{\frac{1}{2}}$, so

$$
\begin{aligned}
\text{Arc length} &= \int_0^2 \sqrt{1 + [f'(x)]^2}\,dx \\
&= \int_0^2 \sqrt{1 + 9x}\,dx
\end{aligned}
$$

which we can solve with a u-substitution.

End of Example 2. □

Example 3 (Arc length)**. Problem:** Find the length of the curve defined by $f(x) = \frac{1}{2}\left(e^x + e^{-x}\right)$ from $x = -1$ to $x = 3$.

Solution: We first claim that the following equality holds:

$$
1 + [f'(x)]^2 = \left(\frac{e^x + e^{-x}}{2}\right)^2. \tag{1}
$$

This equality isn't obvious... it takes some work to deduce, which is shown below:

Proof of Eq. (1). Observe that $f'(x) = \frac{1}{2}(e^x - e^{-x})$. Therefore

$$\begin{aligned}
1 + [f'(x)]^2 &= 1 + \left[\frac{1}{2}\left(e^x - e^{-x}\right)\right]^2 \\
&= 1 + \frac{1}{4}\left(e^{2x} - 2 + e^{-2x}\right) \\
&= \frac{1}{4}e^{2x} + \frac{1}{2} + \frac{1}{4}e^{-2x} && \text{expanding everything out} \\
&= \frac{1}{4}\left(e^{2x} + 2 + e^{-2x}\right) && \text{factor out } \frac{1}{4} \\
&= \frac{1}{4}\left(e^x + e^{-x}\right)^2 \\
&= \left(\frac{e^x + e^{-x}}{2}\right)^2.
\end{aligned}$$

This shows that Eq. (1) holds. □

Now with Eq. (1) in hand, everything simplifies really nicely when we apply the formula from Theorem 1:

$$\begin{aligned}
\text{Arc Length} &= \int_{-1}^{3} \sqrt{1 + [f'(x)]^2}\, dx \\
&= \int_{-1}^{3} \sqrt{\left(\frac{e^x + e^{-x}}{2}\right)^2}\, dx \\
&= \int_{-1}^{3} \frac{e^x + e^{-x}}{2}\, dx \\
&= \frac{1}{2}\int_{-1}^{3} \left(e^x + e^{-x}\right) dx \\
&= \frac{1}{2}\left[e^x - e^{-x}\right]_{x=-1}^{x=3} \\
&= \frac{1}{2}\left[\left(e^3 - e^{-3}\right) - \left(e^{-1} - e^1\right)\right] \\
&\approx 11.2
\end{aligned}$$

End of Example 3. □

Tips

- Check that your answer is positive. Arc length is never negative.
- These problems are usually contrived so that you can actually do the integral. If the integral looks took hard, double check your work. Also, there might be a trick to the integral.

2 2025-01-15 | Week 1 | Lecture 2

This lecture is based on chapters 8.1 and 10.1 in the textbook.

Example 4 (Warm up problem: Chain rule practice)**.** Find the derviative of

$$f(x) = \log\left(\frac{x^2}{\sin x}\right).$$

Solution:

$$\begin{aligned}
f'(x) &= \frac{d}{dx}\left[\log\left(\frac{x^2}{\sin x}\right)\right] \\
&= \frac{\sin x}{x^2}\frac{d}{dx}\left[\frac{x^2}{\sin x}\right] && \text{by the chain rule} \\
&= \frac{\sin x}{x^2}\left(\frac{2x\sin x - x^2\cos x}{(\sin x)^2}\right) && \text{by the quotient rule} \\
&= \frac{2}{x} - \cot x && \text{after simplification.}
\end{aligned}$$

End of Example 4. □

2.1 Arc length (continued)

This section is based on chapter 8.1 in the textbook.

Example 5 (Arc length)**. Problem:** Let

$$f(x) = \frac{x^3}{3} + \frac{1}{4x} + 7,$$

for $1 \leq x \leq 2$. earlier examples.

(*Hint:* This problem is similar to Example 3.)

Solution: We want to apply Theorem 1. To do this, we first compute $1 + [f'(x)]^2$:

$$\begin{aligned}
1 + [f'(x)]^2 &= 1 + \left(x^2 - \frac{1}{4}x^{-2}\right)^2 \\
&= x^4 + \frac{1}{2} + \frac{1}{16}x^{-4} \\
&= \left(x^2 + \frac{1}{4}x^{-2}\right)^2.
\end{aligned}$$

The last step, where we write the equation as a perfect square, is kinda hard to see. But since we're familiar with Example 3, we know to possibly expect something like that.

Now, we can apply the formul from Theorem 1:

$$\begin{aligned}
\text{Arc Length} &= \int_1^2 \sqrt{1+[f'(x)]^2}dx \\
&= \int_1^2 \sqrt{\left(x^2+\frac{1}{4}x^{-2}\right)^2}dx \\
&= \int_1^2 x^2+\frac{1}{4}x^{-2}dx \\
&= \left[\frac{x^3}{3}-\frac{1}{4}x^{-1}\right]_1^2 \\
&= \left(\frac{8}{3}-\frac{1}{8}\right)-\left(\frac{1}{3}-\frac{1}{4}\right) \\
&\approx 2.46.
\end{aligned}$$

End of Example 5. □

2.2 Parametric equations

This section is based on chapter 10.1 in the textbook.

I am an ant walking around the whiteboard, and I'm covered in ink. My path is described in the following way:

- The variable t measures time. We start at $t=0$ and end at time $t=T_{\text{end}}>0$.
- At each moment in time t, the number $x(t)$ represents my x-coordinate
- And the number $y(t)$ represents my y-coordinate.

The *parametric equation* is the picture I draw by walking around the whiteboard.

Definition 6 (Parametric Equations and Curves)**.** Let I be an interval, and let $f : I \to \mathbb{R}$ and $g : I \to R$ be continuous functions whose domain is I. The set of all points of the form

$$(x, y) = (f(t), g(t))$$

as t varies over I, is the *graph* of the *parametric equations* $x=f(t)$ and $y=g(t)$. Here, t is called the *parameter*. A *curve* is a graph along with the parametric equations that define it.

Notation 7. In "set-builder notation," the graph of the parametric equations $x=f(t)$ and $y=g(t)$ is the set

$$\{(f(t), g(t)) : t \in I\}.$$

For those unfamiliar with this notnation: the colon ' :' is read as "such that" and 'ϵ' is an abbreviation of "is an element of"

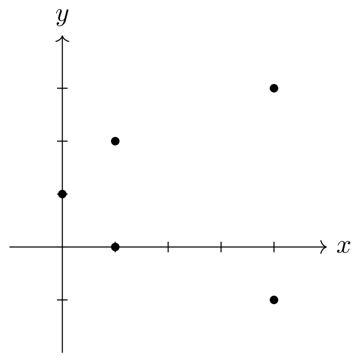
Example 8 (Plotting parametric functions)**.** Plot the graph of the parametric equations

$$\begin{cases} x = t^2 \\ y = t+1 \end{cases}$$

for t in $I=[-2,2]$.

Solution: Start by plotting some points by making a table.

t	$x = t^2$	$y = t + 1$
-2	4	-1
-1	1	0
0	0	1
1	1	2
2	4	3



Then we can draw a curve through the points. (Not shown).

If we replace $I = [-2, 2]$ by $I = \mathbb{R}$, then thie corresponding graph would trace out a sideways opening parabola defined by the equation $x = (y-1)^2$.

Question: can we write y a function of x?

No! No, because the graph doesn't satisfy the vertical line test.

Question: can we express this as an equation involving only x and y?

Yes! $x = (y-1)^2$.

End of Example 8. $\square$

3 2025-01-17 | Week 1 | Lecture 3

This lecture is based on section 10.1 in the textbook

Reading: Please read sections 10.1-10.3 in the text book.

3.1 Plotting parametric equations

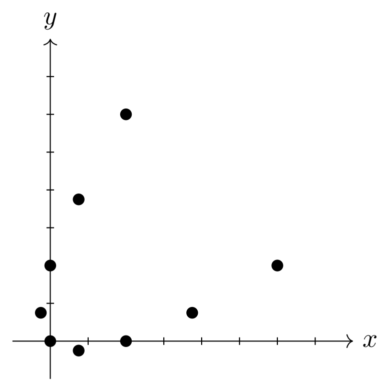
Example 9. *Question:* Plot the graph of the parametric equations

$$(x, y) = (t^2 + t, t^2 - t)$$

as the parameter t ranges over $I = [-2, 2]$.

Solution: Again, we plot some points by making a table.

t	$x = t^2 + t$	$y = t^2 - t$
-2	2	6
$-\frac{3}{2}$	.75	3.75
-1	0	2
$-\frac{1}{2}$	$-\frac{1}{4}$	$\frac{3}{4}$
0	0	0
$\frac{1}{2}$	$\frac{3}{4}$	$-\frac{1}{4}$
1	2	0
$\frac{3}{2}$	3.75	$\frac{3}{4}$
2	6	2



The graph of the parametric equations is obtained by drawing the curve through the plotted points.

End of Example 9. □

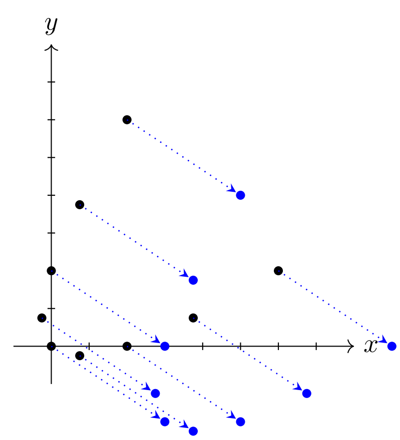
Example 10. Find the parametric equations that shift the graph from Example 9 to the right by 3 and down by 2.

Solution: The graphs of parametric equations are easy to shift. The parameteric equations defining the shifted graph are

$$(x, y) = (t^2 + t + 3, t^2 - t - 2)$$

We've simply added 3 to the x-value and subtracted 2 from the y-value. To see why this is the case, observe what happens to the points in the plot from Example 9:

t	$x = t^2 + t + 3$	$y = t^2 - t - 2$
-2	$2 + 3$	$6 - 2$
$-\frac{3}{2}$	$.75 + 3$	$3.75 - 2$
-1	$0 + 3$	$2 - 2$
$-\frac{1}{2}$	$-\frac{1}{4} + 3$	$\frac{3}{4} - 2$
0	$0 + 3$	$0 - 2$
$\frac{1}{2}$	$\frac{3}{4} + 3$	$-\frac{1}{4} - 2$
1	$2 + 3$	$0 - 2$
$\frac{3}{2}$	$3.75 + 3$	$\frac{3}{4} - 2$
2	$6 + 3$	$2 - 2$



Here we have shifted every point to the right by 3 and down by 2, as shown by the arrows. Drawing a curve through the blue points now gives us the curve of the new, shifted, parametric equations. (Not shown in figure).

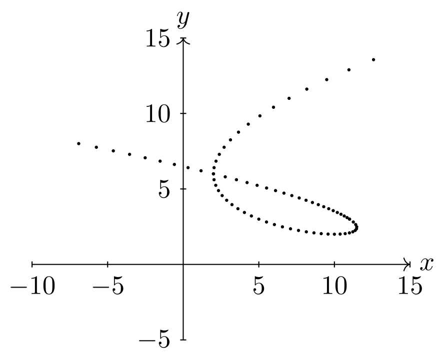
End of Example 10. □

Parametric functions have the advantage that we can use them to plot curves which cannot be plotted by "$y = f(x)$" type graphs.

Example 11. Here's an example of a plot of the parametric curve

$$\begin{pmatrix} x \\ y \end{pmatrix} = \begin{pmatrix} t^3 - 5t^2 + 3t + 11 \\ t^2 - 2t + 3 \end{pmatrix}$$

This one would be tedious to plot by hand, so instead I plotted a bunch of points with a computer, with t varying on the interval $I = [-1.5, 4.5]$.



The parametric equation can be thought of as tracing the trajectory of a particle as it moves over time. In this example, t represents time. At time $t = -1.5$, the particle is at point $(x, y) = (-8.125, 8.25)$, and starts heading southeast. It then turns around and loops over its previous path, before heading northeast.

It would be impossible to represent this trajectory using a function $y = f(x)$, becausethe curve fails the vertical line test. This is one advantage of parametric curves.

The particle has the same position at times $t = -1$ and $t = 3$. At these times, the particle is at the point

$$(x, y) = (2, 6).$$

Thus, it is at this point that the curve intersects itself, a phenomenon known as a *singularity*.

End of Example 11. □

3.2 Converting between parametric equations and Cartesian equations

In contrast to parametric equations, an equation is said to be in *rectangular form* (aka: *Cartesian form*) if it takes the form $y = f(x)$ (or, less commonly, $x = f(y)$) for some function f. We've just seen in Example 11 that parametric equations offer significantly more flexibility when graphing.

In fact, it is always possible to convert an equation from rectangular form into parametric form. We illustrate this with the next example:

Example 12 (Converting from rectangular form to parametric form)**.** Consider the equation $y = x^2$, which is in rectangular form. This is an upward-opening parabola. There are many ways to represent this in parametric form. The simplest is with the parametric equations

$$(x, y) = (t, t^2)$$

where t ranges over the interval $(-\infty, \infty)$.

End of Example 12. □

Example 13 (Converting from parametric form to rectangular form)**.** Sometimes (but not always) we can also convert from parametric to rectangular form. Consider the parametric equations

$$\begin{cases} x = t^2 \\ y = 7 + 5t \end{cases}$$

This is done by "eliminating" the parameter t, in the following way:

Since $y = 7 + 5t$, it follows that

$$t = \frac{y-7}{5}.$$

Plugging this into $x = t^2$ gives

$$\begin{aligned} x &= \left(\frac{y-7}{5}\right)^2 \\ &= \frac{1}{25}\left(y^2 - 14y + 49\right) \\ &= \frac{1}{25}y^2 - \frac{14}{25}y + \frac{49}{25}. \end{aligned}$$

This equation has the form $x = f(y)$, and therefore is in rectangular (i.e., Cartesian) form.

End of Example 13. □

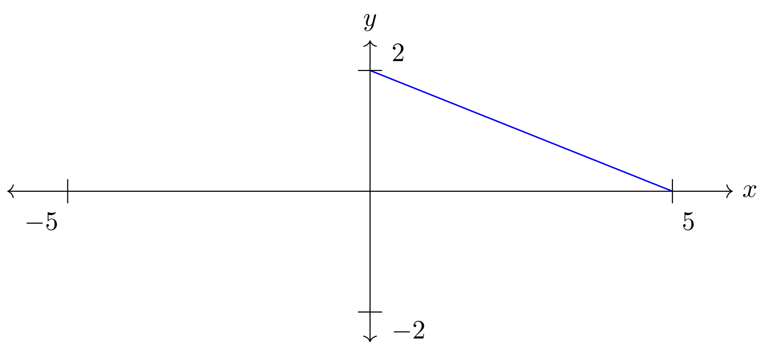
Here's another example

Example 14 (Converting to Cartesian form: eliminating a parameter)**.** Consider the parametric equations

$$\begin{cases} x = 5\cos^2(\theta) \\ y = 2\sin^2(\theta) \end{cases}$$

with $0 \leq \theta \leq 2\pi$.

I plotted this using a computer and got the following. It is a line segment:



Question: Convert these parametric equations into Cartesian form, ie a function of the form $y = f(x)$.

Solution: Before we start, recall the identity $\sin^2(\theta) + \cos^2(\theta) = 1$ for all θ. using this, we have:

$$\begin{aligned} y &= 2\sin^2(\theta) \\ &= 2\left(1 - \cos^2(\theta)\right) \end{aligned}$$

Now, $x = 5\cos^2(\theta)$, so $\cos^2(\theta) = \frac{x}{5}$. Plugging this into the above equation gives:

$$y = 2\left(1 - \frac{x}{5}\right).$$

This is an equation of a line.

But we are not done! We are trying to plot only a line *segment*, not the full line. So we need to specify that we are restricting the line to only those x in the interval $[0, 5]$. So our solution (in Cartesian form) is

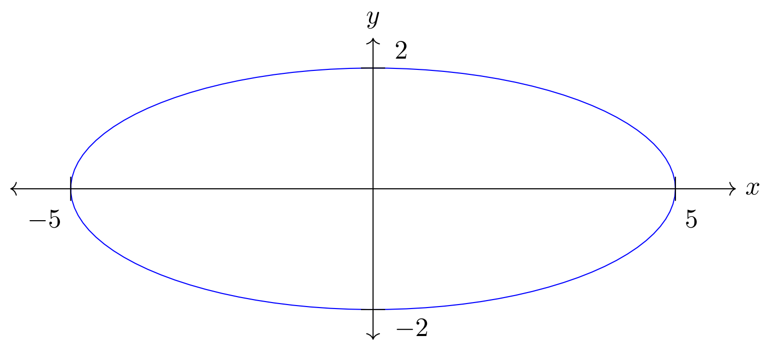
$$\begin{cases} y = 2\left(1 - \frac{x}{5}\right) \\ 0 \leq x \leq 5 \end{cases}$$

End of Example 14. □

Example 15 (An ellipse, parametrized)**.** What if we slightly modify the equations from the Example 14? Consider the parametric equations

$$\begin{cases} x = 5\cos(\theta) \\ y = 2\sin(\theta) \end{cases}$$

with $0 \leq \theta \leq 2\pi$. Here, the only difference from the equations in Example 14 is that we have removed the squares. Now we get a very different looking curve: an ellipse.



End of Example 15. □

4 2025-01-22 | Week 2 | Lecture 4

This lecture covers material from section 10.2 in the textbook.

4.1 Parameterized Ellipses

Example 16 (Eliminating a parameter)**.** Eliminate the parameter t in the parametric system of equations

$$\begin{cases} x = 4\cos(t) + 3 \\ y = 2\sin(t) + 1 \end{cases}$$

Before we start this problem, let's observe that these parametric equations trace out an ellipse that has been shifted RIGHT 3 and UP 1. The horizontal radius is 4 and the vertical radius is 2. So when we convert to Cartesian form by eliminating the paramter t, we should expect to get the equation of this ellipse.

Solution: The trig identity $\sin^2(t) + \cos^2(t) = 1$ served us well in Example 14, so let's try to use that.

First, solve for $\sin(t)$ and $\cos(t)$:

$$\sin(t) = \frac{y-1}{2} \quad \text{and} \quad \cos(t) = \frac{x-3}{4} \tag{2}$$

By the trig identity, we have

$$\cos^2(t) + \sin^2(t) = 1.$$

Substituting in the formulas from Eq. (2) gives:

$$\left(\frac{x-3}{4}\right)^2 + \left(\frac{y-1}{2}\right)^2 = 1,$$

or equivalently

$$\frac{1}{16}(x-3)^2 + \frac{1}{4}(y-1)^2 = 1.$$

This is indeed the equation of an ellipse, with center $(3, 1)$ and with horizontal and vertical axes 4 and 2 respectively.

End of Example 16. □

Remark 17. In Examples 15 and 16, we saw equations of ellipses. In general the parametric equations for an ellipse take the form

$$\begin{cases} x = a\cos(t) + h \\ y = b\sin(t) + k \end{cases}$$

where (h, k) is the center of the ellipse, a is the horizontal radius, and b is the vertical radius.

The Cartesian equations of the same ellipse are

$$\frac{1}{a^2}(x-h)^2 + \frac{1}{b^2}(h-k)^2 = 1.$$

4.2 Smoothness of parameterized curves

Definition 18 (Smooth)**.** A curve C defined by the parametric equations

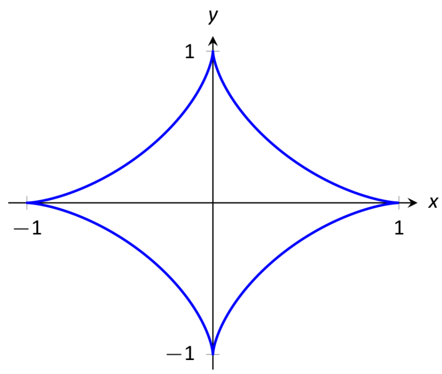
$$\begin{cases} x = f(t) \\ y = g(t) \end{cases}$$

is *smooth* on an interval I if f' and g' are continuous on I and not simultaneously zero on the interior of I. A curve is *piecewise smooth* on I if I can be partitioned into subintervals such that C is smooth on each subinterval.

Example 19 (A non-smooth curve)**.** Consider the equations

$$\begin{cases} x(t) := \cos^3(t) \\ y(t) := \sin^3(t) \end{cases}$$

with t in $I = [0, \infty)$. This has graph



Astroid
$x = \cos^3 t$
$y = \sin^3 t$

This sure doesn't look smooth, because it has pointy cusps! Let's show that using the definition.

First, differentiate:

$$\begin{cases} x'(t) = -3\cos^2(t)\sin(t) \\ y'(t) = 3\sin^2(t)\cos(t) \end{cases}$$

Are these functions ever simultaneously zero? Yes. For example, when $t = 0$. Or $t = \frac{\pi}{2}, \pi, \frac{3\pi}{2}, 2\pi, \ldots$. Therefore the curve is not smooth (although it is piecewise smooth).

End of Example 19. □

4.3 Tangent and normal lines to parameterized curves

Question 20. *Suppose we have a parametric system of equations*

$$\begin{cases} x = f(t) \\ y = g(t) \end{cases}$$

for $t \in I$. How can we find the derivative $\frac{dy}{dx}$?

The answer to this question is the following important idea:

$$\frac{dy}{dx}(t) = \frac{g'(t)}{f'(t)} \tag{3}$$

provided that the following two assumptions hold:

- f and g are both differentiable (otherwise f' and g' wouldn't exist)
- $f'(t) \neq 0$ (otherwise we'd have divsion by zero)

Definition 21 (Tangent Line and Normal Line)**.** Let C be a curve parameterized by

$$\begin{cases} x = f(t) \\ y = g(t) \end{cases}$$

where $f, g : I \to \mathbb{R}$ are differentiable functions. Let s be a point in I.

- The *tangent line* to C at $t = s$ is the line

$$y - g(s) = \frac{g'(s)}{f'(s)}\left(x - f(s)\right),$$

provided that $f'(s) \neq 0$.

- The *normal line* to C at $t = s$ is the line

$$y - g(s) = -\frac{f'(s)}{g'(s)}\left(x - f(s)\right),$$

provided that $g'(s) \neq 0$.

Remark 22. Both the tangent line and the normal line pass through the point $(x, y) = (f(s), g(s))$, which lies on the curve. The only difference is their slope. The tangent line has slope

$$m_{\tan} = \frac{g'(s)}{f'(s)} = \frac{dy}{dx}(s)$$

while the normal line has slope

$$m_{\text{norm}} = -\frac{f'(s)}{g'(s)} = -\frac{1}{\frac{dy}{dx}(s)}.$$

Since

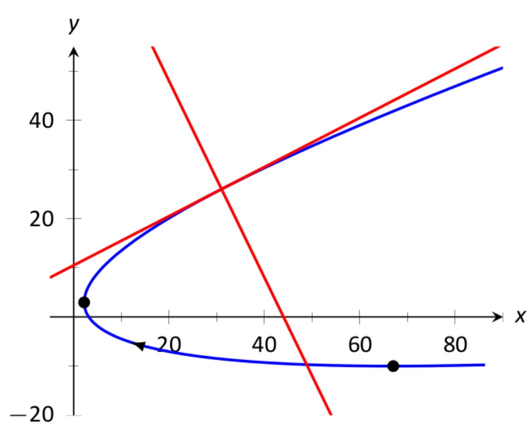
$$m_{\tan} = -\frac{1}{m_{\text{norm}}}$$

it follows that the two lines are perpendicular to each other: the tangent line lies tangent to the curve, while the normal line is perpendicular to it.

Example 23 (Tangent and normal lines)**.** Let C be the curve defined by

$$\begin{cases} x(t) = 5t^2 - 6t + 4 \\ y(t) = t^2 + 6t - 1 \end{cases} \tag{4}$$

Problem: Find the tangent line and normal line to C at $t = 3$.
Solution: For this problem, we have $f(t) = 5t^2 - 6t + 4$ and $g(t) = t^2 + 6t - 1$.
Before we start, this is the picture:



Differentiate:

$$\begin{cases} f'(t) = 10t - t \\ g'(t) = 2t + 6 \end{cases}$$

Therefore

$$\frac{dy}{dx}(t) = \frac{f'(t)}{g'(t)} = \frac{2t+6}{10t-6}. \tag{5}$$

Important note: $\frac{dy}{dx}$ is a function of t. It is not a function of x!

At the point $t = 3$, we have:

- $(x, y) = (31, 26)$ (by Eq. (4))
- The slope is $\frac{dy}{dx}(3) = \frac{1}{2}$ (by Eq. (5))

Combining these two facts, and using Definition 21, we conclude that the tangent line is

$$y - 26 = \frac{1}{2}(x - 31)$$

(this is the equation of a line in point-slope form). Here, we have $m_{\tan} = \frac{1}{2}$. The slope of the normal line is always $-\frac{1}{m_{\tan}}$, which is -2. We conclude that the normal line is

$$y - 26 = -2(x - 31).$$

End of Example 23. □

5 2025-01-24 | Week 2 | Lecture 5

This lecture is based on section 10.2

5.1 Tangent lines and normal lines (continued)

Example 24 (Continuation of Example 23)**.** Consider the curve C from Example 23. Find where C has vertical or horizontal tangent lines.

To find where C has a horizontal tangent line, set $\frac{dy}{dx}(t) = 0$ and solve for 0:

$$\frac{dy}{dx}(t) \overset{set}{=} 0.$$

Substituting in the formula from Eq. (5),

$$\frac{2t+6}{10t-6} = 0.$$

Solving for t gives $t = -3$. When $t = -3$, we have

$$(x, y) = (f(-3), g(-3)) = (67, -10).$$

We have shown that the tangent line is horizontal at this point.

What about a vertical tangent line? **Key idea:** To find where the tangent line is VERTICAL, we need to find the points where the normal line is HORIZONTAL. (This is because the tangent line and normal line are perpendicular to each other). To do this:

$$-\frac{1}{\frac{dy}{dx}(t)} \overset{set}{=} 0.$$

Substituting in the formula from Eq. (5),

$$-\frac{1}{\left(\frac{2t+6}{10t-6}\right)} = 0,$$

or equivalently

$$-\frac{10t-6}{2t+6} = 0.$$

Solving for t gives $t = 3/5$. When $t = 3/5$, the corresponding (x, y) point is $(2.2, 2.96)$. We have shown that the tangent line is vertical at this point.

End of Example 24. □

5.2 Arc Length of parametrized curves

Theorem 25 (Arc Length of Parametric Curves)**.** *Let*

$$\begin{cases} x = f(t) \\ y = g(t) \end{cases}$$

be a parametric equations defined on an interval $I = [a, b]$. Assume that f and g are differentiable an that f' and g' are continuous on $[a, b]$. Also assume that the graph traces itself only once. The arc length of the curve is

$$L = \int_a^b \sqrt{f'(t)^2 + g'(t)^2} dt.$$

Remark 26. These integrals are often hard to compute.

Example 27 (Arc length of a circle)**.** Consider the circle paramterized by the equationss

$$\begin{cases} x = 3\cos t \\ y = 3\sin t \end{cases}$$

where t ranges over the interval $[0, 2\pi]$. What is the arc length?

Solution: Here $f(t) = 3\cos(t)$ and $g(t) = 3\sin(t)$. Using the formula from Theorem 25

$$\begin{aligned} L &= \int_0^{2\pi} \sqrt{f'(t)^2 + g'(t)^2} dt \\ &= \int_0^{2\pi} \sqrt{(-3\sin t)^2 + (3\cos t)^2} dt \\ &= 3\int_0^{2\pi} \sqrt{\sin^2(t) + \cos^2(t)} dt \\ &= 3\int_0^{2\pi} 1 dt & \text{since } \sin^2(t) + \cos^2(t) = 1 \\ &= 6\pi. \end{aligned}$$

Taking a step back. What was this question really asking? It was asking for the circumference of a circle with radius 3 (since that's the curve traced out by the equations). Using the formula

$$\text{circumference} = 2\pi(\text{radius}),$$

our answer $6\pi^2$ makes sense.

End of Example 27. □

Example 28 (Arc length)**.** What is the arc length of the curve parameterized by the equations

$$\begin{cases} x(t) = \frac{7}{3}t^3 + t \\ y(t) = \frac{2}{3}\left(2t^2 + \frac{1}{4}\right)^{3/2} \end{cases}$$

for $0 \le t \le 2$?

Solution:

Step 1. Compute $x'(t)$ and $y'(t)$:

$$x'(t) = 7t^2 + 1 \quad \text{and} \quad y'(t) = \left(2t^2 + \frac{1}{4}\right)^{\frac{1}{2}} \cdot 4t.$$

Step 2. Compute $[x'(t)]^2 + [y'(t)]^2$ and simplify.

$$\begin{aligned} [x'(t)]^2 + [y'(t)]^2 &= \left[7t^2 + 1\right]^2 + \left[\left(2t^2 + \frac{1}{4}\right)^{\frac{1}{2}} \cdot 4t\right]^2 \\ &= 49t^4 + 14t^2 + 1 + \left(2t^2 + \frac{1}{4}\right) \cdot 16t^2 \\ &= 49t^4 + 14t^2 + 1 + 32t^4 + 4t^2 \\ &= 81t^4 + 18t^2 + 1. \end{aligned}$$

Step 3. Recognize that you've got a perfect square:

$$81t^4 + 18t^2 + 1 = \left(9t^2 + 1\right)^2$$

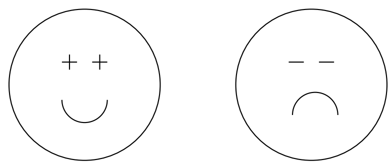
(Step 3 doesn't always happen, but it's really nice when it does!)

Step 4. Plug what you get back into the formula from Theorem 25:

$$
\begin{aligned}
\text{Arc Length} &= \int_0^2 \sqrt{[x'(t)]^2 + [y'(t)]^2}\,dt \\
&= \int_0^2 \sqrt{(9t^2+1)^2}\,dt \\
&= \int_0^2 (9t^2+1)\,dt \\
&= \left[3t^3 + t\right]_0^2 \\
&= 26.
\end{aligned}
$$

We conclude that the arc length is 26.

End of Example 28. $\square$



6 2025-01-27 | Week 3 | Lecture 6

This is based on section 10.2 in the textbook, and section 10.3 (polar coordinates)

6.1 Comment about finite-length parameterized curves

Suppose you've got a curve C

$$\begin{cases} x(t) \\ y(t) \end{cases}$$

where t ranges over an *infinite* interval, say $0 \leq t < \infty$. Is it possible that the curve is finite?

Short answer: yes. Think of the ant. It walks forever, but as long as it keeps slowing down (but never completely stops), it is *possible* that the ant travel only a finite distance.

This should give us a new way to think about the formula

$$\text{Arc length} = \int_a^b \underbrace{\sqrt{[x'(t)]^2 + [y'(t)]^2}}_{\text{instantaneous speed of the ant at time } t} \, dt$$

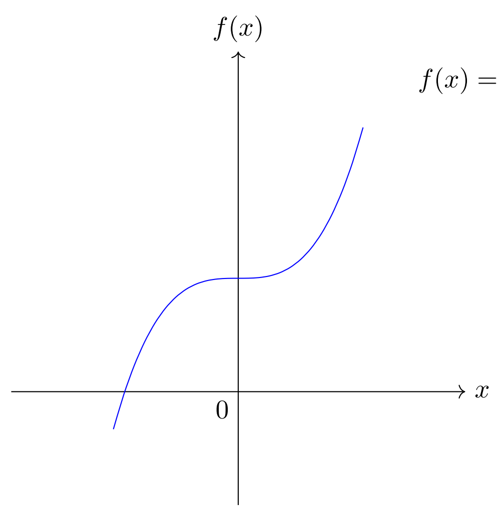
6.2 Concavity of parameterized curves

The concavity of a differentiable function is determined by its second derivative. For example, the graph

$$f(x) = x^2$$

is concave up. Moreover, $f''(x) = 2 > 0$.

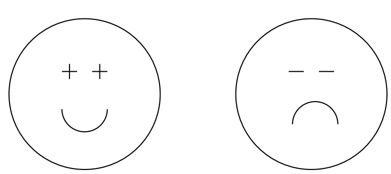
What about $f(x) = x^3 + 1$? This has graph



Tihs graph is concave down when $x < 0$ and concave up when $x > 0$. The concavity is determined by whether the second derivative of f is positive or negative:

$$f''(x) = 6x = \begin{cases} \text{positive} & : \quad x > 0 \\ \text{negative} & : \quad x < 0 \end{cases}$$

This is a general property of twice-differentiable functions: when $f''(x) < 0$, the graph is concave down. When $f''(x) > 0$, the graph is concave up. Just like in the following picture:



With parameterized curves, the story is no different, but we'll need to learn how to compute the second derivative $\frac{d^2y}{dx^2}$, as we now show.

Consider a curve

$$\begin{cases} x = f(t) \\ y = g(t) \end{cases}$$

so that f, g are twice differentiable on the inteval I and $f'(t) \neq 0$ on I.

We have already seen how to compute

$$\frac{dy}{dx} = \frac{dy/dt}{dx/dt} = \frac{g'(t)}{f'(t)}.$$

What about $\frac{d^2y}{dx^2}$? The formula is:

$$\frac{d^2y}{dx^2} = \frac{\frac{d}{dt}\left[\frac{dy}{dx}\right]}{\frac{dx}{dt}} = \frac{\frac{d}{dt}\left[\frac{dy}{dx}\right]}{f'(t)} \tag{6}$$

Example 29 (Concavity of a parameterized curve)**.** Let C be the curve defined by the parametric equations

$$\begin{cases} x(t) = \frac{1}{2}t^2 + \frac{1}{2} \\ y(t) = t - t^3 \end{cases}$$

(a) Find the tangent line to C at $t = 1$.

(b) For what values of t is the curve convex up? For what values is it concave down?

(c) At what points (x, y) does the curve have a horizontal tangent line? Vertical tangent line?

Solution to (a): To apply the line formula $y - y_0 = m(x - x_0)$, we need two things: (1) the point $(x(1), y(1))$ and the slope at $t = 1$.

First observe that

$$(x(1), y(1)) = (1, 0)$$

Second, to compute the slope:

$$\begin{aligned} \frac{dy}{dx} &= \frac{dy/dt}{dx/dt} \\ &= \frac{\frac{d}{dt}\left[t - t^3\right]}{\frac{d}{dt}\left[\frac{1}{2}t^2 + \frac{1}{2}\right]} \\ &= \frac{1 - 3t^2}{t} \end{aligned}$$

So when $t = 1$, we have

$$\frac{dy}{dx}(t) = \frac{dy}{dx}(1) = -2$$

Therefore by the line formula $y - y_0 = m(x - x_0)$, we conclude that the tangent line at $t = 1$ has equation

$$y - 0 = -2(x - 1)$$

or equivalently,

$$y = -2x + 2.$$

Solution to (b). Here, our approach is to compute the second derivatve and then check when it is positive or negative, as this is sufficient to determine whether the curve is concave up or concave down.

To compute the second derivative we'll use the formula:

$$\frac{d^2y}{dx^2} = \frac{\frac{d}{dt}\left[\frac{dy}{dx}\right]}{\frac{dx}{dt}}.$$

Now, in part (a) we've already computed some components of the above equation. Specifically, we know that $\frac{dy}{dx} = \frac{1-3t^2}{t}$ and $\frac{dx}{dt} = t$. Plugging these in, we get

$$\frac{d^2y}{dx^2} = \frac{\frac{d}{dt}\left[\frac{1-3t^2}{t}\right]}{t}$$

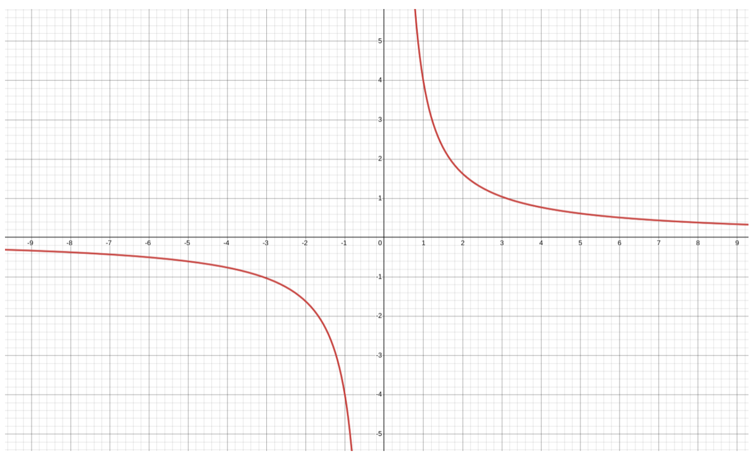
doing the derivative (e.g., using the quotient rule) and simplifying gives

$$\frac{d^2y}{dx^2} = -\frac{1+3t^2}{t^3}$$

One can check that this function is NEGATIVE when $t < 0$, and POSITIVE when $t > 0$. To see this, we can either solve the inequality

$$-\frac{1+3t^2}{t^3} > 0$$

or we can look at the graph of the second derivative:



Solution to (c). To find the horizontal tangent lines, we need check when $\frac{dy}{dx}(t) = 0$. To find the vertical tangent lines, we check when $\frac{dy}{dx}(t) = \pm\infty$ or undefined.

In part (a) we computed

$$\frac{dy}{dx} = \frac{1-3t^2}{t}. \tag{7}$$

By inspection, this is undefined when $t = 0$, since there is a division by zero in that case. Therefore the tangent line is vertical when $t = 0$, which occurs at the point

$$(x(0), y(0)) = \left(\frac{1}{2}, 0\right).$$

On the other hand, we see from Eq. (7) that $\frac{dy}{dx}(t) = 0$ (i.e., the tangent line is horizontal) if and only if

$$1 - 3t^2 = 0$$

or equivalently, when $t = \pm\frac{1}{\sqrt{3}}$. At what points does this occur?

- If $t = \frac{1}{\sqrt{3}}$, then the point is
$$\left(x\left(\frac{1}{\sqrt{3}}\right), y\left(\frac{1}{\sqrt{3}}\right)\right) = \left(\frac{2}{3}, \frac{2}{3\sqrt{3}}\right)$$
- If $t = -\frac{1}{\sqrt{3}}$, then the point is
$$\left(x\left(-\frac{1}{\sqrt{3}}\right), y\left(-\frac{1}{\sqrt{3}}\right)\right) = \left(\frac{2}{3}, -\frac{2}{3\sqrt{3}}\right)$$

We conclude that the curve C has a horizontal tangent line at the points $\left(\frac{2}{3}, \frac{2}{3\sqrt{3}}\right)$ and $\left(\frac{2}{3}, -\frac{2}{3\sqrt{3}}\right)$, and a vertical tangent line at the point $\left(\frac{1}{2}, 0\right)$. This answers part (c).

End of Example 29. $\square$

7 2025-01-29 | Week 3 | Lecture 7

7.1 Concavity of parameterized curves (continued)

Example 30 (Concavity of a parameterized curve)**.** Consider the curve

$$\begin{cases} x(t) = 5t^2 - 6t + 4 \\ y(t) = t^2 + 6t - 1 \end{cases}$$

For what values of t is this curve concave down? For what values is it conccave up?

Solution: To compute the 2nd derivative, we need to begin by computing the first derivative:

$$\frac{dy}{dx} = \frac{g'(t)}{f'(t)} = \frac{2t+6}{10t-6}$$

Now we use the formula from Eq. (6):

$$\begin{aligned}
\frac{d^2y}{dx^2} &= \frac{\frac{d}{dx}\left[\frac{dy}{dx}\right]}{x'(t)} \\
&= \frac{\frac{d}{dt}\left[\frac{2t+6}{10t-6}\right]}{10t-6} \\
&= \frac{\left(\frac{2(10t-6)-10(2t+6)}{(10t-6)^2}\right)}{10t-6} \\
&= -\frac{72}{(10t-6)^3}
\end{aligned}$$

Observe that this is positive when $10t - 6 < 0$, i.e., when $t < \frac{3}{5}$. On the other hand, it is negative when $10t - 6 > 0$, i.e., when $t > \frac{3}{5}$.

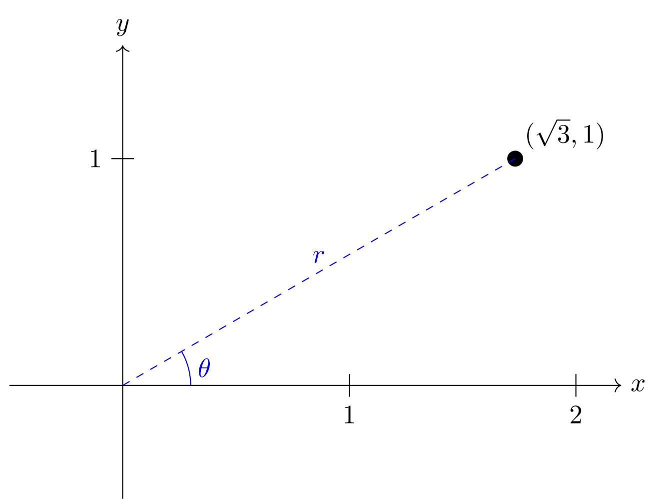
So the curve is concave up when $t < 3/5$ and concave down when $t > 3/5$. To see a graph of this, see Example 23.

End of Example 30. □

7.2 Polar Coordinates

We are now starting 10.3 in the textbook.

Consider the point on the plane whose Cartesian coordinates are $(x, y) = (\sqrt{3}, 1)$:



Here I have also plotted a ray of length r extending from the origin to the point, as well as the angle θ of the ray.

Can we figure out what r and θ are?

- By the Pythagorean theorem, we have

$$r^2 = \left(\sqrt{3}\right)^2 + (1)^2 = 4$$

  so $r = 2$.

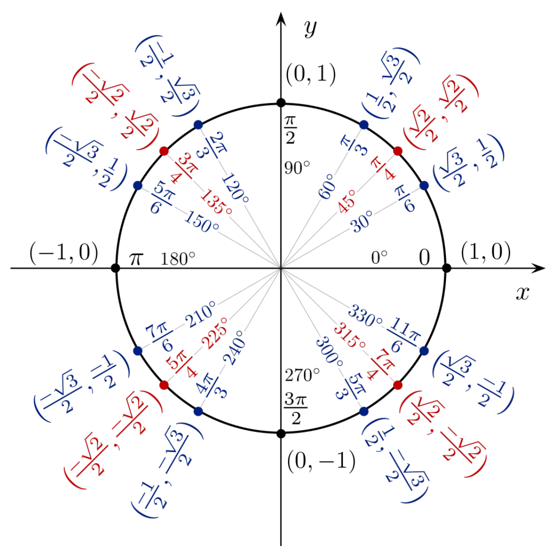
- Observe that $\tan(\theta) = \text{slope} = \frac{\text{rise}}{\text{run}} = \frac{1}{\sqrt{3}}$.

  And with a little bit of thought after looking at a unit circle, we conclude that

$$\theta = \frac{\pi}{6}$$

  since

$$\tan(\theta) = \frac{\sin(\theta)}{\cos(\theta)} = \frac{\left(\frac{1}{2}\right)}{\left(\frac{\sqrt{3}}{2}\right)} = \frac{1}{\sqrt{3}}.$$



Therefore, using the two bullet points above, we can represent the point $(x, y) = (\sqrt{3}, 1)$ by the pair

$$(r, \theta) = \left(2, \frac{\pi}{6}\right).$$

This representation is called the *polar coordinates* of the point, and it can be done for any point in the xy-plane. *polar coordinates.*

Theorem 31 (Key identities for converting between polar and rectangular coordinates)**.** *If (r, θ) is a point in polar coordinates, the rectangular coordinates are given by*

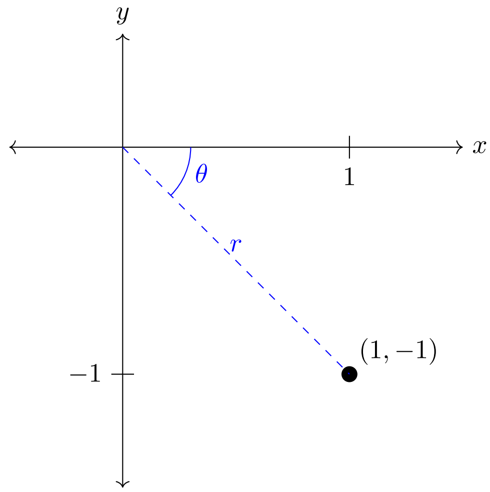
$$x = r\cos\theta \quad \textit{and} \quad y = r\sin\theta \tag{8}$$

On the other hand, given a point (x, y) in rectangular coordinates, the polar coordinates are determined by the identities

$$r^2 = x^2 + y^2 \quad \textit{and} \quad \tan\theta = \frac{y}{x} \tag{9}$$

Eq. (8) allows us to find the Cartesian coordinates of a point when the polar coordinates are known. More generally, can use the identities in Theorem 31 to convert between coordinate systems, as shown in the next examples.

Example 32 (Example 3 in Chapter 10.3)**.** Consider the point on the plane whose Cartesian coordinates are $(1, -1)$. What are the polar coordinates of the point?



Solution: We need to determine r and θ. We use Eq. (9):

- $r = \sqrt{1^2 + (-1)^2} = \sqrt{2}$.
- $\tan\theta = \frac{y}{x} = -1$. This implies that $\theta = -\frac{\pi}{4}$ (or $\theta = \frac{7\pi}{4}$ works too).

End of Example 32. □

Example 33.

- What is the curve represented by the polar equation $r = 2$?
- What abou the the curve $\theta = \frac{\pi}{3}$?

End of Example 33. □

Example 34. Find a polar equation of the form $r = f(\theta)$ for the curve represented by the Cartesian equation $x = 3$.

Solution: Use $x = r\cos(\theta)$:

$$r\cos(\theta) = 3$$

so

$$r = \frac{3}{\cos(\theta)}$$

or equivalently,

$$r = 3\sec(\theta)$$

End of Example 34. □

8 2025-01-31 | Week 3 | Class 8

8.1 Converting between Polar to Cartesian form

Example 35 (Cartesian $\implies$ Polar)**.** Find a polar equation of the form $r = f(\theta)$ for the curve represented by the Cartesian equation $x = -2y^2$.

Solution: Use $x = r\cos(\theta)$ and $y = r\sin(\theta)$:

$$\begin{aligned} r\cos(\theta) &= -2\left(r\sin(\theta)\right)^2 \\ &= -2r^2\sin^2(\theta). \end{aligned}$$

Divide both sides by r:

$$\cos(\theta) = -2r\sin^2(\theta).$$

Solve for r:

$$r = -\frac{\cos(\theta)}{2\sin^2(\theta)}$$

End of Example 35. $\square$

Example 36 (Polar $\implies$ Cartesian)**.** Find a Cartesian equation for the polar curve $r = 2\cos(\theta)$.

Solution: By Eq. (8), we know that

$$x = r\cos(\theta).$$

Multiply both sides of our original equation by r:

$$r^2 = 2r\cos(\theta)$$

Thus

$$r^2 = 2x.$$

By Eq. (9), we know that $r^2 = x^2 + y^2$. Therefore

$$x^2 + y^2 = 2x.$$

Completing the square gives

$$(x-1)^2 + y^2 = 1$$

This is a circle of radius 1 with center at $(1, 0)$.

End of Example 36. $\square$

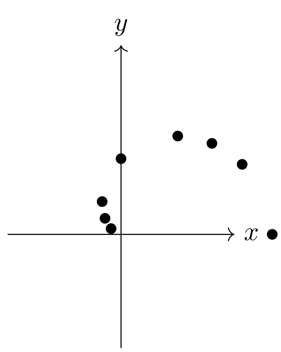
8.2 Sketching polar functions

Example 37. Sketch the polar function

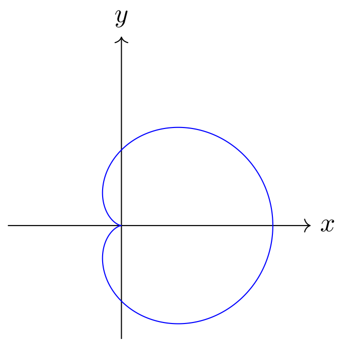
$$\begin{cases} r = 1 + \cos(\theta) \\ 0 \le \theta \le 2\pi \end{cases}$$

Solution: Let's use the method of making a table to plot points. It will be convenient to use the approximations $\sqrt{3} \approx 1.7$ and $\sqrt{2} \approx 1.4$.

θ	$r = 1 + \cos(\theta)$
0	2
$\pi/6$	$1 + \frac{\sqrt{3}}{2} \approx 1.85$
$\pi/4$	$1 + \frac{\sqrt{2}}{2} \approx 1.7$
$\pi/3$	1.5
$\pi/2$	1
$2\pi/3$	$1 - \frac{1}{2} = .5$
$3\pi/4$	$1 - \frac{\sqrt{2}}{2} \approx 1 - 0.7 = .3$
$5\pi/6$	$1 - \frac{\sqrt{3}}{2} \approx 1 - .85 = .15$
π	0



After plotting these points, we make the observation that since $\cos(\theta)$ is an even function, $\cos(-\theta) = \cos(\theta)$. Therefore the graph is symmetric about the horizontal axis. Using this property to sketch the curve, we obtain:



This shape is called a cardiod.

End of Example 37. $\square$

Example 38. Sketch the graph of the polar equation

$$r = \frac{1}{\sin(\theta) + 2\cos(\theta)}.$$

Solution: No way are we making a table for this problem. Maybe we can rewrite it in Cartesian coordinates.

Start by mutlipying both sides by $(\sin\theta + 2\cos\theta)$:

$$r\sin(\theta) + 2r\cos(\theta) = 1$$

Now use Theorem 31, which says $x = r\cos(\theta)$ and $y = r\sin(\theta)$:

$$y + 2x = 1.$$

This is the line $y = -2x + 1$. Easy-peasy to graph. :)

End of Example 38. $\square$

8.3 Finding $\frac{dy}{dx}$ with polar functions

No matter what coordinate system we use, the slope of a tangent line is always given by the quantity $\frac{dy}{dx}$.

We've seen how to compute that in other contexts. How do we compute $\frac{dy}{dx}$ when our equation is a polar function of the form $r = f(\theta)$?

As a function of θ, the (x, y)-coordinates are

$$x = f(\theta)\cos(\theta), \quad y = f(\theta)\sin(\theta).$$

By the product rule,

$$\frac{dx}{d\theta} = f'(\theta)\cos(\theta) - f(\theta)\sin(\theta) \quad \text{and} \quad \frac{dy}{d\theta} = f'(\theta)\sin(\theta) + f(\theta)\cos(\theta).$$

Certainly, we must have:

$$\frac{dy}{dx} = \frac{dy/d\theta}{dx/d\theta}.$$

Therefore,

$$\boxed{\frac{dy}{dx} = \frac{f'(\theta)\sin(\theta) + f(\theta)\cos(\theta)}{f'(\theta)\cos(\theta) - f(\theta)\sin(\theta)}}$$

9 2025-02-03 | Week 4 | Lecture 9

9.1 Derivatives of polar functions

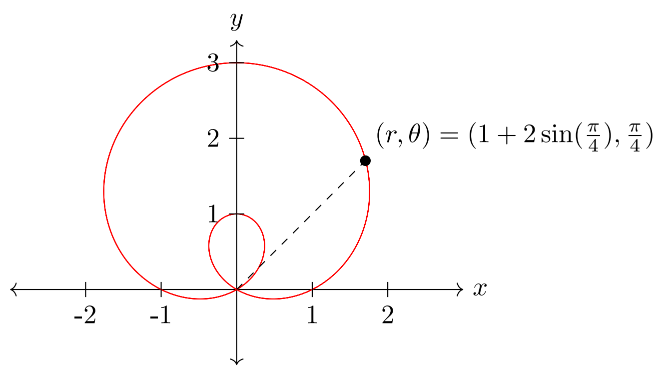
Recall that for a polar curve in the form $r = f(\theta)$, the derivative is

$$\boxed{\frac{dy}{dx} = \frac{f'(\theta)\sin(\theta) + f(\theta)\cos(\theta)}{f'(\theta)\cos(\theta) - f(\theta)\sin(\theta)}.} \tag{10}$$

Example 39. Consider the polar function

$$r = 1 + 2\sin\theta.$$

This polar function has graph:



Questions:

1. Find the equations of the tangent and normal lines to the graph at $\theta = \frac{\pi}{4}$.
2. Find where the graph has vertical and horizontal tangent lines.

Solution: To find the tangent line, first compute $\frac{dy}{dx}$. Since $r(\theta) = 1 + 2\sin(\theta)$, we have

$$r'(\theta) = 2\cos(\theta).$$

Therefore, by the boxed formula above, we have:

$$\begin{aligned}\frac{dy}{dx} &= \frac{r'(\theta)\sin(\theta) + r(\theta)\cos(\theta)}{r'(\theta)\cos(\theta) - r(\theta)\sin(\theta)} \\ &= \frac{2\sin(\theta)\cos(\theta) + (1 + 2\sin\theta)\cos(\theta)}{2\cos^2(\theta) - (1 + 2\sin\theta)\sin\theta}\end{aligned}$$

Plugging in $\theta = \frac{\pi}{4}$, we get

$$\frac{dy}{dx}(\pi/4) = -2\sqrt{2} - 1 \approx -3.8 \tag{11}$$

We now have the slope of the tangent line, but we need it's xy-coordinates.

When $\theta = \frac{\pi}{4}$, we have

$$r = 1 + 2\sin(\frac{\pi}{4}) = 1 + \sqrt{2}.$$

- Using this along with the formula $x = r\sin(\theta)$, observe that when $\theta = \frac{\pi}{4}$, we have:

$$x = (1 + \sqrt{2})\sin(\frac{\pi}{4}) = \left(1 + \sqrt{2}\right)\frac{\sqrt{2}}{2}$$

or approximately

$$x \approx 1.7$$

- Similarly, we can use the formula $y = r\sin(\theta)$ to obtain the value of y whe $\theta = \frac{\pi}{4}$. In that case, we also get $y \approx 1.7$ (work not shown).

So the tangent line is approximately the line

$$y - 1.7 = -3.8(x - 1.7)$$

or

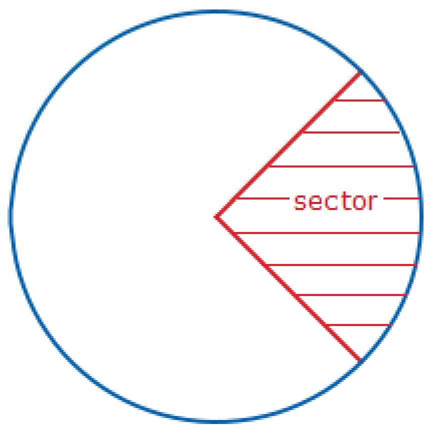
$$y = -3.8x + 8.2$$

End of Example 39. $\square$

9.2 Areas enclosed by polar graphs

Next lecture will discuss computing areas enclosed by polar graphs. The formula for this is the following

A *sector* is the following



Theorem 40 (Area of a sector)**.** *The area of a sector with angle θ is*

$$\textit{area } = \frac{1}{2}\theta r^2$$

For example, if $\theta = 360° = 2\pi$, then

$$\text{area } = \pi r^2,$$

which is the formula for the area of a circle.

Theorem 41. *If $r = f(\theta)$ and f is a continuous and* ***nonnegative*** *function on the interval $[\alpha, \beta]$, where $0 \leq \beta - \alpha \leq 2\pi$ then the area of the region bounded by the lines $r = f(\theta)$ and $\theta = a$ and $\theta = b$ is*

$$\textit{Area} = \frac{1}{2}\int_\alpha^\beta r^2 d\theta = \frac{1}{2}\int_\alpha^\beta [f(\theta)]^2\, d\theta$$

Remark 42. It's really important that you check that the conditions $0 \leq \beta - \alpha \leq 2\pi$ and nonnegativity of f. Otherwise cancelletion or double counting of regions may occur, resulting in the integral not actually equalling the area.

Throughout we will use the following useful proposition.

Proposition 43 (Useful integrals)**.** *It holds that*

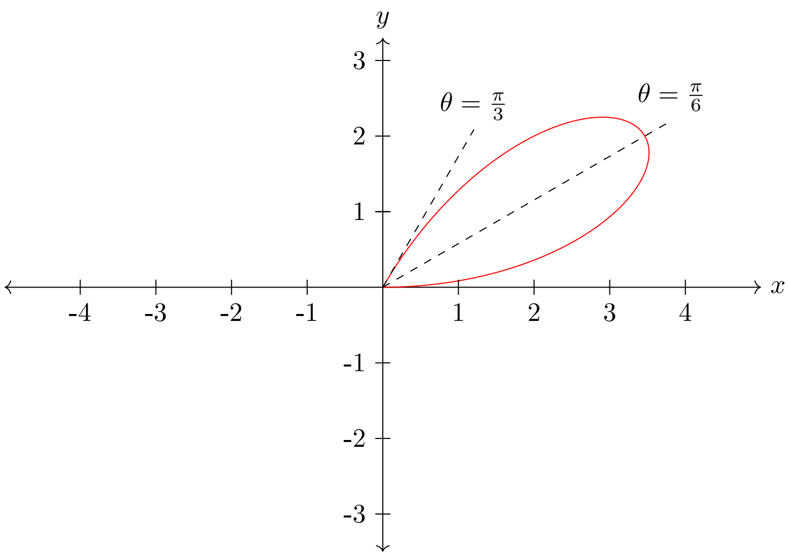
$$\int \cos^2(\theta)d\theta = \frac{\theta}{2} + \frac{\sin(2\theta)}{4} + C$$
$$\int \sin^2(\theta)d\theta = \frac{\theta}{2} - \frac{\sin(2\theta)}{4} + C$$

The integrals of Proposition 43 can be computed using the power-reduction identities $\cos^2(\theta) = \frac{1+\cos(2\theta)}{2}$ and $\sin^2(\theta) = \frac{1-\cos(2\theta)}{2}$.

Example 44 (Area of region enclosed by polar coordinates)**.** Consider the polar function

$$\begin{cases} r = 4\sin(3\theta) \\ 0 \le \theta \le \frac{\pi}{3} \end{cases}$$

This is a petal. As θ ranges from 0 to $\frac{\pi}{3}$, the curve loops counterclockwise from the origin back to the origin. To see why, think about the motion of a particle whose trajectory is traced out by $(r, \theta) = (\sin(3\theta), \theta)$. At $\theta = 0$, the particle starts at the origin. When $\theta = \frac{\pi}{6}$, the particle's distance from the origin is maximized. When $\theta = \frac{\pi}{3}$, the particle is back at the origin. This traces out exactly the full petal.. So we will integrate over the interval $0 \le \theta \le \frac{\pi}{3}$.



$$\begin{aligned}\text{Area of one petal} &= \frac{1}{2}\int_0^{\pi/3} r^2 d\theta \\ &= \frac{1}{2}\int_0^{\pi/3} [4\sin(3\theta)]^2\, d\theta \\ &= 8\int_0^{\pi/3} \sin^2(3\theta)d\theta.\end{aligned}$$

Use the u-substitution $u = 3\theta$, $du = 3d\theta$

$$\begin{aligned}\text{Area of one petal} &= \frac{8}{3}\int_0^{\pi} \sin^2(u)du \\ &= \frac{8}{3}\left[\frac{u}{2} - \frac{\sin(2u)}{4}\right]_{u=0}^{u=\pi} && \text{by Proposition 43} \\ &= \frac{4}{3}\pi.\end{aligned}$$

End of Example 44. □

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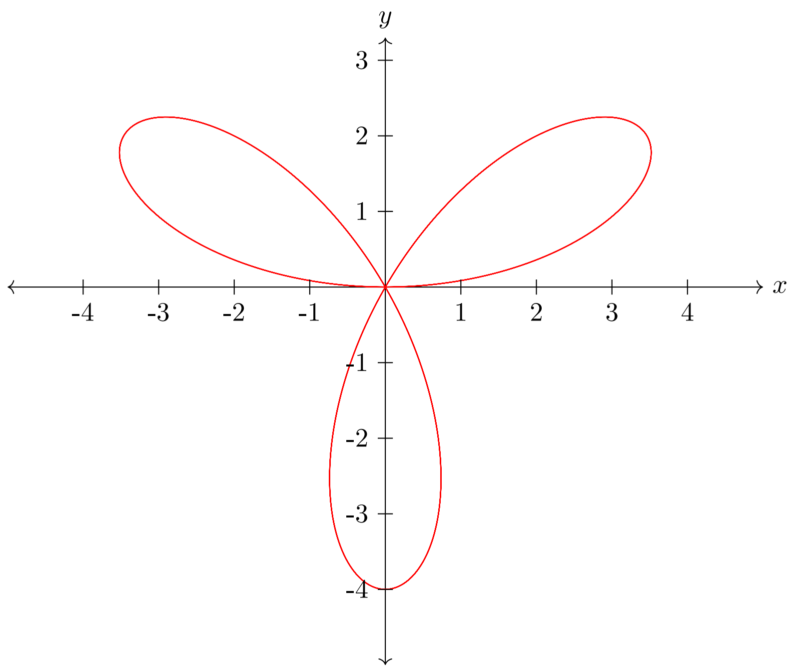
Today we finish section 10.4 in the texbook and start 12.1 and 12.2

The next example illustrates by the nonnegativity assumption of Theorem 41 is so essential

Example 45 (Necessity of the hypotheses of Theorem 41)**.** Consider the polar function

$$\begin{cases} r = 4\sin(3\theta) \\ 0 \le \theta \le 2\pi \end{cases}$$

This is a continuation of the curve from Example 44, but now we allow θ to range all the way to 2π. This gives the following 3-petal flower:



What is the area of the entire bounded region enclosed by the graph?

The first thing we might think to do is to integrate over $0 \le \theta \le 2\pi$, as follows:

$$A = \frac{1}{2}\int_0^{2\pi} r^2 d\theta = \frac{1}{2}\int_0^{2\pi} [4\sin(3\theta)]^2\, d\theta = 8\int_0^{2\pi} \sin^2(3\theta) d\theta.$$

Now do the u-substitution $u = 3\theta$, $du = 3d\theta$:

$$A = \frac{8}{3}\int_0^{6\pi} \sin^2(u) du = \frac{8}{3}\left[\frac{u}{2} - \frac{\sin(2u)}{4}\right]_0^{6\pi} = \frac{8}{3}[3\pi] = 8\pi.$$

Although it seemed like everything worked out nicely, the above answer of 8π is wrong. The problem is that the function $\sin(3\theta)$ is not nonnegative on the interval $0 \le \theta \le 2\pi$.

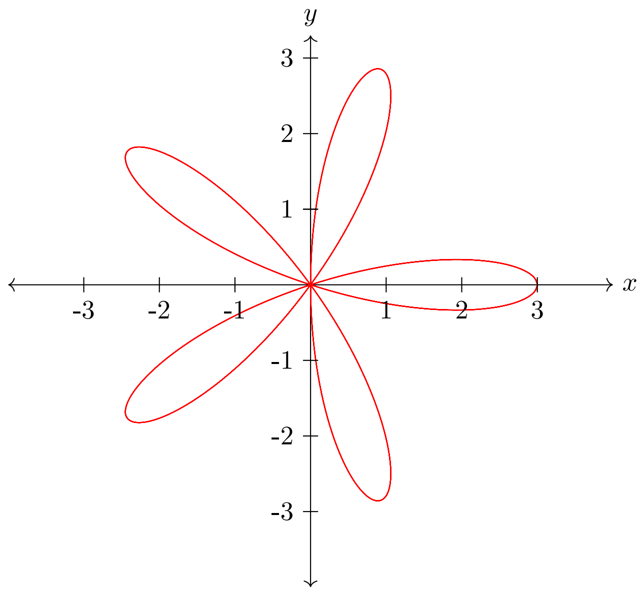
The correct area is 4π, obtained by observing that there are 3 petals and the area of a single petal is $\frac{4}{3}\pi$ by Example 44.

End of Example 45. □

Example 46. Find the area enclosed by one loop of the curve

$$r = 3\cos(5\pi).$$

The plot of this is shown below:



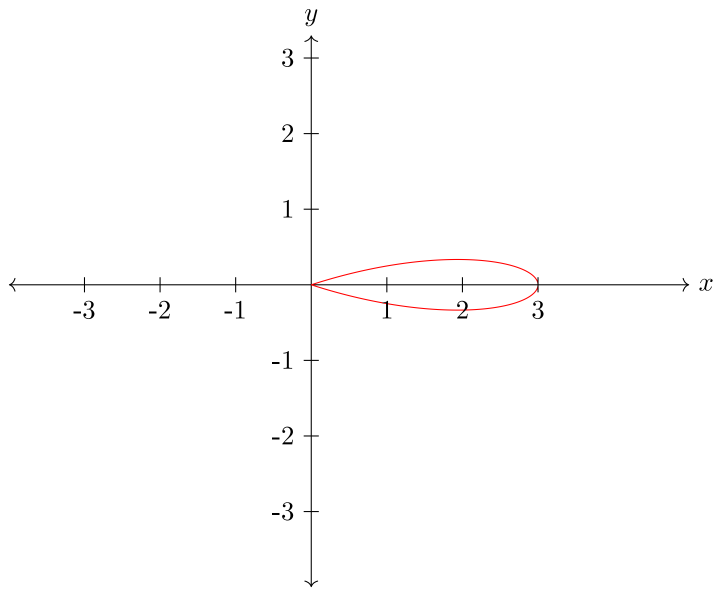
What should we pick as the interval? One way to figure this out is to look at when the polar function crosses the origin, i.e., when $r = 0$:

$$r = 0$$
$$\Updownarrow$$
$$\cos(5\theta) = 0$$
$$\Updownarrow$$
$$5\theta = \pm\frac{\pi}{2}, \pm\frac{3\pi}{2}, \pm\frac{5\pi}{2}, \ldots$$
$$\Updownarrow$$
$$\theta = \pm\frac{\pi}{10}, \pm\frac{3\pi}{10}, \pm\frac{5\pi}{10}, \ldots$$

Here, the smallest angles about $\theta = 0$ are $\theta = \pm\frac{\pi}{10}$. So the equations

$$\begin{cases} r = 3\cos(5\pi) \\ -\frac{\pi}{10} \leq \theta \leq \frac{\pi}{10} \end{cases}$$

traces out the following petal:



The area of enclosed by this loop is

$$\begin{aligned}
A &= \frac{1}{2}\int_{-\pi/10}^{\pi/10} r^2 d\theta \\
&= \frac{1}{2}\int_{-\pi/10}^{\pi/10} \left[3\cos(5\theta)\right]^2 d\theta \\
&= \frac{9}{10}\int_{-\pi/2}^{\pi/2} \cos^2(\theta) d\theta && \text{subtitution: } u = 5\theta \\
&= \frac{9}{10}\left[\frac{u}{2} + \frac{\sin(2u)}{4}\right]_{-\pi/2}^{\pi/2} && \text{by Proposition 43} \\
&= \frac{9}{10}\left[\frac{\pi}{2}\right] \\
&= \frac{9\pi}{20}.
\end{aligned}$$

End of Example 46. □

10.1 3D Space

Based on sections 12.1 in the texbook

3D space is denoted by the symbol $\mathbb{R}^3$. A point in 3D space is specified as an ordered triple

$$(x, y, z)$$

where x, y, z are real numbers. Here, z is the distance above (or below) the xy-plane.

Many familiar equations take on new meaning in 3D space, compared to 2D space.

- Consider the equation

  $$\begin{cases} x^2 + y^2 = 1 \\ z = 3 \end{cases}$$

  This is a unit circle floating 3 units above the xy-plane.

  On the other hand, the equation

  $$x^2 + y^2 = 1$$

  imposes no restriction on the value of z. Consequently, this is the equation of a (hollow) cyclinder.

- The equation
$$x^2 + y^2 + z^2 = 1$$
is the equation of a sphere with radius 1 centered at the origin.

10.2 Introduction to vectors

Based on textbook section 12.2

What is a vector?

`https://www.3blue1brown.com/lessons/vectors`

- **Algebraic definition:** a list of numbers
- **Geometric definition:** the difference between two points
- **Physical definition:** an arrow with direction and magnitude

[picture]

Example 47 (Vector)**.** Consider three points in the plane

$$A = (4, 3), \quad \text{and} \quad B = (1, 2)$$

Then, for example, we have the vector $\overrightarrow{AB} = \langle 4 - 1, 3 - 2 \rangle = \langle 3, 1 \rangle$.

Your textbook uses the angled brackets notation $\langle x, y \rangle$ for vectors. We will sometimes use

$$\begin{bmatrix} x \\ y \end{bmatrix} \quad \text{or} \quad \begin{bmatrix} x \\ y \\ z \end{bmatrix}$$

or

$$(x, y) \quad \text{or} \quad (x, y, z)$$

All of these notations are common. When using the last, try to be clear in your mind (and your writing) whether you are working with a vector or a point.

The book denotes vectors with **bold** letters, like $\mathbf{v}$ or $\mathbf{x}$. In these notes I will usually use different notation: $\vec{v}$ or $\vec{x}$, which is easier to write by hand.

End of Example 47. □

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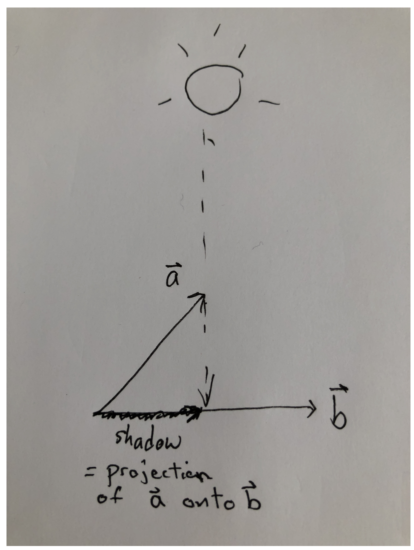
- please skim chapters 12.1 and 12.2
- plase read 12.3, 12.4
- next week's homework will be posted later today

Today: sections 12.3

11.1 Eratosthenes' estimation of the Earth's circumference

To motivate the present discussion of vectors, I presented Eratosthenes' famous calculation in which he correctly estimated the circumference of the Earth over 2200 years ago using nothing but sticks, shadows, and trigonometry.

We are interested in understanding—in a precise, mathematical way—the *shadow* of a vector as it falls onto another vector, as shown:



Eratosthenes' calculation (which isn't written up in these notes) suggests that understanding shadows like this can be a powerful mathematical tool.

The shadow above is an example of a *vector projection.* To understand vector projections, we'll need to have a firm grasp of the following things

- lengths of vectors
- the geometry of vectors and angles between them
- how to scale and rescale vectors
- and some more things that aren't apparent yet.

11.2 Vectors and 3D space

Example 48 (Vector addition and scaling)**.** Let $\vec{u} = \langle 1, 2\rangle$, $\vec{v} = \langle -3, -1\rangle$. Find the sum of the fetors and illustrate geometrically. Then

$$\vec{u} + \vec{v} = \langle -2, 1\rangle .$$

and you can add in any order

$$\vec{v} + \vec{u} = \langle -2, 1\rangle .$$

What about $\langle -1, 4\rangle$ and $\langle 6, -2\rangle$?

We can have 3d vectors, like

$$\langle 3, -2, 5\rangle$$

and multiplying a vector by a number a (called a *scalar*) *scales* the vector by stretching it out or whatever. For example, compare the vectors the vector

$$\vec{u} = \langle 2, 3\rangle \quad \text{and} \quad 2 \cdot \vec{u} = \langle 4, 6\rangle$$

End of Example 48. □

Example 49 (Spheres)**.** A sphere with radius 1 centered at the point $(x, y, z) = (1, 0, 2)$ is given by

$$(x-1)^2 + y^2 + (z-2)^2 = 1$$

This is similar to the equation of a circle.

End of Example 49. □

Proposition 50. *The length of a 2D vector* $\vec{a} = (a_1, a_2)$ *is*

$$|\vec{a}| = \sqrt{a_1^2 + a_2^2}$$

and the length of a 3D vector $\vec{a} = (a_1, a_2, a_3)$ *is*

$$|\vec{a}| = \sqrt{a_1^2 + a_2^2 + a_3^2}$$

Definition 51 (Standard Basis Vectors)**.** In 3 dimensions, the *standard basis vectors* are

$$\vec{\imath} = e_1 = \langle 1, 0, 0\rangle \tag{12}$$
$$\vec{\jmath} = e_2 = \langle 0, 1, 0\rangle \tag{13}$$
$$\vec{k} = e_3 = \langle 0, 0, 1\rangle \tag{14}$$

Sometimes we use the notation $\vec{\imath}, \vec{\jmath}, \vec{k}$, and other times we use the notation e_1, e_2, e_3.

Any vector in 3D can be written as a sum of the standard basis vectors, like this:

$$\begin{aligned}\langle 5, -4, 2\rangle &= 5\vec{\imath} - 4\vec{\jmath} + 2\vec{k} \\ &= 5e_1 - 4e_2 + 2e_3\end{aligned}$$

Example 52. Let $\vec{a} = \vec{\imath} + 2\vec{\jmath} - 3\vec{k}$ and $\vec{b} = 4\vec{\imath} + 7\vec{k}$. Express the vector $2\vec{a} + 3\vec{b}$ in terms of $\vec{\imath}, \vec{\jmath}$, and $\vec{k}$:

$$\begin{aligned}2\vec{a} + 3\vec{b} &= 2\left(\vec{\imath} + 2\vec{\jmath} - 3\vec{k}\right) + 3\left(4\vec{\imath} + 7\vec{k}\right) \\ &= (2+12)\vec{\imath} + 4\vec{\jmath} + (-6+21)\vec{k} \\ &= 14\vec{\imath} + 4\vec{\jmath} + 15\vec{k}.\end{aligned}$$

End of Example 52. □

A *unit vector* is any vector of length 1. So $\vec{\imath}, \vec{\jmath}$, and $\vec{k}$ are all unit vectors. If $\vec{v} \neq 0$, then you can *normalize* $\vec{v}$ by dividing it by its length. That is, let

$$\vec{u} := \frac{v}{|\vec{v}|}.$$

When you do this, the vector $\vec{u}$ is a unit vector.

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12.1 Vector Scaling and Magnitude

The vector with length 0 is called the *zero vector*. It's the vector

$$\vec{0} = \langle 0, 0 \rangle \quad \text{or} \quad \vec{0} = \langle 0, 0, 0 \rangle$$

depending on what space you are working with.

A *unit vector* is any vector of length 1. So $\vec{\imath}, \vec{\jmath}$, and $\vec{k}$ are all unit vectors. If $\vec{v} \neq 0$, then you can *normalize* $\vec{v}$ by dividing it by its length. That is, let

$$\vec{u} := \frac{v}{|\vec{v}|}.$$

When you do this, the vector $\vec{u}$ is a unit vector.

Example 53 (Scaling a vector)**.** In general, one can *scale* a vector by multiplying it by a number. Let

$$\vec{v} = \langle -2, 4, 4 \rangle$$

Then $\vec{v}$ has length

$$|\vec{v}| = \sqrt{(-2)^2 + (4)^2 + (4)^2} = \sqrt{4 + 16 + 16} = \sqrt{36} = 6.$$

If we scale $\vec{v}$ by a factor of 3, then the resulting vector is

$$\vec{u} = 3\vec{v} = \langle -6, 12, 12 \rangle$$

which has three times the length:

$$|\vec{u}| = \sqrt{(-6)^2 + (12)^2 + (12)^2} = \sqrt{36 + 144 + 144} = \sqrt{324} = 18.$$

Similarly, if I scale a vector by 2, then its length gets doubled. If I scale a vector by 1/2, its length gets halved.

End of Example 53. □

Example 54 (Distance between points)**.** The formula for computing the distance between two points in 3D space is related to the formula for computing the length of a 3D vector. Say you have a fly which starts at point $P := (x_1, y_1, z_1)$, flies around around a bit, and then terminates at point $Q := (x_2, y_2, z_2)$. Then the *displacement vector* is

$$\overrightarrow{PQ} = \langle x_2 - x_1, y_2 - y_1, z_2 - z_1 \rangle$$

You can think of this as an arrow drawn from the fly's starting position P to its ending position Q. The distance between points P and Q is nothing other than the magnitude of this vector. Therefore, distance between two points (x_1, y_1, z_1) and (x_2, y_2, z_2) is

$$\text{distance } = \sqrt{(x_2 - x_1)^2 + (y_2 - y_1)^2 + (z_2 - z_1)^2}.$$

For example, say the fly starts at $(3, 2, 9)$ and ends at $(2, -2, 1)$. The distance between these two points is

$$\begin{aligned} \text{distance } &= \sqrt{(3-2)^2 + (2-(-2))^2 + (9-1)^2} \\ &= \sqrt{1^2 + 4^2 + 8^2} \\ &= \sqrt{1 + 16 + 64} \\ &= \sqrt{81} \\ &= 9. \end{aligned}$$

End of Example 54. □

12.2 Connection to Physics: Force

Vectors are used to represent *force*, understood in physics as a physical interaction (e.g. a push or a pull) which causes (or can cause) change in an object's motion, shape, or resting position. Force has magnitude (i.e., length) and a direction.

Example 55 (Forces)**.** If you are floating in the ocean, you get pushed around by lots of forces: (1) gravity, (2) waves, (3) current, (4) wind, (5) forces created by your own motion, (6) buoyant force, and others. Each of these forces can be modeled by a vector. The magnitude of the vector indicates the "strength" of the corresponding force. For example, wind doesn't exert much force on a swimmer, so that vector might be tiny. By comparison the vector for current might be very large.

The *resultant force* is the sum of these vectors, and determines your net motion.

End of Example 55. □

Force is measured in **Newtons**, denoted N. How much is one $1N$? It's approximately equal to the weight that you feel pushing down on your hand when you hold up a stick of butter. (Note: *weight* is a force).

Technically, $N = \text{kg} \cdot \text{m/s}^2$. In words one Newton is the amount of force needed to accelerate one kg of mass at one meter per second per second (e.g., so that after two seconds the mass is moving at two meters per second). This is a bit confusing at first, as it is the product of a unit of mass (kg) with a unit of acceleration (m/s^2). This is due to **Newton's Second Law**, which states

$$\text{force} = \text{mass} \times \text{acceleration}.$$

12.3 Dot Product

Definition 56 (Dot Product)**.** Let $\vec{a}$ and $\vec{b}$ be vectors. The *dot product* of $\vec{a}$ and $\vec{b}$ is

$$\vec{a} \cdot \vec{b} = a_1 b_1 + a_2 b_2 + a_3 b_3.$$

A similar definition holds for vectors of length 2 (or 10, or 35, etc).

Example 57. Let $\vec{a} = \langle 1, -4, 1 \rangle$, $\vec{b} = \langle 0, 2, -2 \rangle$, and $\vec{c} = \langle 5, 2, 1 \rangle$ Then

$$\vec{a} \cdot \vec{b} = (1)(0) + (-4)(2) + (1)(-2) = 0 - 8 - 2 = -10.$$

Similarly,

$$\vec{a} \cdot \vec{c} = 5 - 8 + 1 = -2 \quad \text{and} \quad \vec{b} \cdot \vec{c} = 0 + 4 - 2 = 2.$$

End of Example 57. □

The dot product has important geometric meaning, which we will explore. But first, let's make some observations about the properties of the dot product, summarized in the following proposition:

Proposition 58 (Dot Product Properties)**.** *Let $\vec{a} = (a_1, a_2, a_3)$, and $\vec{b} = (b_1, b_2, b_3)$ and $\vec{c} = (c_1, c_2, c_3)$ be vectors. Then the following properties hold*

- $\vec{0} \cdot \vec{a} = 0$
- $\vec{a} \cdot \vec{b} = \vec{b} + \vec{a}$
- $\vec{a} \cdot \vec{a} = a_1^2 + a_2^2 + a_3^2 = |\vec{a}|^2$
- $\vec{a} \cdot (\vec{b} + \vec{c}) = \vec{a} \cdot \vec{b} + \vec{a} \cdot \vec{c}$

The next theorem connects the dot product of two vectors with the angle between them.

Theorem 59 (The dot-product/angle relationship)**.** *Let $\vec{a}$ and $\vec{b}$ be vectors. Suppose that the angle between $\vec{a}$ and $\vec{b}$ is θ. Then*

$$\vec{a} \cdot \vec{b} = |\vec{a}||\vec{b}|\cos(\theta).$$

Let's try to unwrap some implications of this theorem:

- When θ is an acute angle, that means $0 < \theta < \frac{\pi}{2}$. In that case, $\cos(\theta) > 0$, so $\vec{a} \cdot \vec{b} > 0$.
- On the other hand, when θ is an obtuse angle, that means $\frac{\pi}{2} < \theta < \pi$. In that case, $\cos(\theta) < 0$, and so $\vec{a} \cdot \vec{b} < 0$.
- When we have a right angle, $\theta = \frac{\pi}{2}$, so in that case $\vec{a} \cdot \vec{b} = 0$.

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12.3 (dot product) and 12.4 (cross product)

Theorem 60 (The dot-product/angle relationship)**.** *Let $\vec{a}$ and $\vec{b}$ be vectors. Suppose that the angle between $\vec{a}$ and $\vec{b}$ is θ. Then*

$$\vec{a} \cdot \vec{b} = |\vec{a}||\vec{b}|\cos(\theta).$$

Proposition 61 (Orthogonality condition)**.** *We say that two vectors $\vec{a}$ and $\vec{b}$ are orthogonal (or equivalently, perpendicular) if the angle between them is $\theta = \frac{\pi}{2}$. Theorem 60 implies $\vec{a}$ and $\vec{b}$ are orthogonal precisely when*

$$\vec{a} \cdot \vec{b} = 0.$$

Morover, whether the dot product is positive or negative tells us whether the angle θ is acute or obtuse.

A direct consequence of Theorem 60 and the fact that $|\cos(\theta)| \leq 1$ is the following theorem:

Theorem 62 (Cauchy-Schwarz Inequality)**.**

$$|\vec{a} \cdot \vec{b}| \leq |\vec{a}||\vec{b}|$$

Example 63. Find the angles between the following vectors

- $\vec{a} = \langle 2, 1, 2 \rangle$ and $\vec{b} = \langle 3, 3, 1 \rangle$
- $\vec{a} = \langle 2, 2, -1 \rangle$ and $\vec{b} = \langle 5, -4, 2 \rangle$ (perpendicular)

End of Example 63. □

Example 64. Consider the triangle consisting of the following 3 points:

- $P = (1, -3, -2)$
- $Q = (2, 0, -4)$
- $R = (6, -2, -5)$

Is this triangle a right triangle? Since the triangle lives in 3D spaces, so it's not so easy to simply plot it and look. Use vectors to determine if it is a right triangle.

End of Example 64. □

13.1 Projection

Definition 65 (Vector Projection)**.** Let $\vec{a}, \vec{b}$ be vectors. The *vector projection of $\vec{b}$ onto $\vec{a}$* is

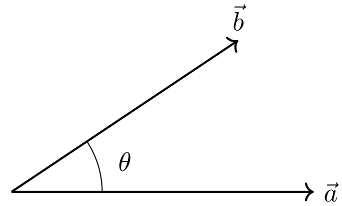
$$\begin{aligned}\vec{P} &:= \left(\frac{\vec{a} \cdot \vec{b}}{|\vec{a}|}\right)\left(\frac{\vec{a}}{|\vec{a}|}\right) \\ &= (\text{length of } P)\,(\text{unit vector})\end{aligned}$$

The book uses the notation $\vec{P} = \text{proj}_{\vec{a}}(\vec{b})$.

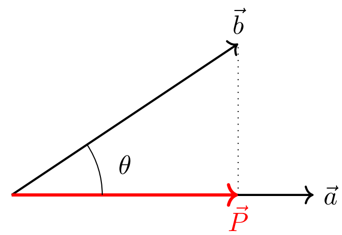
To break this equation down, observe that $\frac{\vec{a}}{|\vec{a}|}$ is a unit vector in the direction of $\vec{a}$. The quantity in parentheses is a number, not a vector. It scales the unit vector $\frac{\vec{a}}{|\vec{a}|}$. This number is the (signed) length of the shadow cast by $\vec{b}$ on $\vec{a}$ when the sun is directly overhead. It is called the *component of $\vec{b}$ along $\vec{a}$*, or alternatively, the *scalar projection of $\vec{b}$ onto $\vec{a}$*.

I think of projections in terms of shadows.

Question: Why is $\frac{\vec{a} \cdot \vec{b}}{|\vec{a}|}$ equal to the (signed) length of $\vec{P}$?
Answer: Suppose we've got two vectors like this:



Now let's draw the projection $\vec{P}$, along with a dotted line perpendicular to $\vec{a}$:



Note that we have a right angle. By trigonometry, we know that

$$\cos(\theta) = \frac{\text{length of adjacent side}}{\text{length of hypotenuse}}$$

In our case, that means

$$\cos(\theta) = \frac{\text{length of } \vec{P}}{\text{length of } \vec{b}} = \frac{|\vec{P}|}{|\vec{b}|} \tag{15}$$

ON THE OTHER HAND, by Theorem 60, we ALSO know that

$$\cos(\theta) = \frac{\vec{a} \cdot \vec{b}}{|\vec{a}||\vec{b}|} \tag{16}$$

Therefore, combining Eqs. (15) and (16), we have:

$$\frac{|\vec{P}|}{|\vec{b}|} = \frac{\vec{a} \cdot \vec{b}}{|\vec{a}||\vec{b}|},$$

which simplifies to

$$\boxed{|\vec{P}| = \frac{\vec{a} \cdot \vec{b}}{|\vec{a}|}}$$

This answers our original question. [To be pedantic, here we have actually only considered the case where θ is an acute angle. The case where θ is an obtuse angle is a little bit different, but not much.]

Example 66. A plane is flying in the direction $\vec{D} = \langle 1, 1 \rangle$. The plane is getting hit by a tailwind with wind velocity $\vec{V} = \langle 100, 0 \rangle$. Find P, the projection of V onto D. Here, V points east and D points northeast.

$$\text{length } |P| = \frac{D \cdot V}{|D|} = \frac{100}{\sqrt{2}}$$

$$\begin{aligned} \text{vector } P &= |P|\left(\frac{D}{|D|}\right) \\ &= \left(\frac{100}{\sqrt{2}}\right)\left(\frac{\langle 1, 1 \rangle}{\sqrt{2}}\right) \\ &= \langle 50, 50 \rangle \end{aligned}$$

End of Example 66. □

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Definition 67 (Vector Projection)**.** Let A, B be vectors. The *vector projection of B along A* is

$$\begin{aligned} P &:= \left(\frac{A \cdot B}{|A|}\right)\left(\frac{A}{|A|}\right) \\ &= (\text{signed length of } P)\,(\text{unit vector}) \end{aligned}$$

The book uses the notation $P = \operatorname{proj}_A(B)$, and $\operatorname{comp}_A B = \frac{A \cdot B}{|A|}$

Example 68. Find the vector projection of $\vec{b} = \langle 1, 1, 2\rangle$ onto $\vec{a} = \langle -2, 3, 1\rangle$.
Solution: We will first compute the length of P:

$$\begin{aligned} \text{signed length of } \overrightarrow{P} &= \frac{\vec{a} \cdot \vec{b}}{|\vec{a}|} \\ &= \frac{(-2)(1) + 3(1) + 1(2)}{\sqrt{(-2)^2 + 3^2 + 1^2}} \\ &= \frac{3}{\sqrt{14}} \end{aligned}$$

Therefore the length of the projection is $\frac{3}{14}$. Note that we computed $|a| = \sqrt{14}$.

The vector projection is therefore

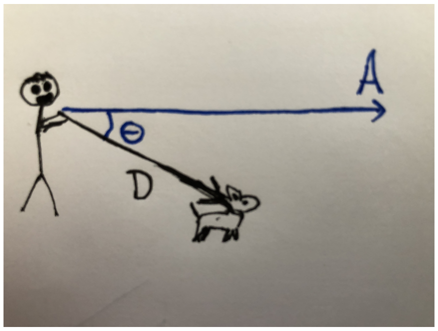
$$\begin{aligned} P &= \left(\frac{3}{\sqrt{14}}\right)\frac{\vec{a}}{|\vec{a}|} \\ &= \left(\frac{3}{\sqrt{14}}\right)\frac{\vec{a}}{\sqrt{14}} \\ &= \frac{3}{14}\vec{a} \\ &= \frac{3}{14}\langle -2, 3, 1\rangle \\ &= \left\langle -\frac{6}{14}, \frac{9}{14}, \frac{3}{14}\right\rangle. \end{aligned}$$

End of Example 68. ☐

14.1 Connection to physics: Effective force in a direction

If a force $\overrightarrow{F}$ is applied to a particle moving along a path, we may wish to know the magnitude of the force in the direction of motion. If A is a direction, then the projection of F along A is the *effective force* in the direction A.

Example 69. Suppose you are talking a dog on a walk and that you are facing in direction $\overrightarrow{A}$. Suppose the dog lunges in some other direction. Let $\overrightarrow{D}$ be force exerted by the pull of the leash. Then the effective force with whic the dog pulls you forward is the projection of D along A.



For example, if $A = \{1, 1, 0\}$ and $D = \langle 13, 12, -5\rangle$, then the projection is

$$\begin{aligned} P &= \left(\frac{A \cdot D}{|A|}\right)\left(\frac{A}{|A|}\right) \\ &= \left(\frac{\langle 13 + 12 + 0\rangle}{\sqrt{2}}\right)\left(\frac{\langle 1, 1, 0\rangle}{\sqrt{2}}\right) \\ &= \left(\frac{25}{\sqrt{2}}\right)\left\langle \frac{1}{\sqrt{2}}, \frac{1}{\sqrt{2}}, 0\right\rangle \\ &= \left\langle \frac{25}{2}, \frac{25}{2}, 0\right\rangle. \end{aligned}$$

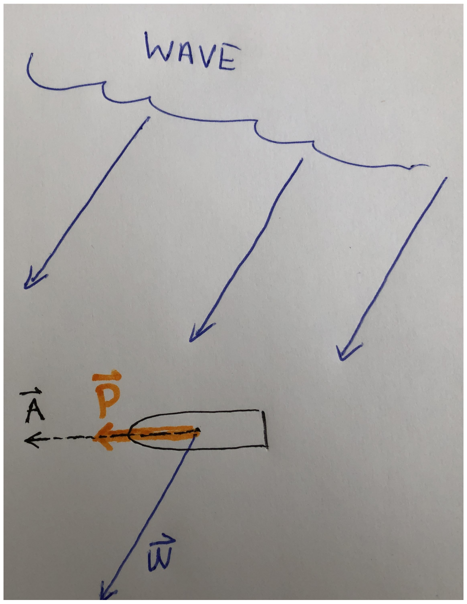
The length of P is

$$|P| = \frac{A \cdot D}{|A|} = \frac{25}{\sqrt{2}} \approx 17.7.$$

So the dog pulls you forward with about 17.7 N of force. That's not enough to topple you unless you are either very frail, or a toddler.

End of Example 69. □

Example 70. Suppose we have a surfer pointed in direction A, who catches a wave which exerts a force (represented by the vector W) on the surfboard, as shown:



Let's assume that $A = \langle -4, -1\rangle$ and that $W = \langle -700, -600\rangle$. The length of vector A doesn't matter; this vector is just here to indicate the direction of the surfboard. We are assuming that the surfboard starts out stationary. On the other hand, -700 and -600 are measured in Newtons. (These vectors aren't quite to scale in the above picture). The effective force of of the wave moving surfboard forward in the A direction is the projection of W along A, which we will denote by P.

To compute this, we will first do some preliminary computations:

$$|A| = \sqrt{(-4)^2 + (-1)^2} = \sqrt{17},$$

and

$$A \cdot W = \langle -4, -1\rangle \cdot \langle -700, -600\rangle = 2800 + 600 = 3400.$$

Then

$$\begin{aligned}
P &= \left(\frac{A \cdot W}{|A|}\right)\left(\frac{A}{|A|}\right) && \text{by Definition 67.}\\
&= \left(\frac{\langle -4,-1\rangle \cdot \langle -700,-600\rangle}{\sqrt{17}}\right)\left(\frac{\langle -4,-1\rangle}{\sqrt{17}}\right)\\
&= \left(\frac{3400}{\sqrt{17}}\right)\left(\frac{\langle -4,-1\rangle}{\sqrt{17}}\right)\\
&= \frac{3400}{17}\langle -4,-1\rangle\\
&= 200\langle -4,-1\rangle\\
&= \langle -800,-200\rangle .
\end{aligned}$$

Here, the intial force of the wave was W, which had a magnitude of $|W| = \sqrt{(-700)^2 + (-600)^2} = \sqrt{100^2\left(7^2+6^2\right)} = 100\sqrt{85} \approx 920\text{N}$. The effective force in the directio of motion, was $P = \langle -800,-200\rangle = 100\langle -8,-2\rangle$, which had a slightly lower magnitude of $|P| = 100\sqrt{64+4} \approx 820\text{N}$.

Question: How much does the surfboard accelerate in direction **A**? Assume the surfer weighs 70kg.

By Newton's second law,

$$F = ma$$

and therefore,

$$\begin{aligned}
a &= \frac{F}{m}\\
&= \frac{820\text{ N}}{70\text{ kg}}
\end{aligned}$$

Recall that $\text{N} = \text{kg} \times \frac{\text{m}}{\text{s}^2}$, so $\frac{\text{N}}{\text{kg}} = \frac{\text{m}}{\text{s}^2}$

$$a = \frac{820}{70}\frac{\text{m}}{\text{s}^2} \approx 11.7\text{m/s}^2$$

This means that if the force lasts for half a second, the surfer will be accelerated to about $\frac{11.7}{2} = 5.85$ meters per second.

End of Example 70. □

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Section 12.4

15.1 Cross Product

The idea of "slope" allows us to describe the tilt of a line. We need something analogous for describing the tilt of a plane. This is done with by describing a vector that is *perpendicular* to the plane, called the *normal vector*. So we need a way to find vectors that are perpendicular to things. In 3D, the simplest way to do this is with the following operation (it's not really simple, sorry):

Definition 71 (Cross Product)**.** Let $\vec{a} = \langle a_1, a_2, a_3 \rangle$ and $\vec{b} = \langle b_1, b_2, b_3 \rangle$. The *cross product* of $\vec{a}$ and $\vec{b}$, denoted $\vec{a} \times \vec{b}$ is the vector

$$\vec{a} \times \vec{b} = \langle a_2b_3 - a_3b_2, \quad a_3b_1 - a_1b_3, \quad a_1b_2 - a_2b_1 \rangle .$$

Fact 1: This new vector is orthogonal to both $\vec{a}$ and $\vec{b}$.
Fact 2: There is no cross product for 2D.

There is a trick to computing cross products, and remembering how to compute them. This is called the *method of cofactor expansion*, which I will demonstrate now. This method touches on important ideas that arise in linear algebra, which gives it additional value in its own right. Given a two-by two matrix

$$\begin{bmatrix} a & b \\ c & d \end{bmatrix}$$

the determinant of the matrix, which we denote by $\begin{vmatrix} a & b \\ c & d \end{vmatrix}$, is the quantity

$$\begin{vmatrix} a & b \\ c & d \end{vmatrix} = ad - bc.$$

The cross product $\vec{a} \times \vec{b}$ can be computed as

$$\begin{vmatrix} \vec{\imath} & \vec{\jmath} & \vec{k} \\ a_1 & a_2 & a_3 \\ b_1 & b_2 & b_3 \end{vmatrix} = \vec{\imath} \begin{vmatrix} a_2 & a_3 \\ b_2 & b_3 \end{vmatrix} - \vec{\jmath} \begin{vmatrix} a_1 & a_3 \\ b_1 & b_3 \end{vmatrix} + \vec{k} \begin{vmatrix} a_1 & a_2 \\ b_1 & b_2 \end{vmatrix}$$

Note that the sign *alternates:* the term with the $\vec{\jmath}$ is subtracted, not added like the first and third term.

Example 72. Suppose $\vec{a} = \langle 1, 3, 4 \rangle$ and $\vec{b} = \langle 2, 7, -5 \rangle$. The cross product is

$$\begin{aligned} \begin{vmatrix} \vec{\imath} & \vec{\jmath} & \vec{k} \\ 1 & 3 & 4 \\ 2 & 7 & -5 \end{vmatrix} &= \vec{\imath} \begin{vmatrix} 3 & 4 \\ 7 & -5 \end{vmatrix} - \vec{\jmath} \begin{vmatrix} 1 & 4 \\ 2 & -5 \end{vmatrix} + \vec{k} \begin{vmatrix} 1 & 3 \\ 2 & 7 \end{vmatrix} \\ &= \vec{\imath}(-15 - 28) - \vec{\jmath}(-5 - 8) + \vec{k}(7 - 6) \\ &= \vec{\imath}(-43) - \vec{\jmath}(-13) + \vec{k}(1) \\ &= \langle -43, 13, 1 \rangle \end{aligned}$$

Check our answer: Is this new vector indeed orthogonal to both $\vec{a}$ and $\vec{b}$? To check this, we can use the fact that two vectors A and B are orthogonal precisely when $A \cdot B = 0$. Indeed, doing the dot product computation gives

$$\vec{a} \cdot \langle -43, 13, 1 \rangle = \langle 1, 3, 4 \rangle \cdot \langle -43, 13, 1 \rangle = -43 + 39 + 4 = 0$$

Since this dot product is zero, we conclude that our new vector $\vec{a} \times \vec{b}$ is orthogonal to $\vec{a}$. Similarly, we have

$$\vec{b} \cdot \langle -43, 13, 1 \rangle = \langle 2, 7, -5 \rangle \cdot \langle -43, 13, 1 \rangle = -86 + 91 - 5 = 0,$$

and this implies that $\vec{a} \times \vec{b}$ is orthogonal to $\vec{b}$.

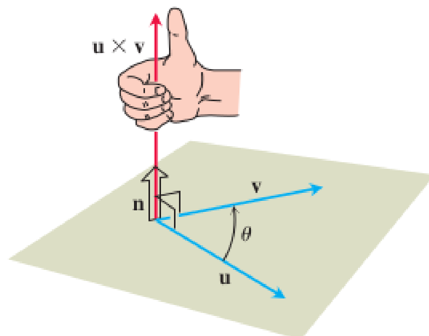
End of Example 72. □

The next theorem establishes a geometric interpretation of the cross product:

Theorem 73. *Let $\vec{u}$ and $\vec{v}$ be 3d vectors. If $\vec{u}$ and $\vec{v}$ are not parallel, then the vectors determine a plane. The plane is*

$$P = \left\{(x, y, z) \in \mathbb{R}^3 : (x, y, z) = c_1\vec{u} + c_2\vec{v} \text{ for some } c_1, c_2 \in R^2\right\}$$

In words, P is set of all linear combinations of the vectors $\vec{u}$ and $\vec{v}$. We select a unit vector $\vec{n}$ called the unit normal vector according to the right hand rule as follows.



In words, if your fingers curl from $\vec{u}$ to $\vec{v}$, then $\vec{n}$ points in the direction of your thumb. Then

$$\vec{u} \times \vec{v} = (|\vec{u}||\vec{v}|\sin(\theta))\,\vec{n}. \tag{17}$$

where $\theta \in [0, \pi]$ is the angle between the two vectors $\vec{u}$ and $\vec{v}$.

This geometric interpretation comes with a number of implications:

(A) Eq. (17) implies that two vectors $\vec{u}$ and $\vec{v}$ are parallel (i.e., so that $\theta = 0$), precisely when

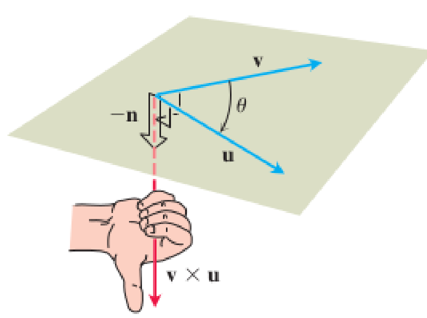
$$\vec{u} \times \vec{v} = 0.$$

(B) We verify the fact, stated earlier, that $\vec{u} \times \vec{v}$ is orthogonal to both $\vec{u}$ and $\vec{v}$.

(C) The cross product is *anticommutative*:

$$\vec{v} \times \vec{u} = -(\vec{u} \times \vec{v})$$

Because when computing $\vec{v} \times \vec{u}$, your fingers have to sweep from $\vec{v}$ to $\vec{u}$, which causes your thumb to point the other way, as shown:



And geometrically, reversing the direction of vector is the same as mupltiplying it by -1. This is is why

$$\vec{v} \times \vec{u} = -(\vec{u} \times \vec{v})$$

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Recall we had the following theorem:

Theorem 74. *If u and v are vectors in 3-space with an angle of $\theta \in [0, \pi]$, then*

$$u \times v = (|u||v|\sin(\theta))\vec{n}$$

This is a nice geometric result. Two more implications are as follows:

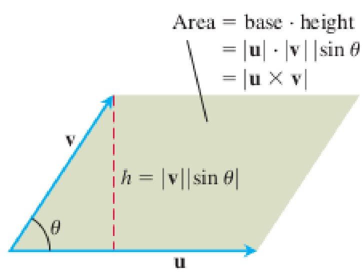
1. If r, s are scalars then we can pull them out:

$$(s\vec{u}) \times (r\vec{v}) = (sr)(\vec{u} \times \vec{v})$$

2. We get a nice way to compute the *magnitude* of the cross product:

$$|\vec{u} \times \vec{v}| = |u||v|\sin(\theta),$$

and we get a neat interpretation too: the magnitude of the cross product equals the area of the parallelogram:



Example 75. Find the area of the triangle with vertices $P = (1, 4, 6)$, $Q = (-2, 5, -1)$ and $R = (1, -1, 1)$
Solution: this is example 4, p858

End of Example 75. ☐

Example 76 (Application to physics: torque)**.** Suppose we have a wrench, whose handle is represented by a vector $\vec{r}$. Turning the wrench will tighten a bolt or something. The tendency of an object to rotate is represented by a vector called the torque vector, denoted $\vec{\tau}$, which is similar to—but *not* the same as—a force vector.

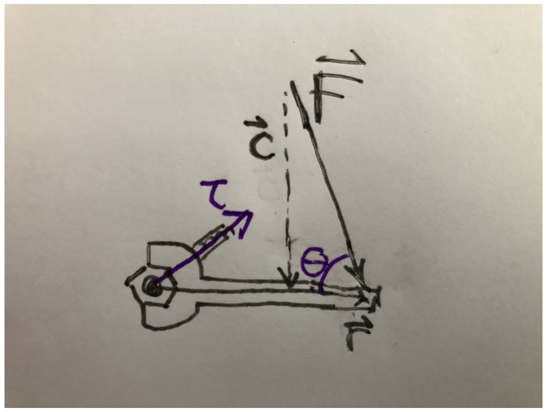
The magnitude of τ, $|\tau|$ determines how much push there is to rotate:

$$\text{bigger } |\tau| = \text{more push to rotate}$$

This is similar to a force vector. What is different is that for a force vector $\vec{F}$, the direction of $\vec{F}$ indicates the direction in which motion would occur. This is NOT the case for torque. For torque, as we will see, the direction instead indicates the following two things:

- the axis of rotation
- whether the rotation is clockwise or anticlockwise.

Suppose that the bolt is stuck, so I take a hammer and hit the end of my wrench. The hammer strike is represented by a force vector F, as shown:



Notice that I didn't have great aim, so although my hammer hit the tip of the wrench, the hammer didn't come down exactly perpendicular to the wrench. Instead, it was at a bit of an angle θ.

The direction of τ is always parallel to the axis of rotation—so τ will be a vector which passes through the bolt. Whether it points up or down is always determined by whether the rotation is clockwise or anticlockwise, according to the right-hand rule. [1]

In this setting, if we think about this phsically, we will conclude that the magnitude of the torque should be determined by two things:

- The length of the wrench handle $|\vec{r}|$.
- How much of the force gets applied in the direction perpendicular to $\vec{r}$. In the picture above, this is the magnitude of the vector $\vec{c}$. The vector $\vec{c}$ is the component of $\vec{F}$ which is perpendicular to $\vec{r}$. In particular we can compute $|\vec{c}|$ by observing that

$$\sin(\theta) = \frac{\text{opposite}}{\text{hypotenuse}}\frac{|\vec{c}|}{|\vec{F}|}$$

so that

$$|\vec{c}| = |\vec{F}|\sin(\theta).$$

It turns out that the cross product is the right thing to use to represent the torque. That is, we define

$$\vec{\tau} = \vec{r} \times \vec{F}$$

In particular, by Eq. (17), the magnitude of the torque is

$$\begin{aligned} |\vec{\tau}| &= |\vec{r} \times \vec{F}| \\ &= \underbrace{|r|}_{\substack{\text{length of} \\ \text{wrench}}} \underbrace{|F|\sin(\theta)}_{|\vec{c}|} \end{aligned}$$

End of Example 76. □

[1] i.e., if you curl the fingers of your right hand along the axis in the direction of the rotation, then τ should point in the direction of your thumb. Whether τ points up or down has nothing to do with whether the bolt gets looser or tighter.. the direction only indicates whether the rotation is clockwise or anticlockwise.

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17.1 Torque Example

Recall that $A \times B = (|A||B|\sin(\theta))\,\vec{n}$.

Suppose we are attempting to turn a bolt using a wrench of length $r = .25\texttt{m}$ and we apply a force $\vec{F}$ of $40\texttt{N}$ at an angle of $75°$. What is the magnitude of the torque applied to the bolt?

$$\begin{aligned}|\tau| &= |\vec{r} \times \vec{F}| \\ &= |r||F|\sin(\theta) \\ &= (0.25\texttt{m})(40\texttt{N})\sin(75°) \\ &\approx 9.66\texttt{Nm} \\ &= 9.66\texttt{J}\end{aligned}$$

where we used the fact that `1 Joule = 1 Netwon-meter`.

The torque vector is

$$\tau \approx 9.66\vec{n}$$

What direction does $\vec{n}$ point? If we are tightening the bolt then the rotation is clockwise, and the torque vector points away from us. (This matches with the righty-tighty lefty-loosy convention that almost all bolts and screws obey.)

17.2 Lines in any dimension

12.5 in Stewart

In this section, we'll use the notation $\mathbb{R}^n$ to mean n-dimensional space, which is the set of points of the form

$$(x_1, x_2, \ldots, x_n)$$

where $x_1, \ldots, x_n$ are all real numbers. You are already familiar with $\mathbb{R}^2$ and $\mathbb{R}^3$, where we usually use (x, y) or (x, y, z) rather than (x_1, x_2) or (x_1, x_2, x_3).

Let $\vec{v} = (v_1, \ldots, v_n)$ is a nonzero vector in n-dimensional space. You can think of $n = 2$ or $n = 3$, but what follows will hold for any positive integer n. Then for any scalar t, the vector $t\vec{v}$ has the same direction as v when $t > 0$ and opposite direction when $t < 0$. Therefore the set of points

$$\{t\vec{v} : -\infty < t < \infty\}$$

is a line through the origin. We we add a vector $\vec{p} = \langle x_0, y_0, z_0\rangle$ to all of those points, we obtain

$$\{t\vec{v} + \vec{p} : -\infty < t < \infty\}$$

which is a line through $\vec{p}$ in the direction of $\vec{v}$.

Definition 77 (vector equation of a line)**.** Given a vector $\vec{p}$ and a nonzero vector $\vec{v}$ in $\mathbb{R}^n$, the set of all points $\vec{y}$ in $\mathbb{R}^n$ such that

$$\vec{y} = t\vec{v} + \vec{p} \tag{18}$$

where $-\infty < t < \infty$ is called the *line through $\vec{p}$ in the direction of $\vec{v}$*.

If we write $\vec{y} = (y_1, y_2, \ldots, y_n)$ and $\vec{v} = (v_1, v_2, \ldots, v_n)$ and $\vec{p} = (p_1, p_2, \ldots, p_n)$, then Eq. (18) can be written as

$$\begin{bmatrix} y_1 \\ y_2 \\ \vdots \\ y_n \end{bmatrix} = t \begin{bmatrix} v_1 \\ v_2 \\ \vdots \\ v_n \end{bmatrix} + \begin{bmatrix} p_1 \\ p_2 \\ \vdots \\ p_n \end{bmatrix}$$

This vector equation can be written as

$$\begin{cases} y_1 = v_1 t + p_1 \\ y_2 = v_2 t + p_2 \\ \vdots \\ y_n = v_n t + p_n \\ -\infty < t < \infty \end{cases}$$

which are called the *parametric equations of a line.*

Example 78. Suppose L is the line in $\mathbb{R}^2$ through the point $\vec{p} = (1,2)$ in the direction of $\vec{v} = (1,-3)$. Then the vector equation of the line is

$$\vec{y} = t\langle 1,-3\rangle + (1,3) = \langle t+1, -3t+2\rangle$$

for $-\infty < t < \infty$. If we let $\vec{y} = \langle x, y\rangle$, this is the parametric equations

$$\begin{cases} x = t+1 \\ y = -3t+2. \end{cases}$$

If we eliminate t, we get the equation

$$y = -3x + 5.$$

End of Example 78. $\square$

Example 79. Consider the following two points in 3D space

$$A = (2,4,-3) \quad \text{and} \quad B = (3,-1,1)$$

(a) Find the equation of the line which passes through A and B

(b) Find the equation of the line segment which connects A and B.

Solution to (a): We will follow the ideas presented in Definition 77. Let $\vec{v} = \overrightarrow{AB} = \langle 1,-5,4\rangle$. We need a vector $\vec{p}$ on the line, so we can take either $\vec{p} = A$ or $\vec{p} = B$ since both of these points is on the line. Let's take $\vec{p} = A$. Then the vector equation of the line is

$$\langle y_1, y_2, y_3\rangle = t\langle 1,-5,4\rangle + \langle 2,4,-3\rangle$$

and the parameteric equations of the line are

$$\begin{cases} y_1 = t+2 \\ y_2 = -5t+4 \\ y_3 = 4t-3 \\ -\infty < t < \infty \end{cases}$$

Since we are working in 3 dimensions, we may wish to write x, y, z rather than y_1, y_2, y_3, in which case our equations become

$$\langle x, y, z\rangle = t\langle 1,-5,4\rangle + \langle 2,4,-3\rangle$$

and

$$\begin{cases} x = t+2 \\ y = -5t+4 \\ z = 4t-3 \\ -\infty < t < \infty \end{cases}$$

Solution to (b): Since $\overrightarrow{AB} = (B - A)$, observe that our equation is of the form

$$\langle x, y, z\rangle = \overrightarrow{AB}t + A$$

or

$$\langle x, y, z \rangle = (B - A)t + A$$

When $t = 0$, we have

$$\langle x, y, z \rangle = A.$$

In other wordss, when $t = 0$, the ant is at point A. As t increases, the ant walks in the direction $\overrightarrow{AB}$, i.e., towards B.

Moreover, when $t = 1$, we have

$$\langle x, y, z \rangle = (B - A)(1) + A = B.$$

so the ant arrives at point B at time $t = 1$. So the equation of the line segement between A and B is

$$\begin{cases} \langle x, y, z \rangle = t \langle 1, -5, 4 \rangle + \langle 2, 4, -3 \rangle \\ 0 \leq t \leq 1 \end{cases}$$

This example also illustrates the more general fact that the *line segment* between two points A and B is given by the parametric equation

$$\begin{aligned} \langle x, y, z \rangle &= (B - A)t + A \\ &= tB + (1 - t)A \end{aligned}$$

for $0 \leq t \leq 1$.

(This equation makes sense: e.g. if $t = 1/2$ then we get the midpoint between A and B.)

End of Example 79. □

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18.1 Planes in 3D

In the last lecture we saw that a *line* in n-dimensional space can be represented as the set of points $\vec{y} = (y_1, \ldots, y_n)$ such that

$$\begin{cases} \vec{y} = t\vec{v} + \vec{p} \\ -\infty < t < \infty \end{cases}$$

where $\vec{v} = \langle v_1, \ldots, v_n \rangle$ is a nonzero vector and $\vec{p} = \langle p_1, \ldots, p_n \rangle$ represents any point in n-dimensional space.

If we have another nonzero vector $\vec{u}$ which isn't parallel to $\vec{v}$, then the set of points $\vec{y}$ satisfying the equation

$$\begin{cases} \vec{y} = t\vec{v} + s\vec{u} + \vec{p} \\ -\infty < s, t < \infty \end{cases} \tag{19}$$

is a *plane*. This is called the parametric equation of a plane.

We'll focus on understanding planes in 3-dimensional space. A plane $\mathbb{P}$ in 3D space is uniquely determined by the following two things:

- A normal vector $\vec{n} = (A, B, C)$, which determines the tilt of the plane (Note: A, B, C are real numbers)
- A point $\vec{p} = (x_0, y_0, z_0)$ which the plane passes through, to determined the location in 3D space where the plane lies.

The plane $\mathbb{P}$ consists of the set of points (x, y, z) satisfying the following equation:

$$A(x - x_0) + B(y - y_0) + C(z - z_0) = 0 \tag{20}$$

Eq. (20) is called the *normal form* equation of the plane. Frequently this is written as

$$Ax + By + Cz = D$$

where $D = Ax_0 + By_0 + Cz_0$.

We can also write Eq. (20) as the following dot product:

$$\underbrace{\langle A, B, C \rangle}_{\text{this is } \vec{n}} \cdot \langle x - x_0, y - y_0, z - z_0 \rangle = 0$$

or

$$\vec{n} \cdot \begin{bmatrix} x - x_0 \\ y - y_0 \\ z - z_0 \end{bmatrix} = 0$$

Example 80. Find the normal equation of the plane through the point $\vec{p} = (2, 4, -1)$ with normal vector $\vec{n} = \langle 2, 3, 4 \rangle$. Find the intercepts and sketch the plane.

Solution: By Eq. (20), with $A = 2$, $B = 3$, $C = 4$, we have:

$$2(x - 2) + 3(y - 4) + 4(z + 1) = 0.$$

We are done. We could also write this as:

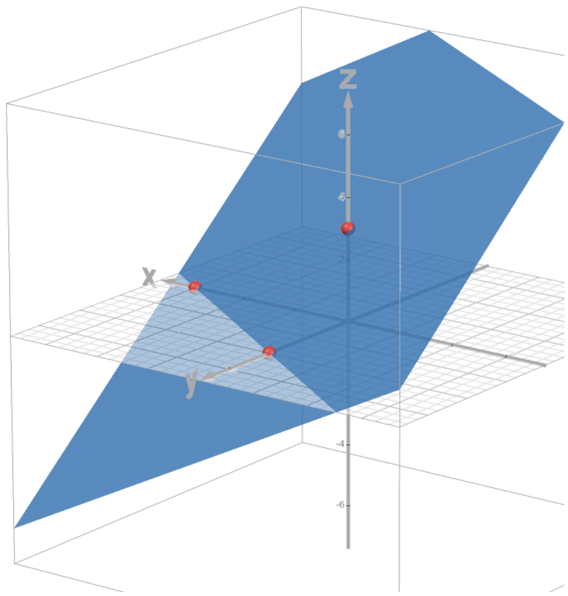
$$2x + 3y + 4z = 12. \tag{21}$$

To find the intercepts, we consider what happens when we set two of the variables x, y, z equal to zero at a time:

- To find the x-interecept, set $y = z = 0$ in Eq. (21). In that case, we get $x = 6$. Hence the x-intercept is $(6, 0, 0)$.
- To find the y-interecept, set $x = z = 0$ in Eq. (21). In that case, we get $y = 4$. Hence the y-intercept is $(0, 4, 0)$.

- To find the z-interecept, set $x = y = 0$ in Eq. (21). In that case, we get $z = 3$. Hence the z-intercept is $(0, 0, 3)$.

Recall that any two points uniqqly determine a line. A similar fact holds for planes:
Fact: A plane is uniquely determined by any three points (as long as they don't all lie in a line).
So the plane can be plotted as follows:



(see: `https://www.desmos.com/3d/brbgxh7qz7`)

End of Example 80. $\square$

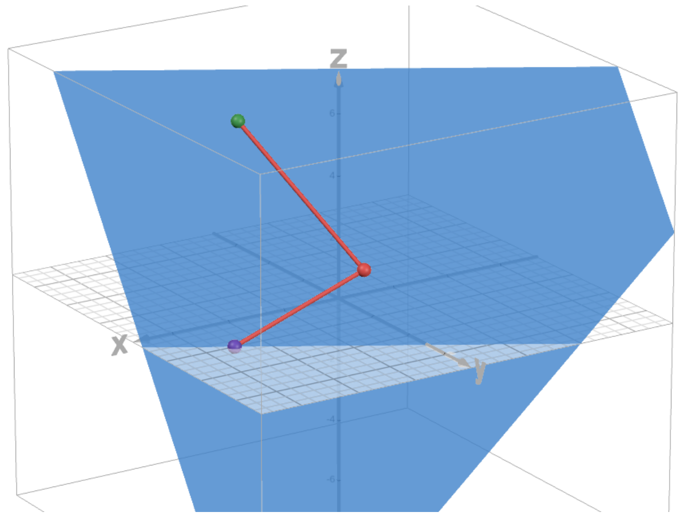
Example 81. Consider the three points $P = (1, 3, 2)$, $Q = (3, -1, 6)$, and $R = (5, 2, 0)$. These three points are not colinear (i.e., they don't all lie in a line), and hence there is a single, unique plane passing through all three of them. Find the equation of the plane passing through P, Q, and R.
Solution:

- $\overrightarrow{PQ} = \langle 2, -4, 4\rangle$
- $\overrightarrow{PR} = \langle 4, -1, -2\rangle$

We need a normal vector $\vec{n}$ which is orthogonal to both $\overrightarrow{PQ}$ and $\overrightarrow{PR}$. We can use the cross product. Let

$$\begin{aligned}
\vec{n} &= \overrightarrow{PQ} \times \overrightarrow{PR} \\
&= \begin{vmatrix} \vec{\imath} & \vec{\jmath} & \vec{k} \\ 2 & -4 & 4 \\ 4 & -1 & -2 \end{vmatrix} \\
&= \vec{\imath}(8 + 4) - \vec{\jmath}(-4 - 16) + \vec{k}(-2 + 16) \\
&= 12\vec{\imath} + 20\vec{\jmath} + 14\vec{k} \\
&= \langle 12, 20, 14\rangle
\end{aligned}$$

Here's a picture:



(see `https://www.desmos.com/3d/sqdoxhvsng`)

The parametric equation is

$$\begin{cases} \vec{y} = t\overrightarrow{PQ} + s\overrightarrow{PR} + \vec{P} \\ -\infty < t, s < \infty \end{cases}$$

and letting $\vec{y} = \langle x, y, z \rangle$, we can write this out at

$$\begin{bmatrix} x \\ y \\ z \end{bmatrix} = t \begin{bmatrix} 2 \\ -4 \\ 4 \end{bmatrix} + s \begin{bmatrix} 4 \\ -1 \\ -2 \end{bmatrix} + \begin{bmatrix} 1 \\ 3 \\ 2 \end{bmatrix}$$

End of Example 81. □

Example 82. Find the angle between the two planes $x + y + z = 1$ and $x - 2y + 3x = 1$.

Solution: It will be enough to find the angle between the two normal vectors. We can use the dot product. We'll get $\cos(\theta) = \frac{2}{\sqrt{41}}$. The angle with this cosine is about 72 degrees.

End of Example 82. □

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Section 12.6: Quadric surfaces

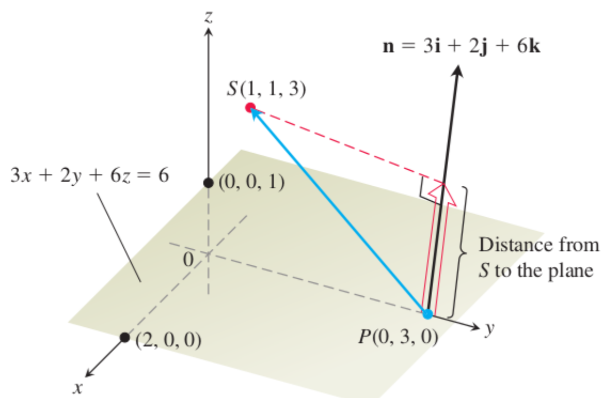
19.1 More plane examples

Example 83. Find the distance from the point $S = (1, 1, 3)$ to the plane $3x + 2y + 6z = 6$.

Solution:

Step 1. Find *any* point on the plane. For example, $P = (0, 3, 0)$ lies on the plane.

Step 2. Draw it



Step 3. Observe that the distance is length of the projection of $\overrightarrow{PS}$ along $\vec{n}$:

$$\text{Projection} = \left(\frac{\overrightarrow{PS} \cdot \vec{n}}{|\vec{n}|}\right)\left(\frac{\vec{n}}{|\vec{n}|}\right)$$

This has length

$$\text{length} = \frac{\vec{PS} \cdot \vec{n}}{|\vec{n}|}$$

which is

$$\begin{aligned}\text{length} &= \frac{\langle 1, -2, 3\rangle \cdot \langle 3, 2, 6\rangle}{\sqrt{9 + 4 + 36}} \\ &= \frac{17}{\sqrt{49}} \\ &= \frac{17}{7}.\end{aligned}$$

End of Example 83. □

Example 84. Find the equation of the plane which passes through the poitns $(0, -2, 5)$ and $(-1, 3, 1)$ and is perpendicular to the plane $2z = 5x + 4y$.

End of Example 84. □

19.2 Cylinders and quadric surfaces

Let $f = f(x, y, z)$ be a function. A *level set* of f is an set of the from

$$\{(x, y, z) \in \mathbb{R}^3 : f(x, y, z) = c\}$$

where $c \in \mathbb{R}$.

If we take $c = 0$,

$$\left\langle (x, y, z) \in \mathbb{R}^3 : f(x, y, z) = 0 \right\rangle$$

is called the *zero set* of f. For brevity, we usually don't write out all the set notation, and instead just write the equation, like this:

$$f(x, y, z) = 0.$$

Frequently, level sets form interesting shapes or surfaces, especially in higher dimensions. In two dimensions,

- $y - x^2 = 0$ is a parabola
- $2x + y - 3 = 0$ is a line
- $x^2 + y^2 = 0$ is a circle

Here we focus on 3-dimension. Here are some examples in 3-dimensions

- $x^2 + y^2 - 1 = 0$ is the unit cylinder
- $x^2 + y^2 + z^2 - 9 = 0$ is a sphere with radius 3.
- $z = x^2$ is a parabolic cylinder
- $Ax + By + Cz + D = 0$ is a plane

In 3-dimensions, a *quadric surface* is the zero set of a degree 2 polynomial in the variables x, y, and z. In other words, it is the graph of an equation of the form

$$Ax^2 + By^2 + Cz^2 + Dxy + Eyz + Fxz + Gx + Hy + Iz + J = 0,$$

wher $A, B, \ldots, J$ are real numbers. This forms a surface. There are, obviously, lots of possibilities so we can't do them all, but we'll do some special cases that will provide a rich source of examples of surfaces in 3D that we can work with.

Before we start we recall some things from 2-dimensional space (see section 10.5 in the texbook):

Proposition 85 (Conic sections in 2D)**.**

1. *The equations*

$$\frac{x^2}{a^2} + \frac{y^2}{b^2} = 1$$

   is an ellipse passing through the points $(a, b), (a, -b), (-a, b)$ *and* $(-a, -b)$.

2. *The equation*

$$\frac{x^2}{a^2} - \frac{y^2}{b^2} = 1$$

   is a hyperbola which opens to the sides with vertices $(\pm a, 0)$ *and asymptotes* $y = \pm \frac{b}{a} x$.

3. *The equation*

$$\frac{y^2}{a^2} - \frac{x^2}{b^2} = 1$$

   is a hyperbola which opens up and down, with vertices $(0, \pm b)$ *and asymptotes* $y = \pm \frac{a}{b} x$

Example 86 (Quadric Form, Special case #1: Ellipsoid)**.** Consider the quadric form

$$\frac{x^2}{a^2} + \frac{y^2}{b^2} + \frac{z^2}{c^2} - 1 = 0.$$

For example,

$$x^2 + \frac{y^2}{9} + \frac{z^2}{4} - 1 = 0$$

What is this shape?

To help determine, we will consider the *traces*, which are the cross-sections of the surface obtained by intersecting it with planes parallel to the coordinate planes.

For example, we could intersect the surface with the plane $z = k$, where k is a number. This plane is parallel to the xy-plane. That gives

$$x^2 + \frac{y^2}{9} + \frac{k^2}{4} - 1 = 0$$

or

$$x^2 + \frac{y^2}{9} = 1 - \frac{k^2}{4}.$$

As long as $k^2 < 4$, we can write this as

$$\frac{x^2}{1 - \frac{k^2}{4}} + \frac{y^2}{9(1 - \frac{k^2}{4})} = 1$$

which is the equation of an ellipse by Proposition 85.

All of the traces are ellipses. (You should check this). Ths is an *ellipsoid*.

End of Example 86. □

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Continue 12.6 - quadric forms

Example 87 (Quadric Form, Special case #2: Elliptic Paraboloid)**.** Consider the surface defined by the equation

$$Ax^2 + By^2 + z = 0.$$

For example,

$$z = 4x^2 + y^2.$$

What is this shape?

Again, we consider the traces. If $x = 0$, then we get $z = y^2$, a parabola. So the yz-plane intersects at a parabola. If we put $x = k$ for any constant k, we get

$$z = y^2 + 4k^2.$$

This means that any plane parallel to the yz-plane intersects our surface in a parabola.

The traces $y = k$ also gives parabola.

The traces $z = k$ gives something different:

$$4x^2 + y^2 = k$$

When $k < 0$, there are no real numbers which satisfy the equation. But when $k > 0$, this is an ellipse:

$$\frac{x^2}{(k/4)} + \frac{y^2}{k} = 1.$$

As k increases, this elipse gets bigger. This shape is called an *elliptic paraboloid.*

End of Example 87. ☐

Example 88 (Quadric Form, Special case #3: Hyperbolic Paraboloid)**.**

$$\frac{z}{c} = \frac{x^2}{a^2} - \frac{y^2}{b^2}$$

for example

$$z = \frac{x^2}{9} - \frac{y^2}{4}$$

- some traces are hyperbolas
- some traces parabolas

It looks like a pringle. Check it out on Desmos (along with some cross sections) here:

`https://www.desmos.com/3d/xuaxssutef`

End of Example 88. ☐

20.1 Vector functions and space curves

Please read sections 13.1 and 13.2

We are interested in describing **motion along a curve**.

A *vector-valued* function is a function whose domain is a set of real numbers and whose range is a set of vectors. We are most interested in the 3-dimensional case:

$$\vec{r}(t) = \langle f(t), g(t), h(t) \rangle = \begin{bmatrix} f(t) \\ g(t) \\ h(t) \end{bmatrix}$$

We've already seen these, for lines. Recall that if $\vec{v}$ is any nonzero vector, and $\vec{p} = (x_0, y_0, z_0)$ is a point, then

$$\vec{r}(t) = \vec{v}t + \vec{p}$$

describes the motion of a particle along a line (in the direction of $\vec{v}$, through the point $\vec{p}$). For example

$$\vec{r}(t) = \langle 1, -5, 4 \rangle t + \langle 2, 4, -3 \rangle \tag{22}$$

We define the *limit* of such a function in the following way:

$$\lim_{t \to a} \vec{r}(t) = \begin{bmatrix} \lim_{t \to a} f(t) \\ \lim_{t \to a} g(t) \\ \lim_{t \to a} h(t) \end{bmatrix}$$

For example if

$$\vec{r}(t) = \begin{bmatrix} 2 + t^3 \\ \frac{\sin(t)}{t} \\ te^{-t} \end{bmatrix}$$

then

$$\lim_{t \to 0} \vec{r}(t) = \begin{bmatrix} 2 \\ 1 \\ 0 \end{bmatrix}$$

Let a be a real number. A vector-valued function $\vec{r}$ is said to be *continuous at a* if

$$\lim_{t \to a} \vec{r}(t) = \vec{r}(a)$$

In other words, when the component function f, g, and h are all continuous.

Vector valued functions define curves through space. The curve is the set of points satisfying the parametric equation

$$\begin{cases} x = f(t) \\ y = g(t) \\ z = h(t) \end{cases}$$

for all t in the domain of $\vec{r}$.

The *derivative* of a vector-valued function $\vec{r}(t)$ is defined as

$$\vec{r}'(t) = \langle f'(t), g'(t), h'(t) \rangle$$

provided that this limit exists.

Important: The derivative of $\vec{r}(t)$ in Eq. (22) is $\vec{v}$. The particle moves at a constant velocity $\vec{v}$.

Observations

- $\vec{r}'$ is vector-valued function
- for a particular particular value of t, the quantity $\vec{r}(t)$ is the **tangent vector** to the curve at the point $\vec{r}(t)$. We will frequently work with the *unit tangent vector*, which is

$$\vec{T}(t) = \frac{\vec{r}'(t)}{|\vec{r}'(t)|}$$

- The unit tangent vector is the derivative, normalized

The *definite integral* of a vector function $\vec{r}(t)$ is defined as

$$\int_a^b \vec{r}(t)dt = \left\langle \int_a^b f(t)dt, \int_a^b g(t)dt, \int_a^b h(t)dt \right\rangle$$

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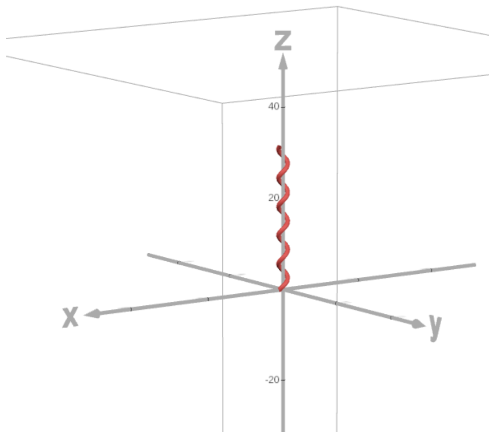
Please read chapter 13, sections 1-4.

Example 89 (Helix)**.**

$$\vec{r}(t) = \langle \cos(t), \sin(t), t\rangle$$

for $0 \le t < 10\pi$.



The derivative at time t is

$$\vec{r}'(t) = \langle -\sin(t), \cos(t), 1\rangle$$

The magnitude of the derivative is

$$|\vec{r}'(t)| = \sqrt{(-\sin t)^2 + (\cos t)^2 + 1} = \sqrt{2}.$$

The unit tangent vector is

$$\vec{T}(t) = \frac{\vec{r}'(t)}{|\vec{r}'(t)|} = \left\langle -\frac{\sin(t)}{\sqrt{2}}, \frac{\cos(t)}{\sqrt{2}}, \frac{1}{\sqrt{2}}\right\rangle$$

Suppose that the coordinate axes are measured in meters and time is measured in seconds. If $\vec{r}(t)$ describes the motion of a particle, then the particle starts at the origin, and travels upwards, rotating around the z-axis exactly 5 times before coming to a halt after $t = 10\pi$ seconds. The velocity of the particle at time t seconds is given by $\vec{r}'(t)$. Velocity has two components: a magnitude and a direction. The magnitude is known as *speed*. In this case, the speed is a constant $\sqrt{2}$ meters per second. The unit tangent vector tells us the direction that the particle is headed at a given instant, but tells us nothing about the speed.

Question: How far did the particle travel? Here, I am asking for arc length. The formula is

$$\text{Arc length} = \int_a^b \sqrt{\left(\frac{dx}{dt}\right)^2 + \left(\frac{dy}{dt}\right)^2 + \left(\frac{dz}{dt}\right)^2}\,dt$$

This can be written more compactly as

$$\boxed{\text{Arc length} = \int_a^b |\vec{r}\,'(t)|dt.}$$

In our case, $a = 0$, $b = 10\pi$, and $|r'(t)| = \sqrt{2}$. So

$$\text{Arc length} = \int_0^{10\pi} \sqrt{2}dt = 10\pi \cdot \sqrt{2} \approx 44 \text{ meters.}$$

The particle traveled $10\pi \cdot \sqrt{2}$ meters. This makes sense: after all, the particle traveled at a constant velocity of $\sqrt{2}$ meters per second for 10π seconds. So of course it traveled $10\pi\sqrt{2}$ meters.

End of Example 89. □

More generally the *arc length function* is defined as

$$s(t) = \int_a^t |r'(u)| du,$$

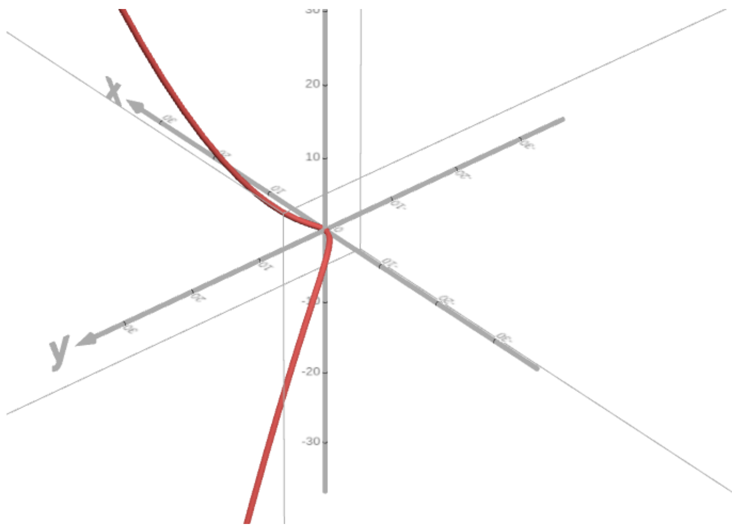
for $t \geq a$, and where a is the starting point of the particle.

In the previous example, we has $a = 0$ and $|r'(t)| = \sqrt{2}$ for all t. So

$$s(t) = \int_0^t \sqrt{2} du = \sqrt{2}t$$

This approach can also handle particles travelling at non-constant velocity, as the next example shows

Example 90 (Twisted Cubic)**.** Let $\vec{r}(t) = \left\langle 2t, t^2, \frac{1}{3}t^3 \right\rangle$



Then

$$r'(t) = \left\langle 2, 2t, t^2 \right\rangle.$$

The speed is

$$|r'(t)| = \sqrt{4 + 4t^2 + t^4} = \sqrt{(2 + t^2)^2} = 2 + t^2.$$

So the particle is accelerating as $t \to \infty$ or $t \to -\infty$.

The distance traveled by the particle from $t = 0$ to $t = 3$ is

$$\begin{aligned} \text{Arc length} &= \int_0^3 |r'(t)| dt \\ &= \int_0^3 (2 + t^2) dt \\ &= \left[2t + \frac{t^3}{3} \right]_0^3 \\ &= (6 + 9) - (0 + 0) \\ &= 15. \end{aligned}$$

The direction of the particle, without respect to the speed is given by the unit tangent vector

$$\vec{T}(t) = \frac{r'(t)}{|r'(t)|} = \left\langle \frac{2}{2 + t^2}, \frac{2t}{2 + t^2}, \frac{t^2}{2 + t^2} \right\rangle$$

End of Example 90. □

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Section 13.3 Recall the *unit tangent vector* of a parametrized curve $\vec{r}(t)$ is the vector

$$\vec{T}(t) = \frac{\vec{r}'(t)}{|\vec{r}\,'(t)|}$$

It's what you get when you normalize the velocity.

The **curvature** of a curve at a point is a nonnegative number which measures how quickly the the curve changes direction at that point. Essentially, it works like this

- straight line = zero curvature
- wide turns = low curvature
- sharp turns = high curvature

The direction is given by $\vec{T}(t)$. When a curve is fairly straight, $\vec{T}(t)$ shouldn't change much over time. But it should change direction quickly when the curve bends or twists sharply.

To be precise, we define *curvature* to be

$$\kappa = \left|\frac{d\vec{T}}{ds}\right|$$

where T is the unit tangent vector, and s is the arc length function.

By the chain rule,

$$\frac{d\vec{T}}{dt} = \frac{d\vec{T}}{ds}\frac{ds}{dt}$$

or on other words

$$T'(t) = \frac{d\vec{T}}{ds}s'(t).$$

Rearranging this, we get:

$$\frac{d\vec{T}}{ds} = \frac{1}{s'(t)}T'(t)$$

Next, recall that $\frac{ds}{dt} = s'(t) = |\vec{r}\,'(t)|$. This gives

$$\frac{d\vec{T}}{ds} = \frac{T'(t)}{|r'(t)|}$$

Taking absolute valus of both sides gives the following important **curvature formula:**

$$\boxed{\kappa(t) = \frac{|T'(t)|}{|r'(t)|}} \tag{23}$$

Example 91 (Curvature of a line)**.** Consider a line in 3d space. This takes the form

$$r(t) = vt + p$$

We can write out the above equation as:

$$\begin{bmatrix} x \\ y \\ z \end{bmatrix} = \begin{bmatrix} v_1 \\ v_2 \\ v_3 \end{bmatrix} t + \begin{bmatrix} x_0 \\ y_0 \\ z_0 \end{bmatrix}$$

where $v = \langle v_1, v_2, v_3\rangle$ is a nonzero vector giving the direction of the line, and $p = (x_0, y_0, z_0)$ is a point the line passes through.

Question: What is the curvature of the line?

We want to apply Eq. (23). First observe that

$$r'(t) = v$$

Therefore

$$T(t) = \frac{v}{|v|} = \left\langle \frac{v_1}{|v|}, \frac{v_2}{|v|}, \frac{v_3}{|v|} \right\rangle.$$

This does not depend on t. Therefore

$$T'(t) = \langle 0, 0, 0 \rangle$$

Therefore by Eq. (23),

$$\kappa(t) = 0$$

for all t.

This makes sense: a straight line never changes direction. So of course under any reasonable definition of "curvature", the curvature of a straight line should be zero.

End of Example 91. □

Example 92 (Curvature of a circle)**.** Consider a circle of radius $a > 0$. We can parametrized this with

$$\vec{r}(t) = (a\cos(t), a\sin(t))$$

What "should" the curvature be? It should be

- constant
- inversely related to a

Indeed, we can compute this and we get $\kappa(t) = \frac{1}{a}$ for all t. See example 3 on page 904 of the text.

End of Example 92. □

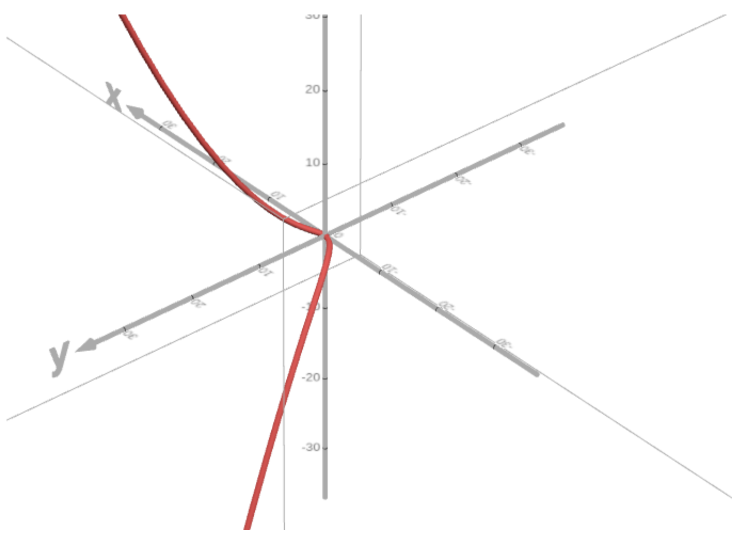
Proposition 93 (A very useful formula for curvature)**.**

$$\kappa(t) = \frac{|r'(t) \times r''(t)|}{|r'(t)|^3}$$

Example 94 (twisted cubic curvature)**.** Let

$$\vec{r}(t) = \left\langle 2t, t^2, \frac{1}{3}t^3 \right\rangle,$$

for $-\infty < t < \infty$. This is an example of a twisted cubic, which we saw in Example 90. Recall this has plot



From this picture, we might reasonably conclude that the curvature should be high near the origin (when t is near zero) and then it should be low far away from the origin (when t is large) because there the graph appears to straighten out. Let's try to compute the curvature:

We already showed that

$$r'(t) = \langle 2, 2t, t^2 \rangle$$

and so

$$|r'(t)| = \sqrt{4 + 4t^2 + t^4} = 2 + t^2$$

Therefore the unit tangent vector (i.e., the normalized velocity of the particle) is

$$\begin{aligned} T(t) &= \frac{r'(t)}{|r'(t)|} \\ &= \left\langle \frac{2}{2+t^2}, \frac{2t}{2+t^2}, \frac{t^2}{2+t^2} \right\rangle \end{aligned}$$

Using Eq. (23),

$$\kappa(t) = \frac{|T'(t)|}{|r'(t)|}$$

But this is going to be a lot of work to take all those derivatives. So instead, let's use Proposition 93, which says that

$$\kappa(t) = \frac{|r'(t) \times r''(t)|}{|r'(t)|^3}$$

- Numerator

$$\begin{aligned} r'(t) \times r''(t) &= \begin{vmatrix} \vec{\imath} & \vec{\jmath} & \vec{k} \\ 2 & 2t & t^2 \\ 0 & 2 & 2t \end{vmatrix} \\ &= \langle 2t^2, -4t, 4 \rangle \end{aligned}$$

  And this implies that

$$|r'(t) \times r''(t)| = \sqrt{4t^4 + 16t^2 + 16} = 2\sqrt{t^4 + 4t^2 + 4} = 2\sqrt{(t^2+2)^2} = 2(t^2 + 2)$$

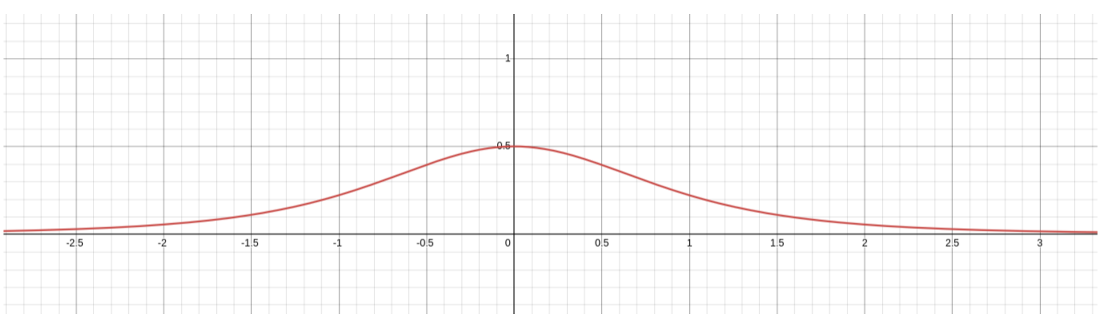
- Denominator. Since $r'(t) = \langle 2, 2t, t^2 \rangle$, we have

$$|r'(t)| = \sqrt{4 + 4t^2 + t^4} = \sqrt{(2+t^2)^2} = t^2 + 2$$

Therefore

$$\kappa(t) = \frac{2}{(2+t^2)^2}$$

This is pretty complicated-looking function, but when we graph $\kappa(t)$, we see that it's actually really simple:



As we expected, the curvature has a bump near zero, and then asymptotically approaches zero as $|t| \to \infty$.

End of Example 94. □

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The next proposition summarizes an important geometric fact

Proposition 95. *Suppose C is any curve on the surface of a sphere in 3d space. The sphere has some radius $c > 0$, which is a constant. Suppose C is parameterized by the vector function*

$$\vec{r}(t) = \begin{bmatrix} x(t) \\ y(t) \\ z(t) \end{bmatrix}$$

for $t \in (-\infty, \infty)$.

Claim: *$\vec{r}'(t)$ and $\vec{r}(t)$ are orthogonal.*

Why?

Observe that

$$\vec{r}(t) \cdot \vec{r}(t) = |\vec{r}(t)|^2 = c^2$$

Differentiate both sides and use the chain rule. You get that $\vec{r}\,'(t) \cdot \vec{r}(t) = 0$, which implies the two vectors are orthogonal

The direction that a curve is turning is given by the *principal unit normal vector $N(t)$*, defined as

$$\vec{N}(t) = \frac{\vec{T}'(t)}{|\vec{T}'(t)|}$$

This vector is normal to $T(t)$. This is because $T(t)$ is a vector on the unit sphere and therefore is perpendicular to its derivative, $\vec{T}'(t)$.

23.1 Motion in space: velocity and acceleration

Example 96. A stone is dropped form a 1600 foot high building.

(a) Find, as functions of time, its position and velocity.

(b) When does it hit the ground?

(c) How fast is it going when at that time?

You may assume that acceleration due to gravity is $32\text{ft}/\text{sec}^2$.
Part (a): Let

$$\begin{aligned} s(t) &= \text{position} \\ v(t) &= \text{velocity} \\ a(t) &= \text{acceleration} \end{aligned}$$

Then $a(t) = \frac{dv}{dt} = -32$ (negative because the direction is downward). So

$$v(t) = \int a(t)dt = \int -32dt = -32t + C_0$$

Since the stone was dropped, not thrown, its initial velocity was zero, i.e.,

$$v(0) = 0.$$

Therefore $C_0 = 0$, and it follows that

$$\boxed{v(t) = -32t}$$

Next we integrate velocity to find position:

$$s(t) = \int v(t)dt = \int -32tdt = -16t^2 + C_1$$

Since the stone started at height 1600, we have

$$s(0) = 1600.$$

Therefore

$$100 = -16(0)^2 + C_1$$

so $C_1 = 1600$. Thus,

$$\boxed{s(t) = -16t^2 + 1600.}$$

Part (b): Let t^* be the time that the stone hits the ground. It must be the case that

$$s(t^*) = 0.$$

By our formula for $s(t)$, we have

$$0 = -16(t^*)^2 + 1600$$

and solving for t^*, we conclude that $\boxed{\text{the stone hits the ground at time } t^* = 10}$.

Part (c): This part is asking what the speed is at time $t^* = 10$. At that time, the velocity is $v(10) = -32(10) = -320$ feet per second.

Therefore $\boxed{\text{the stone is traveling at 320 feet per second when it hits the ground.}}$

End of Example 96. ☐

We want to do the same thing in 3 dimensions. We will use the vector equations

$$\vec{r}(t) = \vec{r}(t_0) + \int_{t_0}^{t} \vec{v}(u)du \quad \text{and} \quad \vec{v}(t) = \vec{v}(t_0) + \int_{t_0}^{t} a(u)du$$

or equivalently,

$$\vec{r}\,'(t) = \vec{v}(t) \quad \text{and} \quad \vec{v}\,'(t) = \vec{a}(t)$$

Example 97. Find the velocity, acceleration, and speed of a particle with position vector $r(t) = \left\langle t^2, e^t, te^t \right\rangle$

End of Example 97. ☐

Example 98. A moving particle starts at initial position $r(0) = \langle 1, 0, 0 \rangle$. It has initial velocity $v(0) = \langle 1, -1, 1 \rangle$. Its acceleration is $a(t) = \langle 4t, 6t, 1 \rangle$. Find its velocity and position at time t.

End of Example 98. ☐

Next example will require the *Netwon's Second Law:* If a force $F(t)$ acts on an object of mass m producing an acceleration $a(t)$, then

$$\vec{F}(t) = m\vec{a}(t)$$

Example 99. A cannon is fired with initial velocity $\vec{v}_0 = \left\langle 100\sqrt{3}, 100 \right\rangle$. This corresponds to being fired at an intial angle of elevation $\theta = \frac{\pi}{6}$ with the cannonball traveling at an initial speed of 200 meters per second. Assume the canonball is fired from an elevation of 3 meters.

Some questions

- How long is the cannonball airborne?
- How far does the canonball travel?
- How fast is it going when it hits the ground?
- Can we characterize the trajectory of the canonball?

Solution: We have downard force of gravity

$$F = ma = \langle 0, -mg \rangle$$

where $g = |a| \approx 9.8 m/s^2$ So

$$a(t) = \langle 0, -g \rangle$$

Therefore

$$\begin{aligned}
v(t) &= \vec{v}(0) + \int_0^t a(u)du \\
&= \left\langle 100\sqrt{3}, 100 \right\rangle + \langle 0, -gt \rangle \\
&= \left\langle 100\sqrt{3}, 100 - gt \right\rangle
\end{aligned}$$

Integrating again to get position,

$$\begin{aligned}
r(t) &= r(0) + \int_0^t v(u)du \\
&= \langle 0, 3 \rangle + \left\langle \int_0^t 100\sqrt{3}dt, \int_0^t (100 - gt)dt \right\rangle \\
&= \langle 0, 3 \rangle + \left\langle 100\sqrt{3}t, 100t - \frac{1}{2}gt^2 \right\rangle \\
&= \left\langle 100\sqrt{3}t, 3 + 100t - \frac{1}{2}gt^2 \right\rangle
\end{aligned}$$

Or in other words, taking $g \approx 9.8 m/s^2$, the xy-coordinates of the cannonball (measured in meters) at time t seconds is given by the parametric equations

$$\begin{cases} x(t) = 100\sqrt{3}t \\ y(t) = 3 + 100t - 4.9t^2 \end{cases} \tag{24}$$

The cannonball is airborne as long as $y(t) > 0$. To find the time when $y(t) = 0$, we need to solve the quadratic equation

$$-4.9t^2 + 100t + 3 = 0$$

This has two solutions: $t \approx -0.4$ and $t \approx 20.8$. Since $t = -0.4$ doesn't make sense for this problem, we conclude the canonball remains airborne for about 20.8 seconds. At $t = 20.8$, the cannonball hits the ground; when that happens, its x-value is

$$x(20.8) \approx 3607 \text{ meters}$$

At that moment, its velocity is

$$v(20.8) = \begin{bmatrix} 100\sqrt{3} \\ 100 - 9.8(20.8) \end{bmatrix} = \begin{bmatrix} 100\sqrt{3} \\ -103.84 \end{bmatrix}$$

so its speed is

$$|v(20.8)| = \sqrt{(100\sqrt{3})^2 + (-103.84)^2} \approx 202 m/s$$

So it's slowed down a lot.

We can eliminate t in Eq. (24) to obtain the equation

$$y = -\frac{4.9}{30000}x^2 + \left(\frac{1}{100\sqrt{3}}\right)x + 3$$

which is a parabola.

End of Example 99. □

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24.1 Section 14.1: Functions of Several Variables

Please read section 14.1. I will assume that you are familiar with the material therein.

A function of one variable takes in one real number and spits out a real number. A function of two variables takes in two real numbers and spits out a real number.

Given a function of two variables $f(x, y)$, we say that x, y are the *independent variables*, and $z = f(x, y)$ is the dependent variable. The domain of f is some subset of the xy-plane. We can denote this by D_f.

For example, the function

$$f(x, y) = \frac{\sqrt{x + y + 1}}{x - 1}$$

has domain

$$D_f = \left\{(x, y) \in \mathbb{R}^2 : x \neq 1\right\}$$

What about the domain of the following function?

$$f(x, y) = x \log\left(x^2 - y\right)$$

We can find the domain D_f of f by recalling that $\log(u)$ is defined only for $u > 0$. Thus, we require that $x^2 - y > 0$, or $y < x^2$. [Plot this].

The *range* of f is the set of values that $f(x, y)$ can take. That is, it is the set

$$\{f(x, y) : (x, y) \in D_f\}$$

Let f be a function of two variables. For any constant k in the range of f, the *k-level curve* of a function of two variables $f = f(x, y)$ is the set

$$\{(x, y) \in D_f : f(x, y) = k\}.$$

In words, this is the set of points (x, y) in the domain of f satisfying

$$f(x, y) = k$$

for fixed value of k. Often, this traces out a curve in the xy-plane.

24.2 Section 14.2: Limits and Continuity

How should we define limits and continuity for functions of several variables?

If $P = (p_1, p_2), Q = (q_1, q_2)$ are two points, let's denote

$$\operatorname{dist}(P, Q) = \sqrt{(p_1 - q_1)^2 + (p_2 - q_2)^2}$$

as the **distance** between P and Q.

Intuitively, for a function of two variables, we want to define the limit $L = \lim_{(x,y)\to(a,b)} f(x, y)$ as the number which the function value $f(x, y)$ approaches as the point (x, y) gets closer and closer to the point (a, b).

For a function g of a single variable, we recall a condition involving the left and right-hand limits:

$$\lim_{x\to a} g(x) \text{ exists}$$

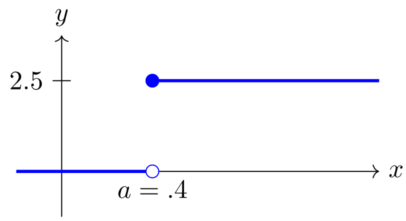
if and only if

$$\lim_{x\to a^+} g(x) \text{ and } \lim_{x\to a^-} g(x) \text{ exist and are equal}$$

The issue was that sometimes a function $g(x)$ will approach different values depending at a point a on whether $x \to a$ from the left or from the right. For example, consider the function

$$g(x) = \begin{cases} 0 & : \quad x < .4 \\ 2.5 & : \quad x \geq .4 \end{cases}$$

which has graph



in this case, in this case, we have

$$\lim_{x \to 0.4^-} g(x) = 0$$

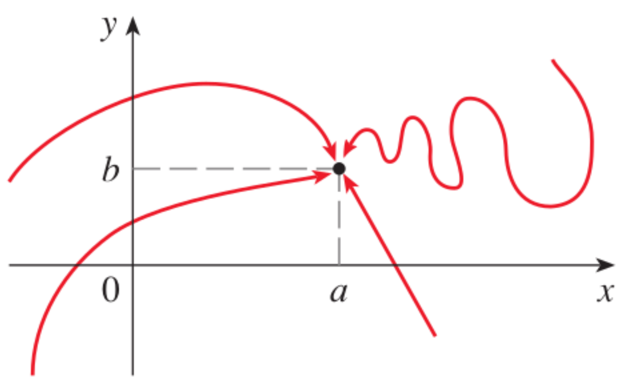
and

$$\lim_{x \to 0.4^+} g(x) = 2.5$$

but these are not equal, so

$$\lim_{x \to 0.4} g(x) \text{ does not exist.}$$

Something similar can happen with functions of two variables, but now, instead of having only two paths of approach (left and right) we now have infinitely many paths by which (x, y) can approach (a, b):



Definition 100 (limit)**.** Let $f = f(x, y)$ be a function of two variables with domain D_f, and we consider some point (a, b) in the plane. Consider a sequence of points

$$(x_1, y_1), (x_2, y_2), (x_3, y_3), \ldots \tag{25}$$

such that each point (x_n, y_n) is in D_f, and such that the points are getting closer to (a, b) in the sense that

$$\text{dist}((x_n, y_n), (a, b)) \to 0 \text{ as } n \to \infty.$$

If

$$\lim_{n\to\infty} f(x_n, y_n) = L.$$

for ANY SUCH SEQUENCE from Eq. (25), then we say that *L is the limit of $f(x, y)$ as $(x, y) \to (a, b)$,* and we write

$$\lim_{(x,y)\to(a,b)} f(x, y) = L$$

or equivalently,

$$f(x, y) \to L \text{ as } (x, y) \to (a, b).$$

On the other hand, if the above condition is not satisfied, then we say that the limit *does not exist (DNE).* In particular, if there are two different sequences

$$(x_1, y_1), (x_2, y_2), (x_3, y_3), \ldots$$

and

$$(\tilde{x}_1, \tilde{y}_1), (\tilde{x}_2, \tilde{y}_2), (\tilde{x}_3, \tilde{y}_3), \ldots$$

with

$$\lim_{n\to\infty} f(x_n, y_n) \neq \lim_{n\to\infty} f(\tilde{x}_n, \tilde{y}_n)$$

then we say that the limit $\lim_{(x,y)\to(a,b)} f(x, y)$ does not exist. [This is directly analogous to the limit of a single-variable function not existing if the left and right hand limits don't agree.]

Moreover, we say that $f(x, y)$ is *continuous at (a, b)* if

$$\lim_{(x,y)\to(a,b)} f(x, y) \text{ exists and equals } f(a, b)$$

Intuitively, this notion of continuity amounts to the idea that if we perturb the point (a, b) by an infinitesimal change in the x and y (call them dx and dy), then the value of the function changes by at most an infinitesmial amount:

$$f(a + dx, b + dy) \text{ is infinitesimally close to the original point } f(a, b).$$

It is hard to show that a function of two variables is continous at a point, because doing so requires considering all possible paths of approach. So there are lots of ways things can go wrong. A classic example:

Example 101. Let

$$f(x, y) = \frac{x^2 - y^2}{x^2 + y^2}.$$

Does the limit

$$\lim_{(x,y)\to(0,0)} f(x, y)$$

exist?

Answer: No, for the following reason:

- Approach along the x-axis (i.e., when $y = 0$) and you get

$$f(x, y) \to 1 \text{ as } (x, y) \to (0, 0) \text{ along the } x\text{-axis}$$

- Approach along the y-axis, and you get -1.

Since these are not equal, the limit does not exist.

End of Example 101. □

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Section 14.2

Example 102. Does the limit

$$\lim_{(x,y)\to(0,0)} \frac{xy}{x^2+y^2}$$

exist?

No. Consider the paths along the x-axis. And then consider along the line $y = x$.

End of Example 102. □

Example 103. What about the limit as $(x, y) \to (0, 0)$ of the function

$$f(x,y) = \frac{xy^2}{x^2+y^4}?$$

No. First, let's try approaching along any line $y = mx$. In that case we get

$$\begin{aligned}
\lim_{\substack{(x,y)\to(0,0)\\ y=mx}} f(x,y) &= \lim_{x\to 0} f(x, mx) \\
&= \lim_{x\to 0} \frac{m^2x^3}{x^2+m^4x^4} \\
&= \lim_{x\to 0} \frac{m^2x}{1+m^4x^2} \\
&= 0
\end{aligned}$$

Then, we try doing along the parabola $x = y^2$:

$$\begin{aligned}
\lim_{\substack{(x,y)\to(0,0)\\ x=y^2}} f(x,y) &= \lim_{y\to 0} f(y^2, y) \\
&= \lim_{y\to 0} \frac{y^4}{y^4+y^4} \\
&= \frac{1}{2}.
\end{aligned}$$

The limits along the different paths are not equal. Therefore the limit $\lim_{(x,y)\to(0,0)} f(x, y)$ does not exist.

End of Example 103. □

Example 104. What about the limit as $(x, y) \to (0, 0)$ of

$$f(x,y) = \frac{3x^2y}{x^2+y^2}$$

Yes. The limit is zero. We will use the squeeze theorem. First observe that

$$\begin{aligned}
|f(x,y)| &= \left|\frac{3x^2y}{x^2+y^2}\right| \\
&\le 3|y|\left(\frac{x^2}{x^2+y^2}\right) \\
&\le 3|y| \\
&\le 3\sqrt{x^2+y^2} \\
&= 3\mathrm{dist}((x,y),(0,0))
\end{aligned}$$

Therefore, as $(x, y) \to (0, 0)$, it follows that $|f(x, y)| \to 0$. This doesn't depend on the path by which (x, y) might approach $(0, 0)$. After all, recall what it means for $(x, y) \to (0, 0)$; it means that $\mathrm{dist}((x, y), (0, 0)) \to 0$.

Since $|f(x, y)| \to 0$ as $(x, y) \to 0$, it follows that $\lim_{(x,y)\to(0,0)} f(x, y) \to 0$ by the squeeze theorem, since

$$-|f(x, y)| \leq f(x, y) \leq |f(x, y)|.$$

End of Example 104. $\square$

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26.1 Continuity

Section 14.2

A function of two variables f is *continuous at (a, b)* if

$$\lim_{(x,y)\to(a,b)} f(x, y) \text{ exists and equals } f(a, b)$$

A function is *continuous* if it is continuous at every point in its domain.

Intuitively, this notion of continuity amounts to the idea that if we perturb the point (a, b) by an infinitesimal change in the x and y (call them dx and dy), then the value of the function changes by at most an infinitesmial amount:

$$f(a + dx, b + dy) \text{ is infinitesimally close to the original point } f(a, b).$$

As a practical matter, if we know that f is continuous at a point (a, b), then we can evaluate the limit

$$\lim_{(x,y)\to(a,b)} f(x, y)$$

by simply plugging in $x = a$ and $y = b$. For example,

Example 105 (Polynomial functions)**.** Let

$$f(x, y) = x^2y^3 - x^3y^2 + 3x + 2y$$

Suppose we want to know

$$\lim_{(x,y)\to(1,2)} f(x, y).$$

Fact: polynomials are always continuous.

Since f is a polynomial (of two variables, x and y), it is continuous. Therefore, we can just plug in $x = 1$ and $y = 2$ to find the limit:

$$\lim_{(x,y)\to(1,2)} f(x, y) = f(1, 2) = 8 - 4 + 3 + 4 = 11.$$

Polynomials of multiple variables are a deep an interesting field of study.

End of Example 105. □

Fact: We can combine continuous functions together (e.g., by adding/subtracting them, multipying/dividing them, and by function composition), and the resulting function will be continuous on its domain.

Example 106 (Rational functions)**.** Consider the function

$$f(x, y) = \frac{x^2 - y^2}{x^2 + y^2}.$$

This is an example of a *rational function*, which is a quotient of two polynomials. At what points in the xy-plane is the function f continuous?

- f is a rational function, and therefore it is continuous at all points on its domain, which is

$$D_f = \left\{(x, y) \in \mathbb{R}^2 : (x, y) \neq (0, 0)\right\}$$

- f is not defined at $(0, 0)$ and is therefore not continuous at that point.

What if we modifed f so that it was defined at the point $(0,0)$? That is, consider the function

$$g(x,y) = \begin{cases} \frac{x^2-y^2}{x^2+y^2} & : \quad (x,y) \neq (0,0) \\ 0 & : \quad (x,y) = (0,0) \end{cases}$$

In this case, g is *still* not continuous at $(0,0)$, because the limit doesn't exist (we showed this in Example 101).

End of Example 106. □

Example 107. Consider the function

$$f(x,y) = \begin{cases} \frac{3x^2y}{x^2+y^2} & : \quad (x,y) \neq (0,0) \\ 0 & : \quad (x,y) = (0,0) \end{cases}$$

Is this function continuous?

- It is continuous at all points $(x,y) \neq (0,0)$ because it is a rational function there
- At the point $(0,0)$ it is continuous, since we showed in an Example 104 that

$$\lim_{(x,y)\to(0,0)} f(x,y) = 0.$$

End of Example 107. □

Example 108. Does $\lim_{(x,y)\to(0,0)} \frac{xy}{\sin(xy)}$ exist?

End of Example 108. □

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We left off the last class with the following question posed

Example 109. Does $\lim_{(x,y)\to(0,0)} \frac{xy}{\sin(xy)}$ exist?

Recall that

$$\lim_{t\to 0} \frac{\sin(t)}{t} = 1$$

Solution: Use polar. Observe that for any path with $(x, y) \to (0, 0)$, we have

$$\lim_{(x,y)\to(0,0)} xy = 0.$$

This is because the function $h(x, y) = xy$ is is a polynomial, and is therefore continuous. Hence, we can do the change of variable $t = xy$:

$$\begin{aligned}\lim_{(x,y)\to(0,0)} \frac{xy}{\sin(xy)} &= \lim_{t\to 0} \frac{\sin(t)}{t} \\ &= 1.\end{aligned}$$

We are done. The limit exists.

End of Example 109. □

Example 110. Does

$$\lim_{(x,y)\to(0,0)} \frac{e^{-x^2-y^2} - 1}{x^2 + y^2}$$

exist?

Solution: Use polar. Recall that we can convert to polar by writing

$$x = r\cos(\theta) \quad \text{and} \quad y = r\sin(\theta)$$

so that

$$x^2 + y^2 = r^2.$$

This gives

$$\begin{aligned}\lim_{(x,y)\to(0,0)} \frac{e^{-x^2-y^2}}{x^2 + y^2} &= \lim_{r\to 0} \frac{e^{-r^2} - 1}{r^2} \\ &= \lim_{u\to 0} \frac{e^{-u} - 1}{u} \\ &= -1 && \text{by L'Hopital's rule.}\end{aligned}$$

End of Example 110. □

27.1 Partial derivatives

Section 14.3

Definition 111 (Partial derivatives)**.** Let $f = f(x, y)$ be a function of two variables, called x and y. The *partial derivative of f with respect to the variable x* is a function, denoted f_x, defined by

$$f_x(x, y) = \lim_{h\to 0} \frac{f(x + h, y) - f(x, y)}{h}$$

The *partial derivative of f with respect to y* is the function f_y defined by

$$f_y(x, y) = \lim_{h\to 0} \frac{f(x, y + h) - f(x, y)}{h}$$

Alternative notation:

$$\frac{\partial f}{\partial x} = f_x \quad \text{and} \quad \frac{\partial f}{\partial y} = f_y$$

An interpretation of what partial derivatives means will be given later; first let's get familiar with computing them.

Example 112. Let

$$f(x,y) = 4x^2 - 2y^2.$$

Find f_x and f_y. Also find $f_x(2,1)$ and $f_y(2,3)$ **Key idea:** When finding f_x, we treat y like it's a constant.

End of Example 112. □

Example 113. Let

$$f(x,y) = x\cos(y) + y\sin(x)$$

Find f_x and f_y, f_{xx}, f_{yy}, f_{xy}, f_{yx}

End of Example 113. □

Definition 114 (Hessian Matrix)**.** Given a function of two variables $f = f(x,y)$, the *Hessian matrix* is the matrix H_f defined by

$$H_f = \begin{bmatrix} f_{xx} & f_{xy} \\ f_{yx} & f_{yy} \end{bmatrix}$$

For a function of a single variable $g(x)$, recall that the concavity of the graph of g is determined by whether $g''(x)$ is positive or negative. The Hessian plays the same role for functions of 2 variables. This will be explained in detail in a later class.

Example 115. Let

$$f(x,y) = 3x^2 + 2xy^2.$$

Find f_x and f_y. We can also find f_{xx}, f_{yy}, f_{xy}, and f_{yx}.
Solution:

$$\begin{aligned} f_x(x,y) &= 6x + 2y^2 \\ f_{xx}(x,y) &= 6 \\ f_{xy}(x,y) &= 4y \\ f_{yx}(x,y) &= 4y \\ f_{yy}(x,y) &= 4x \end{aligned}$$

The Hessian matrix is

$$H_f = \begin{bmatrix} 6 & 4y \\ 4y & 4x \end{bmatrix}$$

End of Example 115. □

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28.1 How to interpret partial derivatives

Intersections of the surface with planes $x = k$ or $y = k$, where k is a constant. This gives a curve. The partial derivatives are the slopes of this curve.

28.2 A Trick: Implicit Differentiation

Assume that $z = f(x, y)$ is a function of two independent variables x and y. Suppose we don't know what f is, but we do know that x, y and z obey the relation

$$yz - \log(z) - x - y = 0.$$

Find $\frac{\partial z}{\partial x} = f_x(x, y)$.
Solution: Differentiate both sides with respect ot x, pretending that y is a constant and remembering that $z = f(x, y)$:

$$\frac{\partial}{\partial z}\left[yz - \log(z) - x - y\right] = 0.$$

or

$$\frac{\partial}{\partial x}[yz] - \frac{\partial}{\partial x}[\log(z)] - \frac{\partial}{\partial x}[x] - \frac{\partial}{\partial x}[y] = 0.$$

or

$$y\frac{\partial z}{\partial x} - \frac{1}{z}\frac{\partial z}{\partial x} - 1 = 0.$$

Solving for $\frac{\partial z}{\partial x}$ gives

$$\frac{\partial z}{\partial dx} = \frac{1}{y - \frac{1}{z}} = \frac{z}{yz - 1}.$$

28.3 Tangent Planes and Linear Approximations

Section 14.4 Tangent planes and linear approximations
A central idea in calculus is approximating curvy things with straight things. Here we will discuss approximations of curvy surfaces by their tangents planes.

Tangent lines: Let $f = f(x)$ be a function of a single variable. Let C be the curve (the graph) defined by f. And let (x_0, y_0) be a which the curve passes through. The *tangent line* at the point (x_0, y_0) is given by the formula

$$y - y_0 = f'(x_0)(x - x_0).$$

Tangent planes: Let $f = f(x, y)$ be a function of two variables, whose graph defines a smooth surface S. Let (x_0, y_0, z_0) be a point on S. The *tangent plane* at the point (x_0, y_0, z_0) is the plane defined by

$$z - z_0 = f_x(x_0, y_0)(x - x_0) + f_y(x_0, y_0)(y - y_0).$$

This is the plane which satisfies the following two properties

- it passes through the point (x_0, y_0, z_0)
- its tilt matches the surfaces as much as possible (i.e, the slope of the plane in the x-direction is $f_x(x_0, y_0)$, and the slope of the plane in the y-direction is $f_y(x_0, y_0)$).

Recall that planes have equations of the form

$$A(x - x_0) + B(y - y_0) + C(z - z_0) = D$$

so this is really a plane.

Example 116. Let $f(x, y) = 2x^2 + y^2$. Then the surface $z = f(x, y)$ is an elliptic paraboloid (like a bowl) which contains the point $(1, 1, 3)$. Find the tangent plane at the point $(1, 1, 3)$.

End of Example 116. □

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29.1 First Order Taylor Approximations

AKA "linear approximations", Section 14.4 in the textbook

Recall: If $f = f(x)$ is differentiable at the point $a \in \mathbb{R}$, then by Taylor's Theorem we have

$$f(x) \approx \underbrace{f(a) + f'(a)(x-a)}_{\text{1st degree Taylor polynomial}}$$

when $(x, y) \approx (a, b)$. But Taylor's theorem says that if f is *twice* differentiable at $a \in \mathbb{R}$, then an even better approximation holds:

$$f(x) \approx \underbrace{f(a) + f'(a)(x-a) + \frac{f''(a)}{2!}(x-a)^2}_{\text{2nd degree Taylor polynomial}}$$

when $(x, y) \approx (a, b)$. We can keep going if f is three or four or more times differentiable at a. But for most applications, the degree 2 approximation is sufficient. This approximation tends to be very good. For example:

Let $f(x) = e^x$. The second degree taylor approximation at $x = 0$ is:

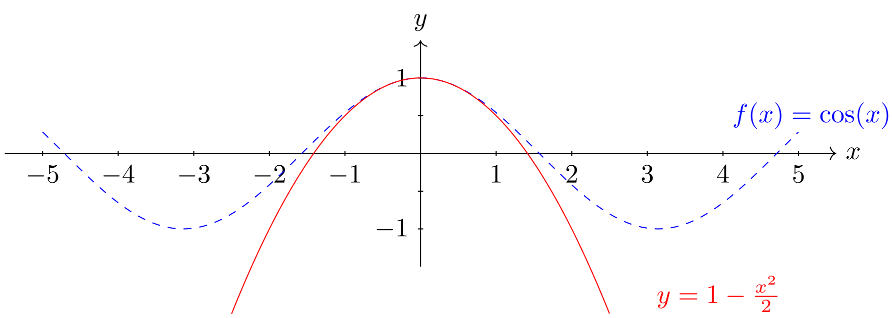
$$e^x \approx 1 + x + \frac{x^2}{2}.$$

If $x = 1/10$, then $e^x \approx 1.1051871$. The approximation on the right hand side is

$$1 + \frac{1}{10} + \frac{1}{200} = 1.105$$

These differ by only about $\frac{1}{100}$th of a percent of the true value. This approximation gets better as $x \to 0$.

Here's another striking example, of the 2nd order taylor approximation of the function $f(x) = \cos(x)$ at $x = 0$:



We can do the same things if $f = f(x, y)$ is a function of two variables.

We can use tangent planes to approximate smooth functions. Suppose that $f = f(x, y)$ is a "smooth" function of two variables. If (x, y) is close to (a, b), then

$$f(x, y) \approx \underbrace{f(a, b) + f_x(a, b)(x-a) + f_y(a, b)(y-b)}_{L(x,y)} \tag{26}$$

The right hand side of Eq. (26) is denoted $L(x, y)$ and goes by many names:

- the *linear approximation* of f at (a, b)
- the *tangent plane approximation* of the f at (a, b)
- the *linearization* of f at (a, b)

- the *first order Taylor approximation* for f near (a, b)

Example 117. Find the linearization of

$$f(x, y) = x^2 - xy + \frac{1}{2}y^2 + 3$$

at the point $(3, 2)$.

End of Example 117. □

Example 118. Find the first order Taylor polynomial for $f(x, y) = 2x^2 + y^2$ at $(1, 1)$ and use it to approximate $f(1.1, 0.95)$.

$$\begin{aligned} L(x, y) = f(1, 1) + f_x(1, 1)(x - 1) + f_y(1, 1)(y - 1) \\ 3 + 4(x - 1) + 2(y - 1) \end{aligned}$$

And

$$\begin{aligned} f(1.1, 0.95) &\approx L(1.1, 0.95) \\ &= 3 + 4(.1) + 2(-.05) \\ &= 3 + .4 - .1 \\ &= 3.3 \end{aligned}$$

(And indeed, we have $f(1.1, 0.95) = 3.3224$ so the approximation is pretty good.)

End of Example 118. □

What does it mean for a function f to be differentiable/smooth?

The quality of this approximation depends on the function f being "smooth", which I haven't defined. Informally, we say that a function $f = f(x, y)$ of two variables is *differentiable* at a point (a, b) if, when we zoom in on the graph of f near (a, b), the graph of f looks like a plane. If that's the case, then we say that f is *differentiable* at (a, b).

More precisely, we say that f is *differentiable* at (a, b) if

$$\lim_{(x,y)\to(a,b)} \frac{|f(x, y) - L(x, y)|}{\text{dist}((x, y), (a, b))} = 0$$

where $\text{dist}((x, y), (a, b)) = \sqrt{(x - a)^2 + (y - b)^2}$.

Intuitively, differentiable functions are functions for which the linearization $L(x, y)$ is a good approximation.

Differentiability can be hard to show.

A useful criterion: if f_x and f_y exist and are continuous at (a, b) then f is differentiable at (a, b).

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Recall the first order Taylor approximation (AKA the linearization) of $f = f(x, y)$ at (a, b) is

$$f(x, y) \approx f(a, b) + f_x(a, b)(x - a) + f_y(a, b)(y - b)$$

Example 119. Show that $f(x, y) = xe^{xy}$ is differentiable at $(1, 0)$, and find its first order Taylor approximation at $(1, 0)$. And then use it to approximate $f(1.1, -0.1)$.
Solution: this is example 2 on page 971.
Note: the true value is $f(1.1, -0.1) = 0.98542$.

End of Example 119. $\square$

Example 120. Verify the linear approximation

$$e^x \cos(xy) \approx 1 + x$$

when (x, y) is near $(0, 0)$.

End of Example 120. $\square$

30.1 Second order Taylor approximation

This part is based on "Discovery Project" on p 1011 of the textbook.
The second degree Taylor approximation of f at (a, b) is

$$f(x,y) \approx f(a,b)+f_x(a,b)(x-a)+f_y(a,b)(y-a)+\frac{1}{2}f_{xx}(a,b)(x-a)^2+f_{xy}(a,b)(x-a)(x-b)+\frac{1}{2}f_{yy}(a,b)(y-b)^2$$

The right hand side, denoted $Q(x, y)$, is sometimes the *quadratic approximation* of f at (a, b).

Example 121. Let $f(x, y) = e^{-x^2-y^2}$. Show that the quadratic approximation at $(0, 0)$ is

$$Q(x, y) = 1 - x^2 - y^2$$

This is a downward opening paraboloid.
Hazard a guess: the function f concave up or concave down at $(0, 0)$?

End of Example 121. $\square$

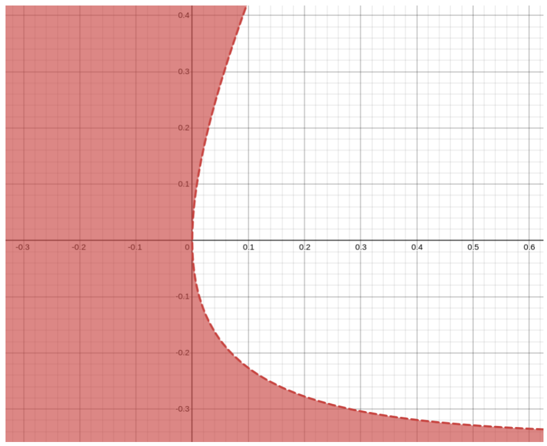
31 2025-04-14 | Week 13 | Lecture 31

31.1 Another example of 2nd order Taylor expansion

This weekend, a collaborator and I were trying to understand the following inequality

$$e^{-4x} - 2e^{-2y} + e^{-4y} \geq 0$$

which has meaning in a statistics problem.



So we decided to look at cutoff curve

$$e^{-4x} - 2e^{-2y} + e^{-4y} = 0$$

We want to know what does this curve look like near the origin?

Let's take the 2nd degree Taylor expansion near $(a, b) = (0, 0)$:

First find the partial derivatives:

$$\begin{aligned} f_x &= -4e^{-4x} & f_y &= 4e^{-2y} - 4e^{-4y} \\ f_{xx} &= 16e^{-4x} & f_{yy} &= -8e^{-2y} + 16e^{-4y} \\ f_{xy} &= 0 & f_{yx} &= 0. \end{aligned}$$

Evaluate them at $(0, 0)$:

$$\begin{aligned} f_x(0,0) &= -4 & f_y &= 0 \\ f_{xx}(0,0) &= 16 & f_{yy}(0,0) &= 8 \\ f_{xy}(0,0) &= 0 & f_{yx}(0,0) &= 0 \\ f(0,0) &= 0. \end{aligned}$$

Next, we plug into the formula for 2nd order taylor expansion. When (x, y) is near $(0, 0)$, we have the following approximation:

$$f(x, y) \approx -4x + 8x^2 + 4y^2$$

So the curve we are interested in is approximately

$$-4x + 8x^2 + 4y^2 = 0$$

What does this look like? Let's complete the square:

$$8(x^2 - \frac{1}{2}x) + 4y^2 = 0$$

$$8(x^2 - \frac{1}{2}x + \frac{1}{16} - \frac{1}{16}) + 4y^2 = 0$$

or

$$8(x^2 - \frac{1}{2}x + \frac{1}{16}) - \frac{1}{2} + 4y^2 = 0$$

$$16\left(x - \frac{1}{4}\right)^2 + 8y^2 = 1$$

This is the equation of an ellipse, centered at $(x, y) = (\frac{1}{4}, 0)$, with horizontal radius $1/4$ and vertical radius $1/\sqrt{8}$.

The above problem would be a good final exam problem.

31.2 Chain Rule

Section 14.5

Theorem 122 (The Chain Rule)**.** *Suppose that $z = f(x, y)$ is a differentiable function of x and y, which are themselves functions of a third variablle t:*

$$\begin{cases} x = x(t) \\ y = y(t) \end{cases}$$

Assume that $x(t), y(t)$ are both differentiable. Then z is a differentiable function of t, and

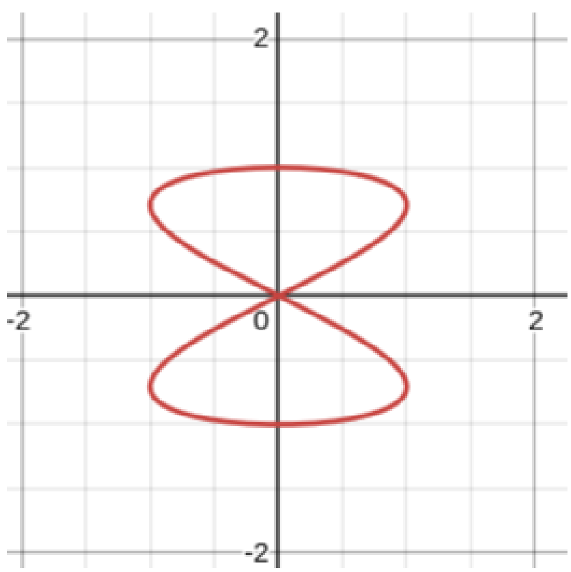
$$\frac{dz}{dt} = \frac{\partial f}{\partial x}\frac{dx}{dt} + \frac{\partial f}{\partial y}\frac{dy}{dt}$$

Let's consider a concrete example:

Example 123. Consider the parameterized curve

$$\begin{cases} x = \sin(2t) \\ y = \cos(t) \end{cases}$$

for $0 \le \tau \le 2\pi$. This is the following figure-8 shaped curve:



In addition, suppose

$$z = x^2 y + 3xy^4.$$

Find $\frac{dz}{dt}$ when $t = 0$.

End of Example 123. $\square$

Example 124. Let $z = e^x \sin(y)$, where $x = st^2$ and $y = s^2 t$. To be clear, we are regarding s, t as independent variables. The variables x, y are functions of both s and t. Find $\frac{\partial z}{\partial s}$ and $\frac{\partial z}{\partial t}$.

Then

$$\begin{aligned}
\frac{\partial z}{\partial s} &= \frac{\partial z}{\partial x}\frac{\partial x}{\partial s} + \frac{\partial z}{\partial y}\frac{\partial y}{\partial s} \\
&= \left(e^x \sin(y)\right)\left(t^2\right) + \left(e^x \cos(y)\right)(2st) \\
&= t^2 e^x \sin(y) + 2ste^x \cos(y) \\
&= t^2 e^{st^2} \sin(s^2 t) + 2ste^{st^2} \cos(s^2 t)
\end{aligned}$$

End of Example 124. $\square$

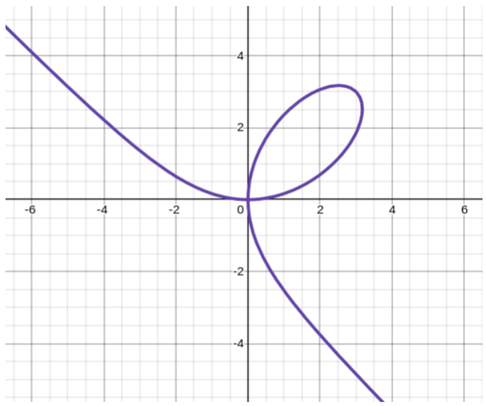
32 2025-04-16 | Week 13 | Lecture 32

32.1 A new perspective (and a new formula for) on implicit differentiation

Suppose we have the equation

$$x^3 + y^3 = 6xy$$

This equation defines the following curve on the plane:



We assume that y is a function of x and we want to compute the slope. In implicit differentiation, we apply the differentiation operator $\frac{d}{dx}$ to both sides, which gives

$$3x^2 + 3y^2\frac{dy}{dx} = 6y + 6x\frac{dy}{dx}$$

Then we solve for $\frac{dy}{dx}$:

$$\begin{aligned} \frac{dy}{dx} &= \frac{3x^2 - 6y}{3y^2 + 6x} \\ &= \frac{x^2 - 2y}{y^2 + 2x} \end{aligned}$$

This is all very nice. Here's another way to do this same problem, using the **Chain Rule:**

First, define a new function F of two variables:

$$F(x, y) = x^3 + y^3 - 6xy$$

So our curve is

$$F(x, y) = 0.$$

Now, assuming that x is an independent variable and that y is a function of x (so that $\frac{d}{dx}[y] = \frac{dy}{dx}$, but $\frac{d}{dx}[x] = 1$), we have

$$\frac{d}{dx}\left[F(x, y)\right] = 0$$

and this gives us

$$\frac{\partial F}{\partial x}\frac{dx}{dx} + \frac{\partial F}{\partial x}\frac{dy}{dx} = 0$$

which simplifies to

$$F_x(x, y) + F_y(x, y)\frac{dy}{dx} = 0$$

and gives us the nice formula

$$\boxed{\frac{dy}{dx} = -\frac{F_x(x,y)}{F_y(x,y)}}$$

Does it really work? Here, we have $F_x(x,y) = 3x^2 - 6xy$ and $F_y(x,y) = 3y^2 - 6x$; plugging these into the above formula gives us the same answer as earlier. Cool. The boxed formula is a little strange at first, because of the negative sign and the way the partial derivatives feel like they are in the wrong place.

32.2 Directional Derivatives

Given a function $z = f(x,y)$, we now know how to find the rate of change of z when moving in the x or y directions (i.e., East or North)—namely, we find f_x and f_y. But what if we wanna know the rate of change some other direction???

For example, suppose that the (x,y)-plane is the ocean, so that each point (x,y) is the coordinates of some location. And suppose that z is the surface temperature of the water at that location. So $z = f(x,y)$ is continuous (no jumps!). But say I want to know the rate of change of the surface temperature if I pilot my boat northwest? Or any other random direction?

Definition 125 (Directional derivative)**.** The *directional derivative* of f at (x,y) in the direction of a vector $\vec{u} = \langle a,b\rangle$ is

$$D_{\vec{u}}f(x,y) = \lim_{h\to 0}\frac{f(x+ha, y+hb) - f(x,y)}{h}$$

provided that this limit exists.

Proposition 126. *If $\bar{u} = \langle a,b\rangle$ is a* ***unit vector****, then*

$$D_{\vec{u}}f(x,y) = af_x(x,y) + bf_y(x,y)$$

(provided that f is differentiable).

In the nicest case, we represent the direction with a vector

$$\vec{u} = \langle\cos(\alpha), \sin(\theta)\rangle$$

in which case

$$D_u f(x,y) = f_x(x,y)\cos(\theta) + f_y(x,y)\sin(\theta)$$

Example 127. Find the directional derivative of

$$f(x,y) = x^3 - 3xy + 4y^2$$

in the direction of $\theta = \frac{\pi}{6}$, i.e., $\vec{u} = \left\langle\cos(\frac{\pi}{6}), \sin(\frac{\pi}{6})\right\rangle = \left\langle\frac{\sqrt{3}}{2}, \frac{1}{2}\right\rangle$

End of Example 127. □

32.3 Gradient Notation

Definition 128. Let $f = f(x,y)$. The *gradient* of f, denote ∇f, is the vector of partial derivatives:

$$\nabla f = \begin{bmatrix} f_x \\ f_y \end{bmatrix}$$

To be clear, ∇f is itself a function x and y:

$$\nabla f(x,y) = \begin{bmatrix} f_x(x,y) \\ f_y(x,y) \end{bmatrix}$$

Using this notation, we have

$$f(x,y) \approx f(a,b) + \nabla f(a,b) \cdot \begin{bmatrix} x-a \\ y-b \end{bmatrix}$$

Example 129. Using gradient notation, find the directional derivative of the function

$$f(x,y) = x^2y^3 - 4y$$

at the point $(2,-1)$ in the direction of hte vector $\vec{v} = 2\vec{\imath} + 5\vec{\jmath}$

End of Example 129. □

33 2025-04-21 | Week 14 | Lecture 33

33.1 Directional Derivatives

Section 14.6

Recall that $\nabla f = \left\langle \frac{\partial f}{\partial x}, \frac{\partial f}{\partial y}, \frac{\partial f}{\partial z} \right\rangle$.

Recall that if f is differentiable, then

$$D_u f(x,y,z) = \nabla f(x,y,z) \cdot u$$

is the directional derivative of f in the direction of the unit vector u.

Example 130. Let $f(x,y,z) = x\sin(yz)$. Find the gradient of f and the directional derivative of f at $v = \langle 1, 2, -1 \rangle$.

End of Example 130. □

33.2 The gradient vector is *the direction of fastest increase*

Suppose an ant is walking along a smooth surface $z = f(x,y)$. Then it starts raining. The ant wants to escape the flood by getting to higher ground as fast as possible. Wherever the ant is on the surface, it needs to walk in the direction which gives the fastest increase in eleveation (i.e., the steepest path).

Recall that for vectors A, B, we have $A \cdot B = |A||B|\cos(\theta)$. We want to find the unit vector u which maximizes $D_u f$. Suppose u and ∇f form an angle of θ. Then

$$\begin{aligned} D_u f &= \nabla f \cdot u \\ &= |\nabla f||u|\cos(\theta) \\ &= |\nabla f|\cos(\theta) && \text{since } |u| = 1. \end{aligned}$$

The right hand side is maximized when $\cos(\theta) = 1$, which occurs when $\theta = 0$. In other words, when u and ∇f point in the same direction. But this implies that u must be the vector

$$u = \frac{\nabla f(x,y)}{|\nabla f(x,y)|}.$$

Hence, if the ant is standing at point (x,y,z), then its next step should be in the direction

Example 131. Suppose $f(x,y) = xe^y$. Find the rate of change of of f at the point $P(2,0)$ in the direction from P to $Q(\frac{1}{2}, 2)$. In what direction does f have maximum rate of change?

End of Example 131. □

Example 132. Suppose $f(x,y) = x\log(yz)$. Find the maximum rate of change of f at the point $(1, 2, \frac{1}{2})$ and the direction it occurs in

End of Example 132. □

34 2025-04-23 | Week 14 | Lecture 34

Review from previous day

Example 133. Temperature at a point (x, y, z) is given by

$$T(x, y, z) = 200e^{-x^2-3y^2-9z^2}$$

where T is measured in Celsius and x, y, z in meters.

(a) Find the rate of change of temperature at the point $P = (2, -1, 2)$ in the direction toward the point $(3, -3, 3)$.

(b) In what direction does V change most rapidly at P?

(c) Whats is the maximum rate of change at P?

End of Example 133. □

34.1 The gradient vector is *orthogonal to level curves*

Recall that the chain rule statese that if $z = f(x(t), y(t))$, then

$$\frac{dz}{dt} = \frac{\partial f}{\partial x}\frac{dx}{dt} + \frac{\partial f}{\partial y}\frac{dy}{dt}$$

Also recall that two vectors are orthogonal if their dot product is zero.

Theorem 134. *Fix $k \in \mathbb{R}$ and consider the level curve C given by the formula*

$$f(x, y) = k.$$

To be precise, C is a curve in the (x, y)-plane consisting of all points (x, y) which satisfy $f(x, y) = k$.

If (a, b) is any point on C, the vector $\nabla f(a, b)$ is orthogonal to the level curve at that point.

Proof. Let $\gamma(t) = \langle x(t), y(t)\rangle$ be a parameterization of C (with t in some interval I). Recall that $\gamma'(t) = \langle x'(t), y'(t)\rangle$ is the tangent vector to the curve. Also, since $\gamma(t)$ is a k-level curve, we have

$$f(\gamma(t)) = f(x(t), y(t)) = k$$

for all t. Therfore $f(\gamma(t))$ is a *constant* function of t. So it's derivative is zero:

$$\frac{d}{dt}\left[f(\gamma(t))\right] = 0 \tag{27}$$

By the chain rule,

$$\begin{aligned}\frac{d}{dt}\left[f(\gamma(t))\right] &= \frac{d}{dt}\left[f(x(t), y(t))\right]\\ &= f_x(x(t), y(t))x'(t) + f_y(x(t), y(t))y'(t)\\ &= \nabla f(\gamma(t)) \cdot \gamma'(t)\end{aligned}$$

Combining this with Eq. (27), it follows that for all t,

$$\nabla f(\gamma(t)) \cdot \gamma'(t) = 0.$$

In other words, at every point of the level curve, the tangent vector and the gradient vector are orthogonal. □

35 2025-04-25 | Week 14 | Lecture 35

This lecture is based on the material of *Section 14.7*.

35.1 “Optimizing” a function of a single variable

Suppose we wish to maximize the function

$$f(x) = \frac{2}{3}x^3 + \frac{5}{2}x^2 - 3x + 1$$

on the interval $-10 \leq x \leq 2$.

Find critical points:

$$\begin{aligned} f'(x) &= 2x^2 + 5x - 3 \\ &= (x+3)(2x-1) \end{aligned}$$

This is equal to zero when $x = -3$ or $x = \frac{1}{2}$. These are the critical points.

Classify critical points:

$$f''(x) = 4x + 5$$

Therefore $f''(-3) < 0$ and $f''(\frac{1}{2}) > 7$. Therefore the function is concave down at $x = -3$ and concave up at $x = \frac{1}{2}$. This allows us to conclude that $x = -3$ is a maximuma nd $x = \frac{1}{2}$ is a minimum. This is the second derivative test.

(If the second derivative were zero, we'd have an inflection point rather than a maximum or minimum.)

Compare with boundardy: We know that $x = -3$ is a local maximum, and $f(-3) = 19$. But is it the global maximum? We need to compare with the endpoint of the domain

- $f(-10) = -335.667$
- $f(2) = 12.333$

We can conlude that $x = -3$ is indeed the global maximum.

We now adapt these ideas to functions of multiple variables.

35.2 Optimizing functions of multiple variables

A function of two variables has a *local maximum* at (a, b) if

$$f(x, y) \leq f(a, b)$$

for all (x, y) within some radius of the point (a, b). The number $f(a, b)$ is the *local maximum value.* The notion of *local minimum* is defined in a similar way.

Fact: Suppose f is a function whose domain is an open set. If f has a local maximum or minimum at (a, b) and the first-order parital derivatives of f exist there, then $f_x(a, b) = 0$ and $f_y(a, b) = 0$. In other words, the tangent plane at the local optimum is horizontal.

A point (a, b) is a *critical point* (or: *stational point*) of f if f_x and f_y are both zero, or if one of them doesn't exist.

Example 135. Let $f(x, y) = x^2 + y^2 - 2x - 6y + 14$. Find the critical points of f.

Also by completing the square, we have:

$$f(x, y) = 4 + (x-1)^2 + (y-3)^2$$

We can observe from this equation that $f(x, y)$ as a minimum at $(x, y) = (1, 3)$ and that the minimum value is 4.

End of Example 135. $\square$

Recall that the Hessian of a function of two variables is the 2×2 matrix

$$H_f = \begin{bmatrix} f_{xx} & f_{xy} \\ f_{yx} & f_{yy} \end{bmatrix}$$

Theorem 136 (Second Derivative Test)**.** *Let $f = f(x, y)$ be a function of two variables, and let (a, b) be a critical point of f. Assume the entries of the Hessian exist and are all continuous, and let*

$$D = \det(H_f(a, b)) = f_{xx}(a, b)f_{yy}(a, b) - [f_{xy}(a, b)]^2 .$$

Then there are three cases

(i) If $D > 0$ and $f_{xx}(a, b) > 0$, then $f(a, b)$ is a local minimum

(ii) If $D > 0$ and $f_{xx}(a, b) < 0$, then $f(a, b)$ is a local maximum

(iii) If $D < 0$ then $f(a, b)$ is a saddle point (i.e., neither a maximum nor a minimum)

Remark 137 (For those taking linear algebra)**.** Whether the function f has a minimum or maximum at (a, b) is determined by the eigenvalues of the matrix $H_f(a, b)$. Condition (i) is equivalent to the eigenvalues of $H_f(a, b)$ both being positive. Condition (ii) is equivalent to the eigenvalues both being negative. Condition (iii) is equivalent to having one negative and one positive eigenvalue.

Example 138. Let

$$f(x, y) = x^2 + xy + y^2 + y.$$

Find the critical points and classify them.

Solution:
Step 1. Find the gradient:

$$\nabla f = \begin{bmatrix} 2x + y \\ x + 2y + 1 \end{bmatrix}$$

Step 2. Set $\nabla f = 0$ and solve for (x, y):

$$2x + y = 0 \quad \text{and} \quad x + 2y + 1 = 0$$

so $y = -2x$ and hence $x = 1/3$, and so $y = -2/3$. We have found one critical point:

$$(x, y) = (\frac{1}{3}, -\frac{2}{3}).$$

Step 3. Classify the critical points using the second derivative test:

$$H_f(x, y) = \begin{bmatrix} 2 & 1 \\ 1 & 2 \end{bmatrix}$$

Therefore $D = H_f(\frac{1}{3}, -\frac{2}{3}) = 4 - 1 = 3 > 0$. Moreover, $f_{xx}(\frac{1}{3}, -\frac{2}{3}) = 2 > 0$, so the second derivative test implies that the function is concave up at the critical point $(\frac{1}{3}, -\frac{2}{3})$. Therefore, the critical point is a local minimum.

End of Example 138. $\square$

Example 139. Find the local maximum and minimum values and saddle points of

$$f(x, y) = x^4 + y^4 - 4xy + 1$$

Solution: Critical points are $(0, 0)$ (saddle point), $(-1, -1)$ (local minimum), and $(1, 1)$ (local minimum).

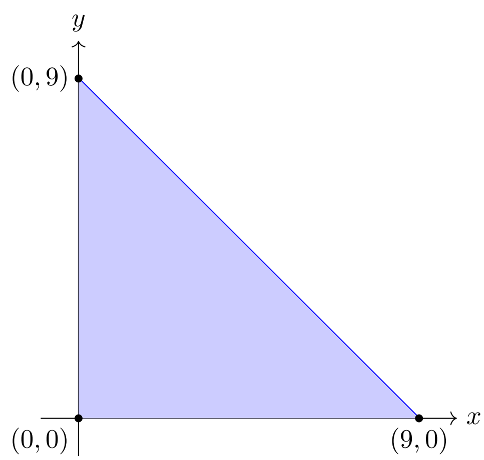
End of Example 139. $\square$

36 2025-04-28 | Week 15 | Lecture 36

36.1 Constrained Optimization

Textbook section 14.7

Example 140. Maximize $f(x, y) = 2 + 2x + 4y - x^2 - y^2$ on the triangle enclosed bythe lines $y = -x + 9$, $y = 0$, and $x = 0$, shown below:



Let's call this triangular region D.

General philosophy for this type of problem: First, maximize the function on the interior of its domain. Then maximize it on the boundary of the domain (which we denoted ∂D). Then compare all the points and pick the one(s) that yeild the biggest value of the function.
Interior: we find the critical points by setting $\Delta f = 0$ and solving for (x, y).

$$\Delta f = \begin{bmatrix} 2 - 2x \\ 4 - 2y \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \end{bmatrix}$$

This gives a single critical point $(x, y) = (1, 2)$. This critical point is in D. (If it hadn't been in D, we would have thrown it out as irrelevelt to the problem at hand.)

We classify the critical point using the second derivative test:

$$H_f(x, y) = \begin{bmatrix} f_{xx} & f_{yx} \\ f_{xy} & f_{yy} \end{bmatrix} = \begin{bmatrix} -2 & 0 \\ 0 & -2 \end{bmatrix}$$

Thus, $D = (-2)(-2) - (0)(0) = 16 > 0$ and $f(1, 2) = -2 < 0$. We conclude that the point $(1, 2)$ is a loal maximum by the second derivative test. The value of the local maximum is

$$f(1, 2) = 7$$

Boundary ∂D: There are three pieces to the buondary, corresponding to the three edges of the triangle.

Let B_1 be the horizontal line segement, B_2 the diagonal segment, and B_3 the vertical segment. We must consider each of these separately.

- For B_1, we have $y = 0$, so the value of the function is

  $$f(x, 0) = 2 + 2x - x^2$$

  with $0 \leq x \leq 9$. This is a downward opening parabola which is maximized when $x = 1$ (you can check this). At the point $(1, 0)$, the function takes value

  $$f(1, 0) = 3$$

- For B_3, we have $x = 0$, so the function value is

$$f(0, y) = 2 + 4y - y^2$$

with $0 \leq y \leq 9$. This is another downward opening parabola, which is maximized when $y = 2$. Thus we have the point $(0, 2)$, which has function value

$$f(0, 2) = 6.$$

- For B_2, we have $y = 9 - x$. On B_2, the value of the function is

$$\begin{aligned} f(x, 9 - x) &= 2 + 2x + 4(9 - x) - x^2 - (9 - x)^2 \\ &- 2x^2 + 16x - 43 \end{aligned}$$

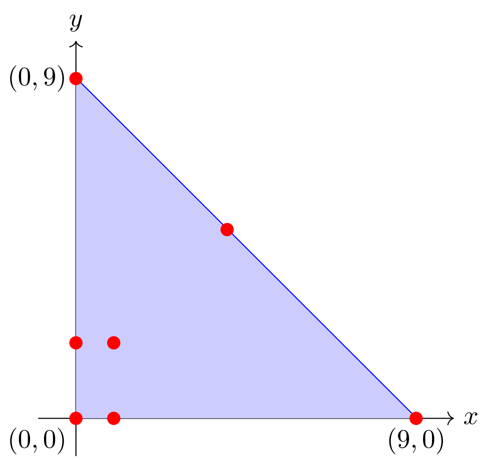
for $0 \leq x \leq 9$. Here, we yet again have a downward opening parabola, and it has maximum when $x = 4$ (so $y = 5$). Thus the function is maximzed on B_2 at the point $(4, 5)$, and the value is

$$f(4, 5) = -11$$

- We also have to check the corners, as these are the "boundaries" of each of the B_1, B_2, and B_3. These are the points $(0, 0), (9, 0), and (0, 9)$. At these three points, the function has values

$$f(0, 0) = 2, \quad f(9, 0) = -61, \quad \text{and} \quad f(0, 9) = -43$$

Putting both interior and boundary analyses together, we have 7 candidates for the global maximum, indicated in red:



We compare the value of the function at each of these points. The point $(1, 2)$ has $f(1, 2) = 7$ which is the largest function value for the points. Thus $(1, 2)$ is the location of the global maximum of the function, and the global maximum of the function is 7.

End of Example 140. □

Example 141. Let $f(x, y) = x^2 + y^2$. Find the global maximum and minimum of f on the region between the line $y = -2$ and the parabola $y = 2 - x^2$.
Similar approach

End of Example 141. □

36.2 Lagrange Multiplier

Example 142. Find the global max and min of the function

$$f(x,y) = x^2 - 2xy + 2y$$

on

$$D = \left\{(x,y) \in \mathbb{R}^2 : 0 \leq x \leq 3, 0 \leq y \leq 2\right\}$$

End of Example 142. □

Example 143. The function $h(x,y) = x - 2y$ represents the height in meters above sea level of some small area of land. Find the highest and lowest points on the ellipse $x^2 + 2y^2 = 4$. Sketch the ellipse and identify the points on your plot.

End of Example 143. □

36.3 Connection to Physics: Work

In everyday language, *work* means the total amount of effort required to perform a task. In physics, it has a techincal meaning which depends on the idea of force. In the simplest case, the amount of *work* W done by a *constant* force F which moves an object a distance d is

$$W = F \times d.$$

Work is measured in **Joules** (J). We have $\mathrm{J} = \mathrm{mN} = \mathrm{kg} \times \mathrm{m}^2/\mathrm{s}^2$. One Joule approximately the amount of work done to raise a stick of butter vertically one meter:

$$W = (\text{force}) \times (\text{distance}) = (1\text{N}) \times (1\text{m}) = 1\text{Nm} = 1\text{J}.$$

This example assumes that the force and object's motion the are lined up in the same direction. But what if the direction of motion is different from the direction of the force. For example, a surfer being pushed diagonally by a wave?

Try to use example 7 from chapter 12.

37 HW?

Example 144. Let

$$f(x, y) = \cos(x^2 y)$$

Find f_x, f_y, f_{xy}, f_{yx}. Also find $f_x(2, 1)$ and $f_y(2, 3)$

End of Example 144. □

Example 145. Let

$$f(x, y) = \sin\left(\frac{x}{1 + y}\right)$$

Find $\frac{\partial f}{\partial x}$ and $\frac{\partial f}{\partial x}$

End of Example 145. □

Example 146. Find f_x, f_z, and f_z if $f(y, z) = e^{xy} \log(z)$

End of Example 146. □

Example 147. Taylor expand

$$f(x, y) = y \sin(x)$$

End of Example 147. □

Example 148. Taylor expand

$$f(x) = \sin(x) \cos(y)$$

End of Example 148. □